PHY 302: Mathematical Methods II

Complete theory, worked examples, practice problems and solved past papers

Compendium booklet (IISER Berhampur)

How to use this booklet

Coloured boxes:

- **Key result** – the formula, definition or theorem to memorise.
- **Intuition** – why it is true.
- **Extra** – background that makes the topic easier or more complete.
- **Common mistake** – traps to avoid.
- **Worked example** – a fully solved example (algebra shown).
- **Past paper question** followed by a **Solution** – every question from the previous mid-sem and end-sem papers (2024, 2025, 2026) is reproduced and solved in the section where it belongs. A table at the end (*Index of past papers*) tells you where each one is.
- **Practice problems** – end-of-chapter exercises with answers (and hints) for self-testing.

Conventions. Fourier transform: $g(\omega) = \int_{-\infty}^{\infty} f(t)\mathrm{e}^{-\mathrm{i}\omega t}\mathrm{d}t$, $f(t) = \frac{1}{2\pi}\int g(\omega)\mathrm{e}^{\mathrm{i}\omega t}\mathrm{d}\omega$ (the convention in which the exam's statement "FT of a constant c is $2\pi c\,\delta(\omega)$" is true). $\mathrm{i}^2 = -1$. Prerequisite: PHY 301 (complex analysis, special functions); the earlier "Complex Analysis" booklet covers residues, Gamma and Bessel functions.

Contents

I Sturm–Liouville Theory and Orthogonal Expansions 6

1 Function spaces, inner products and Hermitian operators 6

2 The Sturm–Liouville problem 6

2.1 The Lagrange identity: where the boundary conditions enter 7

2.2 The three fundamental theorems 7

3 Reducing a general second-order ODE to Sturm–Liouville form 8

4 Eigenvalue problems for $y'' + \lambda y = 0$ 9

5 Orthogonal expansions (generalised Fourier series) 10

5.1 Gram–Schmidt orthogonalisation: how Legendre polynomials arise 10

5.2 Worked expansions 11

II Fourier Series, Fourier Transforms and Laplace Transforms 14

6 Fourier series 14

6.1 Orthogonality of the trigonometric functions 14
6.2 Symmetry shortcuts and half-range series 14
6.3 Complex form, Parseval, term-by-term operations 15
6.4 Worked examples 15

7 Fourier transforms 17

7.1 From the series to the integral 17
7.2 Standard transforms (derivations) 17
7.3 Properties (with proofs) 18
7.4 Past paper: Fourier transform of an integral 19
7.5 Gaussian transform and the uncertainty relation 20
7.6 Fourier transforms in three dimensions and for radial functions 21
7.7 Applications 22

8 Laplace transforms 23

8.1 Table of transforms and short derivations 23
8.2 Properties with proofs 23
8.3 Convolution theorem 25
8.4 Inverse transforms 26
8.5 Solving differential equations 26

III Generalised Functions: the Dirac Delta 28

9 Motivation and definition 28

10 Properties (with proofs) 28

10.1 Distributions: the rigorous viewpoint 29

11 Worked examples and the exam question 29

12 Fourier representation and completeness 30

13 Delta functions in several dimensions 31

14 Principal value and the Sokhotski–Plemelj formula 31

IV Partial Differential Equations **33**

15 Basic notions **33**

16 First-order PDEs and the method of characteristics **33**

17 Classification of second-order PDEs; canonical forms **34**

17.1 d'Alembert's solution of the wave equation 35

18 Separation of variables: the method **36**

19 Laplace's equation in polar coordinates: the disk **36**

20 Cylindrical coordinates and Bessel functions **38**

21 Spherical coordinates and Legendre polynomials **40**

22 Poisson's equation and uniqueness **42**

23 The wave equation **42**

23.1 The circular drum 43

24 The heat (diffusion) equation **43**

25 Transform methods for PDEs **44**

V Green's Functions **45**

26 The idea **45**

27 Green's functions for ordinary differential equations **45**

27.1 Construction for an SL operator 45

27.2 Eigenfunction expansion of G 46

28 Free-space Green's functions for ∇^2 and Helmholtz **46**

29 Green's identities and the representation formula **48**

30 The method of images **49**

30.1 Plane 49

30.2 Sphere 49

30.3 Disk (2D) 50

31 Time-dependent Green's functions **50**

VI Integral Equations **52**

32 Classification **52**

33 Relation to differential equations **52**

34 Methods of solution **53**

34.1 Separable kernels (Fredholm, second kind) . . . 53

34.2 Neumann series (iteration) and the resolvent kernel . . . 53

34.3 Convolution kernels: Laplace and Fourier transforms . . . 54

34.4 Symmetric kernels: Hilbert–Schmidt theory . . . 55

VII Group Theory, Representations and Lie Algebras **57**

35 Groups: definition and examples **57**

35.1 Matrix groups: past paper questions . . . 58

36 Subgroups, cosets, Lagrange's theorem **59**

37 Conjugacy classes, normal subgroups, homomorphisms **59**

38 Permutation groups **60**

38.1 The group S_3 . . . 60

39 Representations of finite groups **61**

39.1 Character table of S_3 . . . 62

40 Continuous groups and Lie groups **63**

40.1 Generators and the Lie algebra . . . 63

40.2 $SO(2) \cong U(1)$ . . . 64

40.3 $SO(3)$ and $su(2)$ . . . 64

40.4 $SU(2)$: the exam question . . . 64

40.5 $SU(2) \to SO(3)$: the double cover . . . 66

41 Representations of $SU(2)$ and the rotation group **66**

41.1 Other Lie groups worth knowing . . . 67

VIII Complex-Variable Techniques: Past-Paper Problems **68**

42 Singularities, principal parts and residues **68**

43 Contour integrals by the residue theorem **69**

44 A real integral with a branch cut (keyhole) **69**

45 Binomial coefficients and a sum by residues **70**

46 Fibonacci numbers and the golden ratio **71**

IX Study Aids **74**

47 Index of past-paper questions **75**

48 Formula sheet **76**

49 Common mistakes to avoid **77**

Part I

Sturm–Liouville Theory and Orthogonal Expansions

1 Function spaces, inner products and Hermitian operators

Why this chapter comes first. Almost every method in this course (Fourier series, Bessel and Legendre expansions, separation of variables for PDEs, Green's functions, integral equations) is the same idea: *an operator that is Hermitian has a complete orthogonal set of eigenfunctions, and arbitrary functions can be expanded in them.* Sturm–Liouville (SL) theory is the precise statement of this for second-order differential operators.

Vectors $\leftrightarrow$ functions.

Finite dimensions	Function space
vector $\boldsymbol{v} = (v_1, \dots, v_N)$	function $f(x)$, $a \le x \le b$
inner product $\boldsymbol{u}^\dagger \boldsymbol{v} = \sum u_i^* v_i$	$\langle f, g\rangle = \displaystyle\int_a^b f^*(x) g(x)\, w(x)\, \mathrm{d}x$, $w(x) > 0$ the **weight**
norm $\|\boldsymbol{v}\|^2 = \boldsymbol{v}^\dagger \boldsymbol{v}$	$\|f\|^2 = \langle f, f\rangle = \int \lvert f\rvert^2 w\, \mathrm{d}x$
matrix A	linear operator $\mathcal{L}$ (e.g. a differential operator)
A Hermitian: $A = A^\dagger$	$\mathcal{L}$ Hermitian: $\langle f, \mathcal{L}g\rangle = \langle \mathcal{L}f, g\rangle$
eigenvector $A\boldsymbol{v} = \lambda \boldsymbol{v}$	eigenfunction $\mathcal{L}y = \lambda y$

Functions with $\|f\| < \infty$ form the Hilbert space $L^2_w[a,b]$. **Orthogonal** means $\langle f, g\rangle = 0$.

Common mistake / check

For differential operators, "Hermitian" depends on the *boundary conditions*: the identity $\langle f, \mathcal{L}g\rangle = \langle \mathcal{L}f, g\rangle$ is obtained by integrating by parts, and the boundary terms must vanish. An operator together with a set of boundary conditions defining its domain is called **self-adjoint**.

2 The Sturm–Liouville problem

Key result

A **Sturm–Liouville equation** on $[a,b]$ is

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(p(x)\frac{\mathrm{d}y}{\mathrm{d}x}\right) + q(x)\, y + \lambda\, w(x)\, y = 0, \qquad p > 0,\ w > 0 \text{ on } (a,b),$$

with p, q, w real. Write $\mathcal{L}y \equiv (py')' + qy$, so the equation is $\mathcal{L}y = -\lambda w y$. A value of λ for which a non-trivial solution satisfying the boundary conditions exists is an **eigenvalue**; the solution is the **eigenfunction**.

2.1 The Lagrange identity: where the boundary conditions enter

For two functions u, v:

$$v\,\mathcal{L}u - u\,\mathcal{L}v = v(pu')' + vqu - u(pv')' - uqv = v(pu')' - u(pv')'.$$

To recognise the right side as a total derivative, expand the two products:

$$\frac{\mathrm{d}}{\mathrm{d}x}\big[p\,v\,u'\big] = v\,(pu')' + p\,u'\,v', \qquad \frac{\mathrm{d}}{\mathrm{d}x}\big[p\,u\,v'\big] = u\,(pv')' + p\,v'\,u'.$$

Subtracting, the terms $pu'v'$ cancel:

$$\frac{\mathrm{d}}{\mathrm{d}x}\big[p(vu' - uv')\big] = v(pu')' - u(pv')' = v\,\mathcal{L}u - u\,\mathcal{L}v.$$

Key result

Lagrange identity (Green's formula in 1D):

$$\int_a^b \big(v\,\mathcal{L}u - u\,\mathcal{L}v\big)\mathrm{d}x = \Big[p\,(vu' - uv')\Big]_a^b.$$

$\mathcal{L}$ is Hermitian on the space of functions satisfying the boundary conditions $\iff$ the right-hand side vanishes for every pair u, v in the space.

Boundary conditions that make the right side vanish:

1. **Dirichlet:** $y(a) = y(b) = 0$ (every term contains u or v at the ends).
2. **Neumann:** $y'(a) = y'(b) = 0$.
3. **Robin / separated:** $\alpha_1 y(a) + \alpha_2 y'(a) = 0$ and $\beta_1 y(b) + \beta_2 y'(b) = 0$ (with real constants). Because u and v obey the same condition, the combination $vu' - uv'$ is zero at that end.
4. **Periodic:** $y(a) = y(b)$, $y'(a) = y'(b)$ and $p(a) = p(b)$.
5. **Singular end point:** if $p(a) = 0$ (e.g. $p = 1 - x^2$ at $x = \pm 1$ or $p = x$ at $x = 0$) require y to be *bounded* (regular) at a. Then $p(vu' - uv') \to 0$.

2.2 The three fundamental theorems

Key result

Let $\mathcal{L}$ be self-adjoint (boundary term zero). Then:

(1) The eigenvalues are real.

(2) Eigenfunctions belonging to different eigenvalues are orthogonal with respect to the weight w: $\displaystyle\int_a^b y_m^*\, y_n\, w\,\mathrm{d}x = 0$ for $\lambda_m \neq \lambda_n$.

(3) Completeness: the eigenvalues form a discrete sequence $\lambda_1 < \lambda_2 < \cdots \to \infty$, and the eigenfunctions $\{y_n\}$ form a complete set: every sufficiently well-behaved function satisfying the boundary conditions can be expanded in them.

Proof of (1). Let $\mathcal{L}y_n = -\lambda_n w y_n$. Complex-conjugate: $\mathcal{L}y_n^* = -\lambda_n^* w y_n^*$ (since p, q, w are real). Apply the Lagrange identity with $v = y_n^*$ and $u = y_n$:

$$\int_a^b \big(y_n^*\mathcal{L}y_n - y_n\mathcal{L}y_n^*\big)\mathrm{d}x = 0 \quad \text{(self-adjoint)}.$$

Substituting the eigenvalue equations,

$$\int_a^b \big(-\lambda_n w y_n^* y_n + \lambda_n^* w y_n y_n^* \big) \mathrm{d}x = (\lambda_n^* - \lambda_n) \int_a^b w |y_n|^2 \mathrm{d}x = 0.$$

Since $w > 0$ and $y_n \not\equiv 0$, $\int w|y_n|^2 > 0$, hence $\lambda_n^* = \lambda_n$. ■

Proof of (2). Let $\mathcal{L}y_m = -\lambda_m w y_m$ and $\mathcal{L}y_n = -\lambda_n w y_n$, with λ_m, λ_n real. Lagrange identity with $v = y_m^*$, $u = y_n$:

$$0 = \int_a^b \big(y_m^* \mathcal{L} y_n - y_n \mathcal{L} y_m^* \big) \mathrm{d}x = \int_a^b \big(-\lambda_n w y_m^* y_n + \lambda_m w y_n y_m^* \big) \mathrm{d}x = (\lambda_m - \lambda_n) \int_a^b w y_m^* y_n \, \mathrm{d}x.$$

If $\lambda_m \neq \lambda_n$ the integral must vanish. ■

Extra: degeneracy, oscillation theorem

For a regular SL problem with *separated* boundary conditions each eigenvalue is non-degenerate (two solutions with the same boundary condition at $x = a$ would have zero Wronskian, hence be proportional). With periodic conditions an eigenvalue can be doubly degenerate ($\sin nx$ and $\cos nx$ share $\lambda = n^2$); one then orthogonalises within the degenerate subspace (Gram–Schmidt). **Oscillation theorem:** the n-th eigenfunction (counting from $n = 1$ for the lowest eigenvalue) has exactly $n - 1$ zeros inside (a, b): higher eigenvalue means more nodes (exactly as for energy levels in quantum mechanics).

Intuition

The SL equation $\mathcal{L}y = -\lambda w y$ is the infinite-dimensional version of $A\boldsymbol{v} = \lambda \boldsymbol{v}$ for a Hermitian matrix. The theorems above are the function-space versions of: real eigenvalues, orthogonal eigenvectors, and a complete eigenbasis. The *weight* w is simply the metric of the inner product.

3 Reducing a general second-order ODE to Sturm–Liouville form

Any equation $a_2(x)y'' + a_1(x)y' + a_0(x)y + \lambda c(x)y = 0$ (with $a_2 \neq 0$) can be put into SL form by multiplying by an **integrating factor** $\mu(x)$. We want $\mu a_2 y'' + \mu a_1 y' = (py')' = py'' + p'y'$, i.e.

$$p = \mu a_2, \qquad p' = \mu a_1 \;\Rightarrow\; \frac{p'}{p} = \frac{a_1}{a_2} \;\Rightarrow\; p = \exp\Big(\int \frac{a_1}{a_2} \mathrm{d}x \Big), \quad \mu = \frac{p}{a_2}.$$

Then $q = \mu a_0$ and $w = \mu c$.

Hermite's equation $y'' - 2xy' + 2ny = 0$

$a_2 = 1$, $a_1 = -2x$, so $p = \exp\big(\int -2x \, \mathrm{d}x\big) = \mathrm{e}^{-x^2}$, $\mu = \mathrm{e}^{-x^2}$, $q = 0$ and $w = \mu \cdot 1 = \mathrm{e}^{-x^2}$ (with $\lambda = 2n$). The SL form: $(\mathrm{e}^{-x^2} y')' + 2n\mathrm{e}^{-x^2} y = 0$. Orthogonality weight e^{-x^2} on $(-\infty, \infty)$, as known.

Laguerre's equation $xy'' + (1 - x)y' + ny = 0$

$a_1/a_2 = (1-x)/x = \frac{1}{x} - 1$, so $p = \exp(\ln x - x) = x\mathrm{e}^{-x}$ and $\mu = p/a_2 = \mathrm{e}^{-x}$: $(x\mathrm{e}^{-x} y')' + n\mathrm{e}^{-x} y = 0$. Weight e^{-x} on $[0, \infty)$.

Equation	p	q	w	λ	Interval	Eigenfunctions
Fourier $y'' + \lambda y = 0$	1	0	1	n^2	$[-\pi, \pi]$	$\cos nx,\ \sin nx$
Legendre	$1 - x^2$	0	1	$l(l+1)$	$[-1, 1]$	$P_l(x)$
Assoc. Legendre	$1 - x^2$	$-\dfrac{m^2}{1-x^2}$	1	$l(l+1)$	$[-1, 1]$	$P_l^m(x)$
Bessel (ν fixed)	x	$-\dfrac{\nu^2}{x}$	x	α^2	$[0, b]$	$J_\nu(\alpha_n x)$
Hermite	e^{-x^2}	0	e^{-x^2}	$2n$	$(-\infty, \infty)$	$H_n(x)$
Laguerre	$x\mathrm{e}^{-x}$	0	e^{-x}	n	$[0, \infty)$	$L_n(x)$
Chebyshev (1st)	$\sqrt{1-x^2}$	0	$\dfrac{1}{\sqrt{1-x^2}}$	n^2	$[-1, 1]$	$T_n(x)$

Orthogonality and norms: $\displaystyle\int_{-1}^{1} P_l P_{l'}\,\mathrm{d}x = \frac{2}{2l+1}\delta_{ll'}$; $\displaystyle\int_0^b x J_\nu(\alpha_m x) J_\nu(\alpha_n x)\mathrm{d}x = \frac{b^2}{2} J_{\nu+1}^2(\alpha_n b)\,\delta_{mn}$ (with $J_\nu(\alpha_n b) = 0$); $\displaystyle\int H_m H_n \mathrm{e}^{-x^2}\mathrm{d}x = 2^n n!\sqrt{\pi}\,\delta_{mn}$; $\displaystyle\int_0^\infty L_m L_n \mathrm{e}^{-x}\mathrm{d}x = \delta_{mn}$.

A worked mini-case: Bessel's equation in SL form

$x^2y'' + xy' + (\alpha^2x^2 - \nu^2)y = 0$. Divide by x: $xy'' + y' + \big(\alpha^2 x - \frac{\nu^2}{x}\big)y = 0$, which is $(xy')' - \frac{\nu^2}{x}y + \alpha^2 xy = 0$. So $p = x$, $q = -\nu^2/x$, $w = x$, and the eigenvalue is α^2. $p(0) = 0$ is a singular endpoint: demand y bounded at 0, so $y = J_\nu(\alpha x)$. On $[0, b]$ with $y(b) = 0$ we need $J_\nu(\alpha b) = 0$, so $\alpha_n = z_{\nu,n}/b$ with $z_{\nu,n}$ the n-th zero of J_ν. Orthogonality has weight $w = x$: this is where the extra x in $\int x J_\nu(\alpha_m x) J_\nu(\alpha_n x)\,\mathrm{d}x$ comes from.

4 Eigenvalue problems for $y'' + \lambda y = 0$

This simplest SL operator ($p = w = 1$, $q = 0$) produces all the elementary Fourier series. Solutions: for $\lambda = k^2 > 0$, $y = A\cos kx + B\sin kx$.

Boundary conditions on $[0, L]$	Eigenvalues λ_n	Eigenfunctions
$y(0) = y(L) = 0$	$\big(\frac{n\pi}{L}\big)^2$, $n = 1, 2, \ldots$	$\sin\frac{n\pi x}{L}$
$y'(0) = y'(L) = 0$	$\big(\frac{n\pi}{L}\big)^2$, $n = 0, 1, 2, \ldots$	$\cos\frac{n\pi x}{L}$
$y(0) = 0,\ y'(L) = 0$	$\big(\frac{(n+1/2)\pi}{L}\big)^2$, $n = 0, 1, \ldots$	$\sin\frac{(n+1/2)\pi x}{L}$
$y'(0) = 0,\ y(L) = 0$	$\big(\frac{(n+1/2)\pi}{L}\big)^2$, $n = 0, 1, \ldots$	$\cos\frac{(n+1/2)\pi x}{L}$
periodic on $[-L, L]$	$\big(\frac{n\pi}{L}\big)^2$, $n = 0, 1, \ldots$ (doubly degenerate for $n \geq 1$)	$\cos\frac{n\pi x}{L},\ \sin\frac{n\pi x}{L}$

Derivation of the first row. $y(0) = 0$ gives $A = 0$; $y(L) = B\sin kL = 0$ gives $kL = n\pi$. **Third row:** $y(0) = 0 \Rightarrow A = 0$; $y'(L) = Bk\cos kL = 0 \Rightarrow kL = (n + \frac{1}{2})\pi$.

Robin condition leads to a transcendental equation

Solve $y'' + \lambda y = 0$, $y(0) = 0$, $y'(1) + hy(1) = 0$ ($h > 0$). With $y = \sin kx$: $k\cos k + h\sin k = 0$, i.e. $\tan k = -k/h$. The eigenvalues $\lambda = k^2$ are the positive roots of this equation (no closed form; for large n, $k_n \approx (n - \frac{1}{2})\pi$). Orthogonality of $\sin k_m x$ and $\sin k_n x$ on $[0, 1]$ still holds, as it must for a self-adjoint problem.

5 Orthogonal expansions (generalised Fourier series)

Let $\{\phi_n\}$ be the orthogonal eigenfunctions of an SL problem. To expand $f(x) = \sum_n c_n\phi_n(x)$, multiply by $w\phi_m^*$ and integrate; by orthogonality only $n = m$ survives:

$$\int_a^b w\phi_m^* f\,\mathrm{d}x = \sum_n c_n \int_a^b w\phi_m^*\phi_n\,\mathrm{d}x = c_m \int_a^b w|\phi_m|^2\mathrm{d}x$$

Key result

$$f(x) = \sum_n c_n\phi_n(x), \qquad c_n = \frac{\langle \phi_n, f\rangle}{\langle \phi_n, \phi_n\rangle} = \frac{\displaystyle\int_a^b w\,\phi_n^* f\,\mathrm{d}x}{\displaystyle\int_a^b w\,|\phi_n|^2\mathrm{d}x}.$$

Completeness relation (the expansion holds for *every* f): $\displaystyle\sum_n \frac{\phi_n(x)\phi_n^*(x')}{\|\phi_n\|^2} = \frac{\delta(x-x')}{w(x)}$.

Best approximation. Let $S_N = \sum_{n\le N} c_n\phi_n$. The error $\|f - \sum_{n\le N} a_n\phi_n\|^2$ is minimised, for fixed N, by choosing $a_n = c_n$:

$$\Big\|f - \sum a_n\phi_n\Big\|^2 = \|f\|^2 - \sum_n |c_n|^2\|\phi_n\|^2 + \sum_n \|\phi_n\|^2|a_n - c_n|^2.$$

(expand the square and complete the square in each a_n). Since the left side is ≥ 0:

Key result

Bessel's inequality: $\displaystyle\sum_n |c_n|^2\|\phi_n\|^2 \le \|f\|^2$. For a *complete* set equality holds (**Parseval's theorem**):

$$\int_a^b w|f|^2\mathrm{d}x = \sum_n |c_n|^2 \int_a^b w|\phi_n|^2\mathrm{d}x.$$

Convergence of the series is **in the mean**: $\|f - S_N\| \to 0$. Pointwise convergence needs extra smoothness; at a jump discontinuity the series converges to the midpoint of the jump.

5.1 Gram–Schmidt orthogonalisation: how Legendre polynomials arise

Given independent functions $1, x, x^2, \ldots$ on $[-1,1]$ with $w = 1$ (so $\langle f, g\rangle = \int_{-1}^1 fg\,\mathrm{d}x$):

$$\begin{aligned}
&\phi_0 = 1, \qquad \langle\phi_0,\phi_0\rangle = 2.\\
&\phi_1 = x - \frac{\langle\phi_0, x\rangle}{\langle\phi_0,\phi_0\rangle}\phi_0 = x - \frac{0}{2} = x \quad (\text{since } \int_{-1}^1 x\,\mathrm{d}x = 0).\\
&\phi_2 = x^2 - \frac{\langle\phi_0, x^2\rangle}{\langle\phi_0,\phi_0\rangle}\phi_0 - \frac{\langle\phi_1, x^2\rangle}{\langle\phi_1,\phi_1\rangle}\phi_1, \text{ with } \langle\phi_0, x^2\rangle = \tfrac{2}{3},\ \langle\phi_1, x^2\rangle = 0,\\
&\quad = x^2 - \frac{2/3}{2} = x^2 - \frac{1}{3}.
\end{aligned}$$

Rescaling so that $P(1) = 1$: $\phi_2 \cdot \frac{3}{2} = \frac{1}{2}(3x^2 - 1) = P_2(x)$. In the same way other weights give other families: weight e^{-x^2} on $\mathbb{R}$ gives Hermite, weight e^{-x} on $[0,\infty)$ gives Laguerre.

5.2 Worked expansions

Legendre series of $f(x) = x^2$

$c_l = \dfrac{2l+1}{2}\displaystyle\int_{-1}^{1} x^2 P_l\,\mathrm{d}x$. $l=0$: $\frac{1}{2}\int x^2 = \frac{1}{2}\cdot\frac{2}{3} = \frac{1}{3}$. $l=2$: $\frac{5}{2}\int_{-1}^{1} x^2\cdot\frac{1}{2}(3x^2-1)\mathrm{d}x = \frac{5}{4}\big[\frac{3}{5}x^5 - \frac{1}{3}x^3\big]_{-1}^{1} = \frac{5}{4}\cdot 2\big(\frac{3}{5}-\frac{1}{3}\big) = \frac{5}{2}\cdot\frac{4}{15} = \frac{2}{3}$. ($l=1,3$: odd integrand, zero; $l\geq 4$: x^2 is a combination of P_0, P_2 only.) So

$$x^2 = \tfrac{1}{3}P_0(x) + \tfrac{2}{3}P_2(x) \quad \text{(check: } \tfrac{1}{3} + \tfrac{2}{3}\cdot\tfrac{1}{2}(3x^2-1) = x^2\ \checkmark\text{)}.$$

Legendre series of $f(x) = |x|$

Even function, so only even l. $c_l = (2l+1)\int_0^1 xP_l(x)\,\mathrm{d}x$.

$$c_0 = \int_0^1 x\,\mathrm{d}x = \tfrac{1}{2},$$
$$c_2 = 5\int_0^1 x\cdot\tfrac{1}{2}(3x^2-1)\mathrm{d}x = \tfrac{5}{2}\Big[\tfrac{3}{4}x^4 - \tfrac{1}{2}x^2\Big]_0^1 = \tfrac{5}{2}\cdot\tfrac{1}{4} = \tfrac{5}{8},$$
$$c_4 = 9\int_0^1 x\cdot\tfrac{1}{8}(35x^4-30x^2+3)\mathrm{d}x = \tfrac{9}{8}\Big(\tfrac{35}{6} - \tfrac{30}{4} + \tfrac{3}{2}\Big) = \tfrac{9}{8}\Big(\tfrac{35}{6} - \tfrac{15}{2} + \tfrac{3}{2}\Big) = \tfrac{9}{8}\Big(-\tfrac{1}{6}\Big) = -\tfrac{3}{16}.$$

So $|x| = \frac{1}{2}P_0 + \frac{5}{8}P_2 - \frac{3}{16}P_4 + \cdots$. At $x=0$: $P_2(0) = -\frac{1}{2}$, $P_4(0) = \frac{3}{8}$: $\frac{1}{2} - \frac{5}{16} - \frac{9}{128} = 0.117$ (it tends to 0 slowly as more terms are added).

Fourier–Bessel series of $f(x) = 1$ on $(0,1)$

Expand $1 = \sum_n c_n J_0(\alpha_n x)$ with $J_0(\alpha_n) = 0$. Weight $w = x$, so

$$c_n = \frac{\int_0^1 xJ_0(\alpha_n x)\,\mathrm{d}x}{\int_0^1 xJ_0^2(\alpha_n x)\,\mathrm{d}x}.$$

Numerator: from $\dfrac{\mathrm{d}}{\mathrm{d}z}[zJ_1(z)] = zJ_0(z)$, $\int_0^1 xJ_0(\alpha x)\,\mathrm{d}x = \dfrac{1}{\alpha^2}\int_0^\alpha zJ_0(z)\,\mathrm{d}z = \dfrac{J_1(\alpha)}{\alpha}$. Denominator: $\frac{1}{2}J_1^2(\alpha_n)$. Therefore

$$c_n = \frac{J_1(\alpha_n)/\alpha_n}{\frac{1}{2}J_1^2(\alpha_n)} = \frac{2}{\alpha_n J_1(\alpha_n)}, \qquad 1 = \sum_{n=1}^{\infty}\frac{2}{\alpha_n J_1(\alpha_n)}J_0(\alpha_n x).$$

Numbers: $\alpha_1 = 2.4048$, $J_1(\alpha_1) = 0.5191 \Rightarrow c_1 = 1.602$; $\alpha_2 = 5.5201$, $J_1 = -0.3403 \Rightarrow c_2 = -1.065$; $\alpha_3 = 8.6537$, $J_1 = 0.2715 \Rightarrow c_3 = 0.851$. This is exactly the expansion needed for heat flow in a cylinder (Part IV).

Parseval check for $f = x$ on $[-1,1]$ in Legendre polynomials

$x = 1\cdot P_1$, so $c_1 = 1$ and all other $c_l = 0$. Parseval: $\int_{-1}^{1} x^2\mathrm{d}x = \frac{2}{3}$ and $\sum c_l^2\frac{2}{2l+1} = 1^2\cdot\frac{2}{3}$. ✓

Past paper: PHY 302 Mid-sem, 3.3.2025, Q5 (10 marks: 5+5)

Show that the linear operator

$$\mathcal{L} = \frac{1}{4}(1+x^2)^2\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{1}{2}x(1+x^2)\frac{\mathrm{d}}{\mathrm{d}x} + a,$$

acting on functions defined on $x \in [-1,1]$ and vanishing at the end points, is Hermitian with respect to the weight function $1/(1+x^2)$ (printed as "$1/(1+x)^2$" in the scan; see the note below). Here $a \in \mathbb{R}$. Now find two even eigenfunctions $f_1(x), f_2(x)$ of $\mathcal{L}f_i = \lambda_i f_i$. *Hint:* put $x = \tan(\theta/2)$; then $x \in [-1,1] \leftrightarrow \theta \in [-\pi/2, \pi/2]$, the boundary conditions become $u(\pm\pi/2) = 0$, and evenness means $u(\theta) = u(-\theta)$.

Solution

Part 1: Hermiticity. Write $\mathcal{L}f = \frac{1}{w}\big(pf'\big)' + af$ and find p, w. Match: $\frac{1}{4}(1+x^2)^2 f'' + \frac{1}{2}x(1+x^2)f' = \frac{1}{4}(1+x^2)\big[(1+x^2)f'' + 2xf'\big] = \frac{1}{4}(1+x^2)\left[(1+x^2)f'\right]'$. So with $p = 1+x^2$,

$$\mathcal{L}f = \frac{1}{w}\big[(1+x^2)f'\big]' + af, \qquad \frac{1}{w} = \frac{1+x^2}{4} \;\Rightarrow\; w = \frac{4}{1+x^2} \propto \frac{1}{1+x^2}.$$

(The constant 4 is irrelevant for Hermiticity.) Then for f, g vanishing at ± 1,

$$\begin{aligned}\langle f, \mathcal{L}g\rangle - \langle \mathcal{L}f, g\rangle &= \int_{-1}^{1} w\big(f^*\mathcal{L}g - g\,\mathcal{L}f^*\big)\mathrm{d}x = \int_{-1}^{1}\Big[f^*\big(pg'\big)' - g\big(pf^{*\prime}\big)'\Big]\mathrm{d}x \\ &= \Big[p\,(f^*g' - gf^{*\prime})\Big]_{-1}^{1} = 0,\end{aligned}$$

since $f(\pm1) = g(\pm1) = 0$ (the a terms cancel because a is real). Hence $\mathcal{L}$ is Hermitian with weight $w = 1/(1+x^2)$.

Part 2: change of variable. Let $x = \tan(\theta/2)$. Then $1 + x^2 = \sec^2(\theta/2) = 1/\cos^2(\theta/2)$ and $\mathrm{d}x = \frac{1}{2}\sec^2(\theta/2)\,\mathrm{d}\theta = \frac{1}{2}(1+x^2)\mathrm{d}\theta$. Therefore

$$\frac{\mathrm{d}}{\mathrm{d}x} = \frac{\mathrm{d}\theta}{\mathrm{d}x}\frac{\mathrm{d}}{\mathrm{d}\theta} = \frac{2}{1+x^2}\frac{\mathrm{d}}{\mathrm{d}\theta} \quad\Longrightarrow\quad (1+x^2)\frac{\mathrm{d}}{\mathrm{d}x} = 2\frac{\mathrm{d}}{\mathrm{d}\theta} \equiv 2\partial_\theta.$$

Now square: $(1+x^2)\dfrac{\mathrm{d}}{\mathrm{d}x}\left[(1+x^2)\dfrac{\mathrm{d}f}{\mathrm{d}x}\right] = (1+x^2)^2 f'' + (1+x^2)(2x)f'$, and the left side is $2\partial_\theta(2\partial_\theta f) = 4\partial_\theta^2 f$. Hence

$$(1+x^2)^2 f'' = 4\partial_\theta^2 f - 2x(1+x^2)f'.$$

Insert in $\mathcal{L}$:

$$\mathcal{L}f = \tfrac{1}{4}\big[4\partial_\theta^2 f - 2x(1+x^2)f'\big] + \tfrac{1}{2}x(1+x^2)f' + af = \partial_\theta^2 f - \tfrac{1}{2}x(1+x^2)f' + \tfrac{1}{2}x(1+x^2)f' + af = \partial_\theta^2 f + af.$$

The operator is simply $\mathcal{L} = \dfrac{\mathrm{d}^2}{\mathrm{d}\theta^2} + a$. (This also explains the weight: $w\,\mathrm{d}x \propto \frac{1}{1+x^2}\cdot\frac{1}{2}(1+x^2)\mathrm{d}\theta = \frac{1}{2}\mathrm{d}\theta$, the flat measure in θ.)

Part 3: eigenfunctions. Solve $u'' + au = \lambda u \Rightarrow u'' + (a-\lambda)u = 0$. Let $\kappa^2 = a - \lambda$. Even solutions: $u = \cos\kappa\theta$. Boundary condition $u(\pm\pi/2) = 0$: $\cos(\kappa\pi/2) = 0 \Rightarrow \kappa\dfrac{\pi}{2} = (2n+1)\dfrac{\pi}{2}$, i.e. $\kappa = 2n+1$, $n = 0, 1, 2, \ldots$ Thus $\lambda_n = a - (2n+1)^2$ and $u_n = \cos\big((2n+1)\theta\big)$.

Recast in x: $\theta = 2\arctan x$, $\cos\theta = \dfrac{1-x^2}{1+x^2}$ (since $\cos\theta = \dfrac{1-\tan^2(\theta/2)}{1+\tan^2(\theta/2)}$).

- $n = 0$: $f_1(x) = \cos\theta = \dfrac{1-x^2}{1+x^2}$, with $\lambda_1 = a - 1$. (Check at $x = \pm1$: 0. ✓)
- $n = 1$: $f_2(x) = \cos 3\theta = \cos\theta(4\cos^2\theta - 3)$. With $c = \dfrac{1-x^2}{1+x^2}$:

$$4c^2 - 3 = \frac{4(1-x^2)^2 - 3(1+x^2)^2}{(1+x^2)^2} = \frac{4 - 8x^2 + 4x^4 - 3 - 6x^2 - 3x^4}{(1+x^2)^2} = \frac{1 - 14x^2 + x^4}{(1+x^2)^2},$$

so $f_2(x) = \dfrac{(1-x^2)(1-14x^2+x^4)}{(1+x^2)^3}$, with $\lambda_2 = a - 9$.

Both vanish at $x = \pm1$ and are even. In general $f_n = \cos[(2n+1)\theta]$ and $\lambda_n = a - (2n+1)^2$.

Note on the exam paper

The weight in the scan reads $1/(1+x)^2$. Only $w \propto 1/(1+x^2)$ makes $\mathcal{L}$ self-adjoint: demanding $w\mathcal{L} = (py')' + \dots$ gives $p = w(1+x^2)^2/4$ and $p' = \tfrac{1}{2}wx(1+x^2)$, hence $w'/w = -2x/(1+x^2)$, i.e. $w = C/(1+x^2)$.

Practice problems

1. Put $y'' + 2y' + (1+\lambda)y = 0$ into SL form and give p, q, w. *Answer:* $p = \mathrm{e}^{2x}$, $q = \mathrm{e}^{2x}$, $w = \mathrm{e}^{2x}$ (the equation is $(\mathrm{e}^{2x}y')' + \mathrm{e}^{2x}y + \lambda \mathrm{e}^{2x}y = 0$).

2. Find the eigenvalues and eigenfunctions of $y'' + \lambda y = 0$ on $[0, \pi]$ with $y(0) = 0$, $y'(\pi) = 0$. *Answer:* $\lambda_n = (n + \tfrac{1}{2})^2$, $y_n = \sin[(n+\tfrac{1}{2})x]$, $n = 0, 1, \dots$

3. Expand $f(x) = 1$ on $(0, \pi)$ in the functions of Problem 2. *Hint:* $c_n = \dfrac{\int_0^\pi \sin[(n+\frac{1}{2})x]\,\mathrm{d}x}{\pi/2}$.

   Answer: $1 = \dfrac{4}{\pi}\sum_{n \ge 0} \dfrac{\sin[(n+\frac{1}{2})x]}{2n+1}$.

4. Use Gram–Schmidt with weight e^{-x} on $[0, \infty)$ on $\{1, x, x^2\}$ and show you obtain 1, $x - 1$, $x^2 - 4x + 2$ (proportional to L_0, L_1, L_2). *Hint:* $\int_0^\infty x^n \mathrm{e}^{-x}\mathrm{d}x = n!$.

5. Expand $f(x) = x^3$ in Legendre polynomials. *Answer:* $x^3 = \tfrac{3}{5}P_1 + \tfrac{2}{5}P_3$.

6. SL eigenvalues are real, but they can be negative. Show that $y'' + \lambda y = 0$, $y(0) = 0$, $y'(1) + hy(1) = 0$ has a negative eigenvalue when $h < -1$. *Answer:* put $\lambda = -\kappa^2$, $y = \sinh \kappa x$; the condition gives $\kappa \cosh \kappa + h \sinh \kappa = 0$, i.e. $\tanh \kappa = \kappa/|h|$, which has a positive root iff $|h| > 1$ (slope of tanh at 0 is 1).

7. Using the Fourier–Bessel result of this chapter, show $\displaystyle\sum_n \frac{2}{\alpha_n J_1(\alpha_n)} = 1$ (set $x = 0$) and check with the first three terms. *Answer:* partial sum $1.602 - 1.065 + 0.851 = 1.388$; the series converges slowly to 1 (conditionally).

Part II

Fourier Series, Fourier Transforms and Laplace Transforms

6 Fourier series

Idea. The functions $\cos nx, \sin nx$ are the eigenfunctions of $y'' + \lambda y = 0$ with periodic boundary conditions, so (by Part I) any reasonable 2π-periodic function can be expanded in them.

6.1 Orthogonality of the trigonometric functions

For integers $m, n \geq 0$ (using $\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$ etc.):

$$\int_{-\pi}^{\pi} \cos mx \cos nx \, \mathrm{d}x = \pi\delta_{mn} \ (m, n \geq 1), \qquad \int_{-\pi}^{\pi} \sin mx \sin nx \, \mathrm{d}x = \pi\delta_{mn} \ (m, n \geq 1),$$

$$\int_{-\pi}^{\pi} \cos mx \sin nx \, \mathrm{d}x = 0 \text{ for all } m, n, \qquad \int_{-\pi}^{\pi} \cos nx \, \mathrm{d}x = 2\pi\delta_{n0}.$$

Key result

A function f of period 2π (satisfying the Dirichlet conditions below) has the **Fourier series**

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \big(a_n \cos nx + b_n \sin nx\big),$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \cos nx \, \mathrm{d}x \ (n \geq 0), \qquad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \sin nx \, \mathrm{d}x \ (n \geq 1).$$

For period $2L$ replace $nx \to n\pi x/L$ and $\int_{-\pi}^{\pi} \to \int_{-L}^{L}$ with prefactor $\frac{1}{L}$.

Derivation of a_n: multiply the series by $\cos mx$ and integrate over a period; only the term $n = m$ survives, giving πa_m. For $m = 0$ the constant term gives $\frac{a_0}{2} \cdot 2\pi = \pi a_0$. The factor $\frac{1}{2}$ in $a_0/2$ makes the same formula work for $n = 0$.

Dirichlet conditions. If f is periodic, single-valued, has a finite number of maxima, minima and finite jumps in a period, and $\int |f| < \infty$, then the series converges to $f(x)$ at points of continuity and to the **midpoint** $\frac{1}{2}[f(x^+) + f(x^-)]$ at a jump.

Gibbs phenomenon. Near a jump the partial sums overshoot by about 9% of the jump, and this overshoot does not decrease as more terms are added (it just becomes narrower).

6.2 Symmetry shortcuts and half-range series

- f **even** ($f(-x) = f(x)$): all $b_n = 0$, $a_n = \frac{2}{\pi}\int_0^{\pi} f \cos nx \, \mathrm{d}x$ (cosine series).
- f **odd**: all $a_n = 0$, $b_n = \frac{2}{\pi}\int_0^{\pi} f \sin nx \, \mathrm{d}x$ (sine series).
- **Half-range expansions:** a function given only on $(0, L)$ can be expanded in a pure sine series (extend as odd) or a pure cosine series (extend as even). These correspond to the Dirichlet and Neumann eigenfunctions of Part I.

6.3 Complex form, Parseval, term-by-term operations

Using $\cos nx = \frac{e^{inx}+e^{-inx}}{2}$, $\sin nx = \frac{e^{inx}-e^{-inx}}{2i}$:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}, \qquad c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx}dx, \qquad c_n = \frac{a_n - ib_n}{2},\ c_{-n} = c_n^* \text{ (real } f).$$

Key result

Parseval: $\frac{1}{\pi}\int_{-\pi}^{\pi} f^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty}(a_n^2 + b_n^2)$, or $\frac{1}{2\pi}\int_{-\pi}^{\pi}|f|^2 dx = \sum_n |c_n|^2$ ("power in time = power in frequency").
A Fourier series can be *integrated* term by term; it can be *differentiated* term by term only if the result satisfies the Dirichlet conditions (continuity of f and periodicity needed).

6.4 Worked examples

Example 1: sawtooth $f(x) = x$ on $(-\pi, \pi)$

Odd, so $a_n = 0$ and $b_n = \frac{2}{\pi}\int_0^{\pi} x\sin nx\,dx$. Integrate by parts ($u = x$, $dv = \sin nx\,dx$):

$$\int_0^{\pi} x\sin nx\,dx = \Big[-\frac{x\cos nx}{n}\Big]_0^{\pi} + \frac{1}{n}\int_0^{\pi}\cos nx\,dx = -\frac{\pi(-1)^n}{n} + \frac{1}{n^2}\big[\sin nx\big]_0^{\pi} = \frac{\pi(-1)^{n+1}}{n}.$$

Hence $b_n = \dfrac{2(-1)^{n+1}}{n}$ and

$$x = 2\Big(\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \cdots\Big).$$

At $x = \pi/2$: $\frac{\pi}{2} = 2(1 - \frac{1}{3} + \frac{1}{5} - \cdots)$, i.e. Leibniz's series $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$. **Parseval:** $\frac{1}{\pi}\int_{-\pi}^{\pi} x^2 dx = \frac{2\pi^2}{3} = \sum b_n^2 = 4\sum \frac{1}{n^2}$, so $\displaystyle\sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}$.

Example 2: $f(x) = x^2$ on $(-\pi, \pi)$

Even, so $b_n = 0$. $a_0 = \frac{2}{\pi}\int_0^{\pi} x^2 dx = \frac{2}{\pi}\cdot\frac{\pi^3}{3} = \frac{2\pi^2}{3}$. For $n \geq 1$, $a_n = \frac{2}{\pi}\int_0^{\pi} x^2\cos nx\,dx$. Integrate by parts twice:

$$\int_0^{\pi} x^2\cos nx\,dx = \Big[\frac{x^2\sin nx}{n}\Big]_0^{\pi} - \frac{2}{n}\int_0^{\pi} x\sin nx\,dx = 0 - \frac{2}{n}\cdot\frac{\pi(-1)^{n+1}}{n} = \frac{2\pi(-1)^n}{n^2}.$$

(using the integral of Example 1). So $a_n = \dfrac{4(-1)^n}{n^2}$:

$$x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos nx.$$

At $x = \pi$ ($\cos n\pi = (-1)^n$): $\pi^2 = \frac{\pi^2}{3} + 4\sum\frac{1}{n^2} \Rightarrow \sum\frac{1}{n^2} = \frac{\pi^2}{6}$ again. At $x = 0$: $0 = \frac{\pi^2}{3} + 4\sum\frac{(-1)^n}{n^2} \Rightarrow \sum\frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$.

Example 3: square wave

$f=-1$ on $(-\pi,0)$, $f=+1$ on $(0,\pi)$. Odd: $b_n=\frac{2}{\pi}\int_0^\pi \sin nx\,dx=\frac{2}{n\pi}(1-(-1)^n)$, which is $\frac{4}{n\pi}$ for odd n and 0 for even n:

$$f(x)=\frac{4}{\pi}\Big(\sin x+\frac{\sin 3x}{3}+\frac{\sin 5x}{5}+\cdots\Big).$$

At the jump $x=0$ the series gives 0, the midpoint. **Parseval:** $\frac{1}{\pi}\int f^2=2=\frac{16}{\pi^2}\sum_{\text{odd}}\frac{1}{n^2}\Rightarrow \sum_{n\text{ odd}}\frac{1}{n^2}=\frac{\pi^2}{8}$.

Example 4: triangle wave $f(x)=|x|$

Even. $a_0=\frac{2}{\pi}\int_0^\pi x\,dx=\pi$. $a_n=\frac{2}{\pi}\int_0^\pi x\cos nx\,dx=\frac{2}{\pi}\Big\{\Big[\frac{x\sin nx}{n}\Big]_0^\pi-\frac{1}{n}\int_0^\pi\sin nx\,dx\Big\}=\frac{2}{\pi}\Big\{0+\frac{1}{n^2}\big[\cos nx\big]_0^\pi\Big\}=\frac{2((-1)^n-1)}{\pi n^2}$, i.e. $-\frac{4}{\pi n^2}$ for odd n, 0 for even n:

$$|x|=\frac{\pi}{2}-\frac{4}{\pi}\Big(\cos x+\frac{\cos 3x}{3^2}+\frac{\cos 5x}{5^2}+\cdots\Big).$$

Example 5: half-range sine series of $f(x)=1$ on $(0,L)$

$b_n=\frac{2}{L}\int_0^L\sin\frac{n\pi x}{L}dx=\frac{2}{n\pi}\big(1-(-1)^n\big)$, so $1=\frac{4}{\pi}\sum_{k=0}^{\infty}\frac{1}{2k+1}\sin\frac{(2k+1)\pi x}{L}$ for $0<x<L$. (The same function with a cosine series is trivially 1.)

Past paper: PHY 302 Mid-sem, 26.02.2026, Q4a (7+5 marks)

Obtain the Fourier series of a 2π-periodic function $f(t)=Pt^2+Qt$ in $-\pi<t<\pi$, where P and Q are arbitrary constants.

Solution

The function is the sum of an even part Pt^2 and an odd part Qt. By linearity, use the results of Examples 1 and 2.

$$a_0=\frac{1}{\pi}\int_{-\pi}^{\pi}(Pt^2+Qt)dt=\frac{P}{\pi}\cdot\frac{2\pi^3}{3}+0=\frac{2P\pi^2}{3},$$
$$a_n=\frac{1}{\pi}\int_{-\pi}^{\pi}(Pt^2+Qt)\cos nt\,dt=\frac{P}{\pi}\int_{-\pi}^{\pi}t^2\cos nt\,dt+0=\frac{4P(-1)^n}{n^2},$$
$$b_n=\frac{1}{\pi}\int_{-\pi}^{\pi}(Pt^2+Qt)\sin nt\,dt=0+\frac{Q}{\pi}\int_{-\pi}^{\pi}t\sin nt\,dt=\frac{2Q(-1)^{n+1}}{n}.$$

(The product of an even function with $\sin nt$ and of an odd function with $\cos nt$ is odd, so those integrals over the symmetric interval vanish.) Therefore

$$\boxed{f(t)=\frac{P\pi^2}{3}+\sum_{n=1}^{\infty}\Big[\frac{4P(-1)^n}{n^2}\cos nt+\frac{2Q(-1)^{n+1}}{n}\sin nt\Big]}$$

Convergence: f is continuous inside $(-\pi,\pi)$; at the end points $t=\pm\pi$ the periodic extension jumps if $Q\neq 0$ (from $P\pi^2+Q\pi$ to $P\pi^2-Q\pi$) and the series converges to the mean $P\pi^2$ (the sine terms vanish at $\pm\pi$): check with $\cos n\pi=(-1)^n$: $\frac{P\pi^2}{3}+4P\sum\frac{1}{n^2}=\frac{P\pi^2}{3}+\frac{2P\pi^2}{3}=P\pi^2$. ✓

Practice problems

1. Expand $f(x) = \pi - x$ on $(0, 2\pi)$ (period 2π). *Answer:* $2\sum_{n\geq 1} \frac{\sin nx}{n}$.
2. Expand $f(x) = x^2$ on $(0, 2\pi)$. *Hint:* $a_0 = \frac{8\pi^2}{3}$, $a_n = \frac{4}{n^2}$, $b_n = -\frac{4\pi}{n}$.
3. Use Parseval's theorem on the triangle wave of Example 4 to show $\sum_{n \text{ odd}} \frac{1}{n^4} = \frac{\pi^4}{96}$.
4. A periodic function has period $2L = 4$ and equals x on $(-2, 2)$. Write its Fourier series. *Answer:* $\frac{4}{\pi}\sum \frac{(-1)^{n+1}}{n} \sin\frac{n\pi x}{2}$.
5. Find the complex Fourier coefficients of $f(x) = \mathrm{e}^x$ on $(-\pi, \pi)$. *Answer:* $c_n = \dfrac{(-1)^n \sinh\pi}{\pi(1 - in)}$.

7 Fourier transforms

7.1 From the series to the integral

Take a function with period $2L$ and write its complex series: $f(x) = \sum_n c_n \mathrm{e}^{\mathrm{i}k_n x}$, $k_n = \frac{n\pi}{L}$, $c_n = \frac{1}{2L}\int_{-L}^{L} f(x')\mathrm{e}^{-\mathrm{i}k_n x'}\mathrm{d}x'$. The spacing $\Delta k = \pi/L$, so $\frac{1}{2L} = \frac{\Delta k}{2\pi}$. Substituting,

$$f(x) = \sum_n \frac{\Delta k}{2\pi}\, \mathrm{e}^{\mathrm{i}k_n x} \int_{-L}^{L} f(x')\, \mathrm{e}^{-\mathrm{i}k_n x'}\mathrm{d}x'.$$

As $L \to \infty$, $\Delta k \to 0$ and the sum becomes an integral:

Key result

Fourier integral theorem / transform pair:

$$g(\omega) = \int_{-\infty}^{\infty} f(t)\, \mathrm{e}^{-\mathrm{i}\omega t}\mathrm{d}t, \qquad f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} g(\omega)\, \mathrm{e}^{\mathrm{i}\omega t}\mathrm{d}\omega.$$

Conditions: f piecewise smooth and absolutely integrable. At a jump $f(t)$ is replaced by the mean of the one-sided limits. (Symmetric convention: $\frac{1}{\sqrt{2\pi}}$ in both formulas.)

Intuition

$g(\omega)$ tells you "how much of the frequency ω" is in f. The transform of a function narrow in t is wide in ω and vice versa; this is the origin of the uncertainty principle (Section 7.5).

Sine and cosine transforms (for functions on $(0, \infty)$; the formula sheet of the exam uses these):

$$F_c(\omega) = \sqrt{\tfrac{2}{\pi}}\int_0^{\infty} f(x)\cos\omega x\,\mathrm{d}x, \qquad F_s(\omega) = \sqrt{\tfrac{2}{\pi}}\int_0^{\infty} f(x)\sin\omega x\,\mathrm{d}x,$$

with inverses $f(x) = \sqrt{\frac{2}{\pi}}\int_0^{\infty} F_{c,s}(\omega)\{\cos,\sin\}\omega x\,\mathrm{d}\omega$. They arise from the full transform of the even / odd extension of f.

7.2 Standard transforms (derivations)

(1) Rectangular pulse $f = 1$ for $|t| < a$, else 0:

$$g(\omega) = \int_{-a}^{a} \mathrm{e}^{-\mathrm{i}\omega t}\mathrm{d}t = \frac{\mathrm{e}^{\mathrm{i}\omega a} - \mathrm{e}^{-\mathrm{i}\omega a}}{\mathrm{i}\omega} = \frac{2\sin\omega a}{\omega}.$$

(2) Two-sided exponential $f = \mathrm{e}^{-a|t|}$, $a > 0$:

$$g(\omega) = \int_{-\infty}^{0} \mathrm{e}^{at - \mathrm{i}\omega t}\mathrm{d}t + \int_{0}^{\infty} \mathrm{e}^{-at - \mathrm{i}\omega t}\mathrm{d}t = \frac{1}{a - \mathrm{i}\omega} + \frac{1}{a + \mathrm{i}\omega} = \frac{2a}{a^2 + \omega^2}.$$

(3) One-sided exponential $f = \mathrm{e}^{-at}H(t)$: $g = \int_0^\infty \mathrm{e}^{-(a+\mathrm{i}\omega)t}\mathrm{d}t = \dfrac{1}{a + \mathrm{i}\omega}$.

(4) Gaussian $f = \mathrm{e}^{-at^2}$ $(a > 0)$: $g(\omega) = \sqrt{\dfrac{\pi}{a}}\,\mathrm{e}^{-\omega^2/4a}$ (derived in the Gaussian example below).

(5) Dirac delta: $f = \delta(t - t_0) \Rightarrow g = \mathrm{e}^{-\mathrm{i}\omega t_0}$ (sifting property); in particular $\delta(t) \leftrightarrow 1$.

(6) Constant: by the inversion theorem applied to (5): $\delta(t) = \frac{1}{2\pi}\int \mathrm{e}^{\mathrm{i}\omega t}\mathrm{d}\omega$, so $\int \mathrm{e}^{\mathrm{i}\omega t}\mathrm{d}\omega = 2\pi\delta(t)$; thus $f = 1 \Rightarrow g = 2\pi\delta(\omega)$. Also $c \cdot 1 \leftrightarrow 2\pi c\,\delta(\omega)$.

(7) Sign and step: $\operatorname{sgn}(t) \leftrightarrow \dfrac{2}{\mathrm{i}\omega}$ (principal value), $H(t) = \frac{1}{2}(1 + \operatorname{sgn} t) \leftrightarrow \pi\delta(\omega) + \dfrac{1}{\mathrm{i}\omega}$.

(8) Cosine: $\cos\omega_0 t \leftrightarrow \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$.

7.3 Properties (with proofs)

Key result

Let $f \leftrightarrow g$ denote a transform pair. Then:

Property	Result
Linearity	$af_1 + bf_2 \leftrightarrow ag_1 + bg_2$
Time shift	$f(t - t_0) \leftrightarrow \mathrm{e}^{-\mathrm{i}\omega t_0} g(\omega)$
Frequency shift	$\mathrm{e}^{\mathrm{i}\omega_0 t} f(t) \leftrightarrow g(\omega - \omega_0)$
Scaling	$f(at) \leftrightarrow \dfrac{1}{\lvert a\rvert} g\!\left(\dfrac{\omega}{a}\right)$
Derivative	$f'(t) \leftrightarrow \mathrm{i}\omega\, g(\omega)$, $\quad f^{(n)} \leftrightarrow (\mathrm{i}\omega)^n g$
Multiply by t	$t\, f(t) \leftrightarrow \mathrm{i}\, g'(\omega)$
Convolution	$(f * h)(t) = \int f(u)h(t-u)\mathrm{d}u \;\leftrightarrow\; g(\omega)\, h_\omega(\omega)$ (product)
Product	$f(t)h(t) \leftrightarrow \frac{1}{2\pi}(g * h_\omega)(\omega)$
Parseval	$\displaystyle\int \lvert f\rvert^2 \mathrm{d}t = \frac{1}{2\pi}\int \lvert g\rvert^2 \mathrm{d}\omega$
Duality	$g(t) \leftrightarrow 2\pi f(-\omega)$

Proofs of the two most used.

Derivative: integrate by parts, $\int f'\mathrm{e}^{-\mathrm{i}\omega t}\mathrm{d}t = \left[f\mathrm{e}^{-\mathrm{i}\omega t}\right]_{-\infty}^{\infty} + \mathrm{i}\omega\int f\mathrm{e}^{-\mathrm{i}\omega t}\mathrm{d}t = \mathrm{i}\omega g$ (boundary term $= 0$ as $f \to 0$ at $\pm\infty$).

Convolution:

$$\int \mathrm{d}t\,\mathrm{e}^{-\mathrm{i}\omega t}\int \mathrm{d}u\, f(u)h(t-u) = \int \mathrm{d}u\, f(u)\mathrm{e}^{-\mathrm{i}\omega u}\int \mathrm{d}t\,\mathrm{e}^{-\mathrm{i}\omega(t-u)}h(t-u) = g(\omega)\, h_\omega(\omega),$$

by swapping the order of integration and substituting $v = t - u$.

Parseval: $\int \lvert f\rvert^2\mathrm{d}t = \int f^*(t)\frac{1}{2\pi}\int g(\omega)\mathrm{e}^{\mathrm{i}\omega t}\mathrm{d}\omega\,\mathrm{d}t = \frac{1}{2\pi}\int g(\omega)\Big[\int f^*\mathrm{e}^{\mathrm{i}\omega t}\mathrm{d}t\Big]\mathrm{d}\omega = \frac{1}{2\pi}\int g\, g^*\mathrm{d}\omega$.

7.4 Past paper: Fourier transform of an integral

Past paper: PHY 302 End-sem, 27.04.2024, Q2 (5 marks)

Prove the following property of the FT under integration:

$$\mathrm{FT}\Big(\int^t f(s)\,\mathrm{d}s\Big) = \frac{1}{\mathrm{i}\omega} g(\omega) + 2\pi c\,\delta(\omega),$$

where $g(\omega)$ is the FT of $f(t)$ and c is the constant of integration associated with the indefinite integral.

Solution

Strategy. Let $F(t)$ be an indefinite integral: $F'(t) = f(t)$. Any two such integrals differ by a constant, so $F(t) = \Phi(t) + c$ where Φ is a *specific* antiderivative and c the constant. The FT of the constant gives the term $2\pi c\,\delta(\omega)$; we must show that Φ contributes $g/(\mathrm{i}\omega)$ and nothing else. Naively, "$\mathrm{FT}(F') = \mathrm{i}\omega\,\mathrm{FT}(F)$ so $\mathrm{FT}(F) = g/\mathrm{i}\omega$" fails at $\omega = 0$ where we divided by zero: that is exactly where the delta function lives.

Step 1: choose the specific antiderivative. Take

$$\Phi(t) = \frac{1}{2}\int_{-\infty}^{\infty} \mathrm{sgn}(t-s)\, f(s)\,\mathrm{d}s = \frac{1}{2}\Big[\int_{-\infty}^{t} f(s)\mathrm{d}s - \int_{t}^{\infty} f(s)\mathrm{d}s\Big]$$

(assuming f integrable). Differentiating, $\Phi'(t) = \frac{1}{2}[f(t)+f(t)] = f(t)$, so Φ is an antiderivative; moreover $\Phi(\pm\infty) = \pm\frac{1}{2}\int f$, so Φ is "centred" with no constant offset (its average of the two end values is zero).

Step 2: transform Φ. $\Phi = \frac{1}{2}(\mathrm{sgn} * f)$, a convolution. Using $\mathrm{FT}[\mathrm{sgn}\,t] = \dfrac{2}{\mathrm{i}\omega}$ (principal value) and the convolution theorem,

$$\mathrm{FT}(\Phi) = \frac{1}{2}\cdot\frac{2}{\mathrm{i}\omega}\cdot g(\omega) = \frac{g(\omega)}{\mathrm{i}\omega}.$$

(Derivation of FT[sgn]: $\mathrm{sgn}\,t = \lim_{\epsilon\to 0}\left[\mathrm{e}^{-\epsilon t}H(t) - \mathrm{e}^{\epsilon t}H(-t)\right]$; transform: $\frac{1}{\epsilon+\mathrm{i}\omega} - \frac{1}{\epsilon-\mathrm{i}\omega} = \frac{-2\mathrm{i}\omega}{\epsilon^2+\omega^2} \to \frac{2}{\mathrm{i}\omega}$ as $\epsilon\to 0$ for $\omega \neq 0$; the combination is odd in ω, so there is no $\delta(\omega)$ term.)

Step 3: add the constant. $\mathrm{FT}(c) = c\int \mathrm{e}^{-\mathrm{i}\omega t}\mathrm{d}t = 2\pi c\,\delta(\omega)$. Hence

$$\boxed{\mathrm{FT}\Big(\int^t f(s)\mathrm{d}s\Big) = \frac{g(\omega)}{\mathrm{i}\omega} + 2\pi c\,\delta(\omega)} \qquad \blacksquare$$

Consistency checks. (i) Differentiate: $\mathrm{FT}(F') = \mathrm{i}\omega\Big[\dfrac{g}{\mathrm{i}\omega} + 2\pi c\delta(\omega)\Big] = g + 2\pi c\,\omega\delta(\omega) = g$, since $\omega\delta(\omega) = 0$. ✓(ii) $f = \delta(t)$ ($g = 1$): the integral $\int_{-\infty}^{t}\delta = H(t) = \Phi + \frac{1}{2}$, so $c = \frac{1}{2}$ and $\mathrm{FT}(H) = \frac{1}{\mathrm{i}\omega} + \pi\delta(\omega)$, matching property (7) above.

7.5 Gaussian transform and the uncertainty relation

Past papers: PHY 302 End-sem 27.04.2024 Q1 (6+2+2+5+5) and End-sem 05.05.2025 Q1 (12+4+4+5+5)

Given the normalised Gaussian distribution $f(t) = \dfrac{1}{\tau\sqrt{2\pi}} \exp\Big(-\dfrac{t^2}{2\tau^2}\Big)$, $-\infty < t < \infty$, find: (a) the Fourier transform $g(\omega)$ of $f(t)$; (b) the averages $\langle t\rangle$ and $\langle\omega\rangle$; (c) Δt and $\Delta\omega$, where $\Delta x = \sqrt{\langle (x - \langle x\rangle)^2\rangle}$ is the root-mean-square deviation.

Solution

(a) Fourier transform. *Method 1 (differential equation, no contour needed).* $f'(t) = -\dfrac{t}{\tau^2} f(t)$, i.e. $tf = -\tau^2 f'$. Transform both sides using the derivative and multiplication-by-t rules:

$$\mathrm{FT}(tf) = \mathrm{i}g'(\omega), \qquad \mathrm{FT}(-\tau^2 f') = -\tau^2\,\mathrm{i}\omega\, g(\omega).$$

Equating: $\mathrm{i}g' = -\mathrm{i}\tau^2\omega g \Rightarrow g' = -\tau^2\omega g \Rightarrow \dfrac{\mathrm{d}\ln g}{\mathrm{d}\omega} = -\tau^2\omega \Rightarrow g(\omega) = g(0)\mathrm{e}^{-\tau^2\omega^2/2}$. The constant is $g(0) = \int f\,\mathrm{d}t = 1$ (normalised). Hence

$$\boxed{g(\omega) = \exp\Big(-\frac{\omega^2\tau^2}{2}\Big)}$$

Method 2 (completing the square). $-\dfrac{t^2}{2\tau^2} - \mathrm{i}\omega t = -\dfrac{1}{2\tau^2}(t + \mathrm{i}\omega\tau^2)^2 - \dfrac{\omega^2\tau^2}{2}$ (expand to check: $(t + \mathrm{i}\omega\tau^2)^2 = t^2 + 2\mathrm{i}\omega\tau^2 t - \omega^2\tau^4$, so $-\frac{1}{2\tau^2}(\ldots) = -\frac{t^2}{2\tau^2} - \mathrm{i}\omega t + \frac{\omega^2\tau^2}{2}$). Then $g = \dfrac{\mathrm{e}^{-\omega^2\tau^2/2}}{\tau\sqrt{2\pi}} \int \mathrm{e}^{-(t+\mathrm{i}\omega\tau^2)^2/2\tau^2}\mathrm{d}t$, and shifting the contour back to the real axis (Cauchy; no singularities, arcs vanish) the integral is the plain Gaussian integral $\tau\sqrt{2\pi}$. Same result.

(b) Averages. *In t:* f is a normalised density, $\int f = 1$. $\langle t\rangle = \int t f(t)\mathrm{d}t = 0$ because the integrand is odd. *In ω:* the function $g(\omega) = \mathrm{e}^{-\omega^2\tau^2/2}$ is even and not normalised ($\int g\,\mathrm{d}\omega = \sqrt{2\pi}/\tau$ by the inversion formula, $2\pi f(0) = \sqrt{2\pi}/\tau$). Use the density $\rho(\omega) = g(\omega)\big/\int g = \dfrac{\tau}{\sqrt{2\pi}}\mathrm{e}^{-\omega^2\tau^2/2}$; then $\langle\omega\rangle = \int \omega\rho\,\mathrm{d}\omega = 0$ (odd integrand).

(c) Widths. Use $\int_{-\infty}^{\infty} \mathrm{e}^{-ax^2}\mathrm{d}x = \sqrt{\pi/a}$ and, differentiating with respect to a, $\int x^2\mathrm{e}^{-ax^2}\mathrm{d}x = \dfrac{\sqrt{\pi}}{2a^{3/2}}$. Therefore for f:

$$\langle t^2\rangle = \frac{1}{\tau\sqrt{2\pi}}\int t^2\mathrm{e}^{-t^2/2\tau^2}\mathrm{d}t = \frac{1}{\tau\sqrt{2\pi}}\cdot\frac{\sqrt{\pi}}{2}\Big(\frac{1}{2\tau^2}\Big)^{-3/2} = \frac{1}{\tau\sqrt{2\pi}}\cdot\frac{\sqrt{\pi}}{2}(2\tau^2)^{3/2} = \frac{2\sqrt{2}\,\tau^3\sqrt{\pi}}{2\tau\sqrt{2}\sqrt{\pi}} = \tau^2.$$

So $\Delta t = \sqrt{\langle t^2\rangle - \langle t\rangle^2} = \tau$. For $\rho(\omega)$ (a Gaussian with $a = \tau^2/2$) the same computation with $\tau \to 1/\tau$ gives $\langle\omega^2\rangle = 1/\tau^2$, so $\Delta\omega = 1/\tau$. Then $\Delta t\,\Delta\omega = 1$.

Alternative ("intensity") interpretation used in physics. If $|f|^2$ and $|g|^2$ are regarded as the distributions of t and ω (as for a wave packet; legitimate by Parseval), then $|f|^2 \propto \mathrm{e}^{-t^2/\tau^2}$ has variance $\tau^2/2$ and $|g|^2 = \mathrm{e}^{-\omega^2\tau^2}$ has variance $\dfrac{1}{2\tau^2}$, giving $\Delta t = \dfrac{\tau}{\sqrt{2}}$, $\Delta\omega = \dfrac{1}{\sqrt{2}\tau}$ and

$$\Delta t\,\Delta\omega = \tfrac{1}{2},$$

the *minimum* allowed by the uncertainty relation. In both readings $\langle t\rangle = \langle\omega\rangle = 0$.

Key result

Uncertainty relation. For f normalised so that $\int |f|^2 \mathrm{d}t = 1$ with $\Delta t, \Delta\omega$ the rms widths of $|f|^2$ and $|g|^2/2\pi$: $\Delta t\,\Delta\omega \geq \frac{1}{2}$, with equality only for Gaussians. *Proof sketch:* for $\langle t\rangle = 0$, $\int |tf|^2\mathrm{d}t \cdot \int |f'|^2\mathrm{d}t \geq \left|\int tf^* f'\mathrm{d}t\right|^2$ (Cauchy–Schwarz) and $\int t\,\frac{1}{2}(|f|^2)'\mathrm{d}t = -\frac{1}{2}\int |f|^2 = -\frac{1}{2}$ (integrate by parts); also $\int |f'|^2\mathrm{d}t = \frac{1}{2\pi}\int \omega^2|g|^2\mathrm{d}\omega$ by Parseval. Combining gives $(\Delta t)^2(\Delta\omega)^2 \geq \frac{1}{4}$. In quantum mechanics with $p = \hbar\omega\ldots$ this is $\Delta x\Delta p \geq \hbar/2$.

7.6 Fourier transforms in three dimensions and for radial functions

Key result

In 3D: $\tilde{f}(\boldsymbol{k}) = \displaystyle\int f(\boldsymbol{r})\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\mathrm{d}^3r$, $f(\boldsymbol{r}) = \displaystyle\int \frac{\mathrm{d}^3k}{(2\pi)^3}\tilde{f}(\boldsymbol{k})\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$. Gradient: $\nabla \leftrightarrow \mathrm{i}\boldsymbol{k}$; Laplacian: $\nabla^2 \leftrightarrow -k^2$.

Past paper: PHY 302 Mid-sem, 26.02.2026, Q6 (5+3 marks)

(a) Use spherical coordinates to determine the general expression for the Fourier transform of an arbitrary function $f(\vec{r})$ depending only on the radial coordinate r. (b) Hence obtain the Fourier transform of $f(r) = \dfrac{\mathrm{e}^{-\alpha r}}{r}$, $\alpha > 0$.

Solution

(a) Choose the polar axis along $\boldsymbol{k}$, so $\boldsymbol{k}\cdot\boldsymbol{r} = kr\cos\theta$ and $\mathrm{d}^3r = r^2\sin\theta\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\phi$:

$$\tilde{f}(k) = \int_0^\infty r^2 f(r)\mathrm{d}r \int_0^{2\pi}\mathrm{d}\phi \int_0^\pi \mathrm{e}^{-\mathrm{i}kr\cos\theta}\sin\theta\,\mathrm{d}\theta.$$

The ϕ-integral gives 2π. For the θ-integral put $u = \cos\theta$ ($\mathrm{d}u = -\sin\theta\,\mathrm{d}\theta$):

$$\int_{-1}^{1}\mathrm{e}^{-\mathrm{i}kru}\mathrm{d}u = \frac{\mathrm{e}^{\mathrm{i}kr} - \mathrm{e}^{-\mathrm{i}kr}}{\mathrm{i}kr} = \frac{2\sin kr}{kr}.$$

Hence

$$\tilde{f}(k) = 2\pi\int_0^\infty r^2 f(r)\frac{2\sin kr}{kr}\mathrm{d}r = \boxed{\frac{4\pi}{k}\int_0^\infty r\,f(r)\sin(kr)\,\mathrm{d}r}$$

(b) With $f = \mathrm{e}^{-\alpha r}/r$, $rf = \mathrm{e}^{-\alpha r}$. Use $\int_0^\infty \mathrm{e}^{-ar}\sin br\,\mathrm{d}r = \dfrac{b}{a^2+b^2}$ (this is the "useful formula" printed in the QM exam too; derive it: $\int_0^\infty \mathrm{e}^{-ar}\mathrm{e}^{\mathrm{i}br}\mathrm{d}r = \frac{1}{a-\mathrm{i}b} = \frac{a+\mathrm{i}b}{a^2+b^2}$, imaginary part $\frac{b}{a^2+b^2}$):

$$\tilde{f}(k) = \frac{4\pi}{k}\cdot\frac{k}{\alpha^2+k^2} = \boxed{\frac{4\pi}{k^2+\alpha^2}}$$

As $\alpha \to 0$, $\tilde{f} \to 4\pi/k^2$: the transform of the Coulomb potential $1/r$. (This is the Yukawa-to-Coulomb limit used in Born scattering.) Check: $(\nabla^2 - \alpha^2)\frac{\mathrm{e}^{-\alpha r}}{r} = -4\pi\delta^3(\boldsymbol{r})$; transforming, $-(k^2+\alpha^2)\tilde{f} = -4\pi$. ✓

7.7 Applications

Damped driven oscillator by Fourier transform

$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = F(t)$. Transform ($\dot{x} \to \mathrm{i}\omega\tilde{x}$): $(-\omega^2 + 2\mathrm{i}\gamma\omega + \omega_0^2)\tilde{x} = \tilde{F}$, so $\tilde{x}(\omega) = \dfrac{\tilde{F}(\omega)}{\omega_0^2 - \omega^2 + 2\mathrm{i}\gamma\omega} \equiv \chi(\omega)\tilde{F}(\omega)$. By the convolution theorem $x(t) = \int G(t-t')F(t')\mathrm{d}t'$ with $G = \frac{1}{2\pi}\int \chi(\omega)\mathrm{e}^{\mathrm{i}\omega t}\mathrm{d}\omega$. The poles of χ are at $\omega = \mathrm{i}\gamma \pm \sqrt{\omega_0^2 - \gamma^2}$, both in the *upper* half-plane. For $t < 0$, $\mathrm{e}^{\mathrm{i}\omega t}$ decays in the lower half-plane, where there are no poles, so $G = 0$ (causality: no response before the force acts). For $t > 0$ close in the upper half-plane and pick the two poles: $G(t) = \dfrac{\mathrm{e}^{-\gamma t}\sin(\Omega t)}{\Omega}H(t)$, $\Omega = \sqrt{\omega_0^2 - \gamma^2}$.

Diffusion on the infinite line

$u_t = \kappa u_{xx}$, $u(x,0) = \varphi(x)$. FT in x: $\tilde{u}_t = -\kappa k^2\tilde{u} \Rightarrow \tilde{u} = \tilde{\varphi}(k)\mathrm{e}^{-\kappa k^2 t}$. By the Gaussian pair, $\mathrm{e}^{-\kappa k^2 t}$ is the transform of $\dfrac{1}{\sqrt{4\pi\kappa t}}\mathrm{e}^{-x^2/4\kappa t}$ (compare $g = \mathrm{e}^{-\omega^2\tau^2/2}$ with $\tau^2 = 2\kappa t$). Convolution gives

$$u(x,t) = \int_{-\infty}^{\infty} \frac{\varphi(y)}{\sqrt{4\pi\kappa t}}\mathrm{e}^{-(x-y)^2/4\kappa t}\mathrm{d}y.$$

Laplace's equation in the upper half-plane

$u_{xx} + u_{yy} = 0$ for $y > 0$, $u(x,0) = f(x)$, bounded. FT in x: $\tilde{u}_{yy} - k^2\tilde{u} = 0 \Rightarrow \tilde{u} = \tilde{f}(k)\mathrm{e}^{-|k|y}$ (bounded as $y \to \infty$). Since $\mathrm{e}^{-y|k|} \leftrightarrow \dfrac{1}{\pi}\dfrac{y}{x^2 + y^2}$ (inverse of the pair $\mathrm{e}^{-a|t|} \leftrightarrow \frac{2a}{a^2+\omega^2}$ by duality),

$$u(x,y) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{y\, f(x')}{(x-x')^2 + y^2}\mathrm{d}x' \quad \text{(Poisson integral for the half-plane).}$$

Practice problems

1. Find the FT of $f(t) = \mathrm{e}^{-at}\cos\omega_0 t\, H(t)$. *Answer:* $\dfrac{a + \mathrm{i}\omega}{(a+\mathrm{i}\omega)^2 + \omega_0^2}$.
2. Using the convolution theorem find $f * f$ for $f = \mathrm{e}^{-at}H(t)$. *Answer:* $t\mathrm{e}^{-at}H(t)$.
3. Find the FT of $f(t) = \dfrac{1}{t^2 + a^2}$. *Answer:* $\dfrac{\pi}{a}\mathrm{e}^{-a|\omega|}$ (duality with the exponential).
4. Show $\int_{-\infty}^{\infty} \dfrac{\sin^2(a\omega)}{\omega^2}\mathrm{d}\omega = \pi a$ using Parseval and the rectangular pulse.
5. Find $\int_{-\infty}^{\infty} \dfrac{\mathrm{d}\omega}{(a^2 + \omega^2)^2}$ using Parseval on the transform pair $\mathrm{e}^{-a|t|} \leftrightarrow \frac{2a}{a^2+\omega^2}$. *Answer:* $\dfrac{\pi}{2a^3}$.
6. A pulse $f(t) = \mathrm{e}^{-t^2/2\tau^2}$: show that the FT width is inversely proportional to the pulse width and state the product of rms widths (intensity version). *Answer:* $\frac{1}{2}$.

8 Laplace transforms

Key result

For $f(t)$ defined for $t \geq 0$ and of exponential order ($|f| \leq Me^{at}$):

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty \mathrm{e}^{-st} f(t)\,\mathrm{d}t, \qquad \operatorname{Re} s > a.$$

Inverse (**Bromwich integral**): $f(t) = \dfrac{1}{2\pi \mathrm{i}} \displaystyle\int_{\gamma-\mathrm{i}\infty}^{\gamma+\mathrm{i}\infty} \mathrm{e}^{st} F(s)\,\mathrm{d}s$, with γ to the right of all singularities of F.

Intuition

The Laplace transform is a Fourier transform with an extra convergence factor $\mathrm{e}^{-\sigma t}$ ($s = \sigma + \mathrm{i}\omega$). It handles functions that do not decay (like e^{at}, t^n), switches on at $t = 0$, and automatically builds in the *initial conditions* of an ODE. This is why it is the standard tool for initial-value problems.

8.1 Table of transforms and short derivations

$f(t)$	$F(s)$	region	$f(t)$	$F(s)$	region
c	c/s	$s > 0$	$\sinh at$	$\dfrac{a}{s^2 - a^2}$	$s > \lvert a\rvert$
ct^n	$\dfrac{cn!}{s^{n+1}}$	$s > 0$	$\cosh at$	$\dfrac{s}{s^2 - a^2}$	$s > \lvert a\rvert$
$\sin bt$	$\dfrac{b}{s^2 + b^2}$	$s > 0$	$\mathrm{e}^{at}\sin bt$	$\dfrac{b}{(s-a)^2 + b^2}$	$s > a$
$\cos bt$	$\dfrac{s}{s^2 + b^2}$	$s > 0$	$\mathrm{e}^{at}\cos bt$	$\dfrac{s-a}{(s-a)^2 + b^2}$	$s > a$
e^{at}	$\dfrac{1}{s-a}$	$s > a$	$t^{1/2}$	$\dfrac{1}{2}\sqrt{\dfrac{\pi}{s^3}}$	$s > 0$
$t^n \mathrm{e}^{at}$	$\dfrac{n!}{(s-a)^{n+1}}$	$s > a$	$t^{-1/2}$	$\sqrt{\dfrac{\pi}{s}}$	$s > 0$
$\delta(t - t_0)$	e^{-st_0}	all s	$H(t - t_0)$	$\dfrac{\mathrm{e}^{-st_0}}{s}$	$s > 0$

(This is the Laplace table printed in the exam formula sheet.) **Derivations:** $\mathcal{L}\{\mathrm{e}^{at}\} = \int_0^\infty \mathrm{e}^{-(s-a)t}\mathrm{d}t = \frac{1}{s-a}$. $\mathcal{L}\{\sin bt\} = \operatorname{Im}\mathcal{L}\{\mathrm{e}^{\mathrm{i}bt}\} = \operatorname{Im}\frac{1}{s-\mathrm{i}b} = \operatorname{Im}\frac{s+\mathrm{i}b}{s^2+b^2} = \frac{b}{s^2+b^2}$ (and similarly cos from the real part). $\mathcal{L}\{t^n\} = \int_0^\infty t^n \mathrm{e}^{-st}\mathrm{d}t = \frac{\Gamma(n+1)}{s^{n+1}}$ (substitute $u = st$); with $n = -\frac{1}{2}$: $\Gamma(\frac{1}{2})s^{-1/2} = \sqrt{\pi/s}$. $\mathcal{L}\{\delta(t-t_0)\} = \int_0^\infty \delta(t-t_0)\mathrm{e}^{-st}\mathrm{d}t = \mathrm{e}^{-st_0}$. $\mathcal{L}\{\sinh at\} = \frac{1}{2}\big[\frac{1}{s-a} - \frac{1}{s+a}\big] = \frac{a}{s^2-a^2}$.

8.2 Properties with proofs

Key result

Property	Result
First shift	$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$
Second shift	$\mathcal{L}\{H(t-a)f(t-a)\} = e^{-as}F(s)$
Derivative	$\mathcal{L}\{f'\} = sF(s) - f(0)$; $\mathcal{L}\{f''\} = s^2F(s) - sf(0) - f'(0)$
Integral	$\mathcal{L}\{\int_0^t f(u)du\} = F(s)/s$
Multiplication by t^n	$\mathcal{L}\{t^n f\} = (-1)^n F^{(n)}(s)$
Division by t	$\mathcal{L}\{f/t\} = \int_s^\infty F(\sigma)d\sigma$
Periodic ($f(t+T) = f(t)$)	$F(s) = \dfrac{1}{1-e^{-sT}}\displaystyle\int_0^T e^{-st}f(t)dt$
Convolution	$\mathcal{L}\{f*g\} = F(s)G(s)$, $(f*g)(t) = \int_0^t f(u)g(t-u)du$
Initial/final value	$\lim_{s\to\infty} sF = f(0^+)$; $\lim_{s\to 0} sF = f(\infty)$ (if it exists)

Past paper: PHY 302 Mid-sem 26.02.2026, Q5a (3 marks)

If $\mathcal{L}[y(t)] = Y(p)$, then obtain $\mathcal{L}[y''(t)]$.

Solution

First the first derivative. Integrate by parts ($u = e^{-pt}$, $dv = y'dt$):

$$\mathcal{L}[y'] = \int_0^\infty e^{-pt}y'dt = \Big[e^{-pt}y\Big]_0^\infty + p\int_0^\infty e^{-pt}y\,dt = -y(0) + pY(p),$$

(the boundary term at infinity vanishes for p large enough). Apply this to y' in place of y:

$$\mathcal{L}[y''] = p\,\mathcal{L}[y'] - y'(0) = p\big(pY - y(0)\big) - y'(0) = \boxed{p^2Y(p) - p\,y(0) - y'(0)}$$

Past paper: PHY 302 Mid-sem 26.02.2026, Q2 (5 marks)

Evaluate the Laplace transform $\mathcal{L}\big[H(t-a)e^{-i\omega(t-a)}\big]$, where $a, \omega > 0$ and $H(t)$ is the Heaviside function.

Solution

Directly: for $t < a$ the integrand is zero, so

$$\int_0^\infty e^{-st}H(t-a)e^{-i\omega(t-a)}dt = \int_a^\infty e^{-st}e^{-i\omega(t-a)}dt.$$

Let $\tau = t - a$ ($t = \tau + a$, limits $0 \to \infty$): $= e^{-sa}\displaystyle\int_0^\infty e^{-s\tau}e^{-i\omega\tau}d\tau = e^{-as}\int_0^\infty e^{-(s+i\omega)\tau}d\tau$. The integral converges for $\mathrm{Re}\,s > 0$ and equals $\dfrac{1}{s+i\omega}$. Hence

$$\boxed{\mathcal{L}\big[H(t-a)e^{-i\omega(t-a)}\big] = \frac{e^{-as}}{s+i\omega}}$$

This is the second shifting theorem with $f(t) = e^{-i\omega t}$, $F(s) = \frac{1}{s+i\omega}$.

8.3 Convolution theorem

Past paper: PHY 302 Mid-sem 26.02.2026, Q3 (7 marks)

State and prove the Convolution Theorem related to the Laplace transform. Provide clear diagrams if required.

Solution

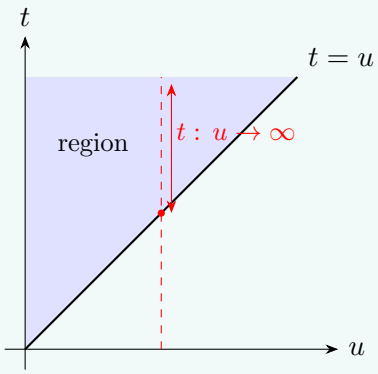
Statement. If $\mathcal{L}\{f\} = F(s)$ and $\mathcal{L}\{g\} = G(s)$, then the Laplace transform of the *convolution* $(f * g)(t) = \int_0^t f(u)\, g(t-u)\, \mathrm{d}u$ is the product:

$$\mathcal{L}\{f * g\} = F(s)\, G(s).$$

Proof.

$$\mathcal{L}\{f * g\} = \int_0^\infty \mathrm{e}^{-st} \Big[\int_0^t f(u) g(t-u) \mathrm{d}u \Big] \mathrm{d}t.$$

This is an integral of $\mathrm{e}^{-st} f(u) g(t-u)$ over the region $0 \le u \le t < \infty$ of the (u,t)-plane: the wedge between the t-axis and the line $t = u$ (shaded). We now integrate in the opposite order, first over t at fixed u (vertical strip: t from u to ∞), then over u from 0 to ∞.



Region of integration $0 \le u \le t$. Horizontal slicing (fixed t, u from 0 to t) is the definition; vertical slicing (fixed u, t from u to ∞) is what we use.

Interchanging the order (allowed since the integrals converge absolutely for $\mathrm{Re}\, s$ large):

$$\mathcal{L}\{f * g\} = \int_0^\infty f(u) \Big[\int_u^\infty \mathrm{e}^{-st} g(t-u) \mathrm{d}t \Big] \mathrm{d}u.$$

In the inner integral put $v = t - u$ ($\mathrm{d}t = \mathrm{d}v$, $v : 0 \to \infty$, $\mathrm{e}^{-st} = \mathrm{e}^{-su}\mathrm{e}^{-sv}$):

$$\int_u^\infty \mathrm{e}^{-st} g(t-u) \mathrm{d}t = \mathrm{e}^{-su} \int_0^\infty \mathrm{e}^{-sv} g(v) \mathrm{d}v = \mathrm{e}^{-su} G(s).$$

Therefore $\mathcal{L}\{f * g\} = G(s) \int_0^\infty f(u) \mathrm{e}^{-su} \mathrm{d}u = F(s) G(s)$. ■

Remarks. The convolution is commutative: $f * g = g * f$ (substitute $u \to t - u$); the theorem gives the inverse transform of a product, $\mathcal{L}^{-1}\{FG\} = f * g$. Example: $\mathcal{L}^{-1}\Big\{ \dfrac{1}{s(s^2+1)} \Big\} = 1 * \sin t = \int_0^t \sin u \, \mathrm{d}u = 1 - \cos t$.

8.4 Inverse transforms

(i) Partial fractions + table. Example: $F(s) = \dfrac{3s+5}{(s+1)(s+2)} = \dfrac{A}{s+1} + \dfrac{B}{s+2}$: $A = \dfrac{3(-1)+5}{(-1+2)} = 2$, $B = \dfrac{3(-2)+5}{(-2+1)} = 1$. So $f(t) = 2\mathrm{e}^{-t} + \mathrm{e}^{-2t}$.
(ii) Residues (Bromwich). Close the contour to the left: $f(t) = \sum_{\text{poles}} \text{Res}\left[F(s)\mathrm{e}^{st}\right]$. For a pole of order k at a: $\text{Res} = \dfrac{1}{(k-1)!}\lim_{s\to a}\dfrac{\mathrm{d}^{k-1}}{\mathrm{d}s^{k-1}}\left[(s-a)^k F(s)\mathrm{e}^{st}\right]$ (as printed on the formula sheet). Example: $F = \dfrac{1}{s(s+1)}$: simple poles at 0 (residue $\mathrm{e}^0/1 = 1$) and -1 (residue $\mathrm{e}^{-t}/(-1) = -\mathrm{e}^{-t}$): $f = 1 - \mathrm{e}^{-t}$. Example with double pole: $F = \dfrac{1}{(s+1)^2}$: $\dfrac{\mathrm{d}}{\mathrm{d}s}\mathrm{e}^{st}\Big|_{s=-1} = t\mathrm{e}^{-t}$.
(iii) Convolution theorem (above).

8.5 Solving differential equations

Past paper: PHY 302 Mid-sem 26.02.2026, Q5b (7 marks)

Use the Laplace transform technique to solve the initial value problem $y''(t) + \omega^2 y(t) = F\,\delta(t - t_0)$, $y(0) = A$, $y'(0) = 0$, where $t_0 \neq 0$ and F, ω, A are real constants.

Solution

Take $t_0 > 0$ (so that the delta function acts within $t > 0$). Transform, using $\mathcal{L}[y''] = p^2Y - py(0) - y'(0) = p^2Y - pA$ and $\mathcal{L}[\delta(t - t_0)] = \mathrm{e}^{-pt_0}$:

$$p^2Y - pA + \omega^2 Y = F\mathrm{e}^{-pt_0} \implies (p^2+\omega^2)Y = pA + F\mathrm{e}^{-pt_0} \implies Y(p) = \frac{Ap}{p^2+\omega^2} + F\frac{\mathrm{e}^{-pt_0}}{p^2+\omega^2}.$$

Invert term by term: $\mathcal{L}^{-1}\left\{\dfrac{p}{p^2+\omega^2}\right\} = \cos\omega t$; for the second term use the second shift theorem with $\mathcal{L}^{-1}\left\{\dfrac{1}{p^2+\omega^2}\right\} = \dfrac{\sin\omega t}{\omega}$:

$$\boxed{y(t) = A\cos\omega t + \frac{F}{\omega}\sin\left[\omega(t-t_0)\right]H(t-t_0)}$$

Interpretation: free oscillation $A\cos\omega t$ for all t; at $t = t_0$ the impulse F changes the velocity by F (jump in y': derivative of the second term at t_0^+ is $F\cos 0 = F$), after which the oscillator continues with the extra term. If $t_0 < 0$ the impulse acts before $t = 0$ and does not affect $t > 0$ in this initial-value formulation (the answer is then just $A\cos\omega t$).

Past paper: PHY 302 Mid-sem 26.02.2026, Q4b (5 marks)

Obtain the general solution of the inhomogeneous ODE $y''(t) + 3y(t) = t^2$.

Solution

Homogeneous part: $y'' + 3y = 0$, characteristic equation $r^2 + 3 = 0 \Rightarrow r = \pm\mathrm{i}\sqrt{3}$, so $y_h = A\cos\sqrt{3}\,t + B\sin\sqrt{3}\,t$.
Particular solution: the right side is a quadratic polynomial, try $y_p = \alpha t^2 + \beta t + \gamma$. Then

$y_p'' = 2\alpha$ and

$y_p''+3y_p = 2\alpha+3\alpha t^2+3\beta t+3\gamma = t^2 \;\Rightarrow\; 3\alpha = 1,\ 3\beta = 0,\ 2\alpha+3\gamma = 0 \;\Rightarrow\; \alpha = \tfrac{1}{3},\ \beta = 0,\ \gamma = -\tfrac{2}{9}$.

$$\boxed{y(t) = A\cos\sqrt{3}\,t + B\sin\sqrt{3}\,t + \frac{t^2}{3} - \frac{2}{9}}$$

(*Laplace check:* $s^2Y - sy_0 - y_0' + 3Y = \frac{2}{s^3} \Rightarrow Y = \frac{sy_0+y_0'}{s^2+3} + \frac{2}{s^3(s^2+3)}$. Partial fractions $\frac{2}{s^3(s^2+3)} = \frac{2}{3s^3} - \frac{2}{9s} + \frac{2s}{9(s^2+3)}$ give $\frac{t^2}{3} - \frac{2}{9} + \frac{2}{9}\cos\sqrt{3}t$, reproducing the same general solution after absorbing constants.)

LRC circuit

Series circuit, $L\ddot{q} + R\dot{q} + q/C = V_0$ for $t > 0$ with $q(0) = \dot{q}(0) = 0$ and constant voltage. $L(s^2Q) + R(sQ) + Q/C = V_0/s \Rightarrow Q = \dfrac{V_0}{s(Ls^2 + Rs + 1/C)}$. For the underdamped case the denominator has complex roots $-\gamma \pm \mathrm{i}\Omega$ and partial fractions give $q(t) = CV_0\Big[1 - \mathrm{e}^{-\gamma t}\Big(\cos\Omega t + \dfrac{\gamma}{\Omega}\sin\Omega t\Big)\Big]$ with $\gamma = \frac{R}{2L}$, $\Omega = \sqrt{\frac{1}{LC} - \gamma^2}$.

Heat flow in a semi-infinite rod

$u_t = \kappa u_{xx}$, $x > 0$, $u(x,0) = 0$, $u(0,t) = u_0$, u bounded. Transform in t: $s\bar{u} = \kappa\bar{u}'' \Rightarrow \bar{u} = C\mathrm{e}^{-x\sqrt{s/\kappa}}$ (bounded). Boundary: $\bar{u}(0,s) = u_0/s \Rightarrow \bar{u} = \dfrac{u_0}{s}\mathrm{e}^{-x\sqrt{s/\kappa}}$, whose inverse is $u(x,t) = u_0\,\mathrm{erfc}\Big(\dfrac{x}{2\sqrt{\kappa t}}\Big)$.

Practice problems

1. Find $\mathcal{L}\{t\sin bt\}$. *Answer:* $\dfrac{2bs}{(s^2+b^2)^2}$ (use $-F'(s)$).

2. Find $\mathcal{L}^{-1}\Big\{\dfrac{s}{(s^2+1)^2}\Big\}$. *Answer:* $\frac{1}{2}t\sin t$ (convolution $\cos t * \sin t$ or derivative rule).

3. Solve $y'' + 4y = \sin 2t$, $y(0) = y'(0) = 0$ by Laplace transforms. *Answer:* $\frac{1}{8}\sin 2t - \frac{1}{4}t\cos 2t$.

4. Solve the integral equation $y(t) = t + \int_0^t \sin(t-u)y(u)\mathrm{d}u$. *Answer:* $y = t + \frac{t^3}{6}$. *Hint:* $Y = 1/s^2 + Y/(s^2+1)$.

5. A rectangular pulse $f = 1$ on $0 < t < 2$, zero afterwards. Find $F(s)$ using Heaviside functions. *Answer:* $\dfrac{1-\mathrm{e}^{-2s}}{s}$.

6. A periodic square wave ($f = 1$ for $0 < t < a$, 0 for $a < t < 2a$, period $2a$): $F(s)$? *Answer:* $\dfrac{1-\mathrm{e}^{-as}}{s(1-\mathrm{e}^{-2as})} = \dfrac{1}{s(1+\mathrm{e}^{-as})}$.

7. Use the final-value theorem for $F(s) = \dfrac{3}{s(s+3)}$ and verify against $f(t) = 1 - \mathrm{e}^{-3t}$. *Answer:* 1.

Part III

Generalised Functions: the Dirac Delta

9 Motivation and definition

Physics repeatedly needs the idealisation of a "point" quantity: the mass density of a point mass, the charge density of a point charge, an instantaneous impulse. These cannot be ordinary functions: they would have to vanish except at one point, yet integrate to a finite number.

Key result

The **Dirac delta** $\delta(x-a)$ is defined by its action under an integral (the **sifting property**):

$$\int_{-\infty}^{\infty} f(x)\,\delta(x-a)\,\mathrm{d}x = f(a) \quad \text{for every continuous } f.$$

In particular $\int \delta(x-a)\mathrm{d}x = 1$. It is a **generalised function (distribution)**: a linear functional $f \mapsto f(a)$ on a space of "test functions". Its dimension is $1/[x]$.

Nascent (approximating) delta functions. $\delta(x) = \lim_{\epsilon\to 0}\delta_\epsilon(x)$ where δ_ϵ is any family of smooth functions of unit area that become narrower and taller:

Gaussian	$\delta_\epsilon(x) = \dfrac{1}{\epsilon\sqrt{\pi}}\mathrm{e}^{-x^2/\epsilon^2}$
Lorentzian	$\delta_\epsilon(x) = \dfrac{1}{\pi}\dfrac{\epsilon}{x^2+\epsilon^2}$
Rectangle	$\delta_\epsilon(x) = \dfrac{1}{2\epsilon}$ for $\lvert x\rvert < \epsilon$, 0 otherwise
Sinc (Dirichlet kernel)	$\delta_L(x) = \dfrac{\sin Lx}{\pi x}$, $L\to\infty$

Check (Lorentzian): $\int \frac{1}{\pi}\frac{\epsilon}{x^2+\epsilon^2}\mathrm{d}x = \frac{1}{\pi}\big[\arctan\frac{x}{\epsilon}\big]_{-\infty}^{\infty} = 1$. For any continuous f, $\int f\,\delta_\epsilon \to f(0)$ since the area concentrates near 0. *Check (sinc):* $\int_{-\infty}^{\infty}\frac{\sin Lx}{\pi x}\mathrm{d}x = \frac{1}{\pi}\int\frac{\sin u}{u}\mathrm{d}u = 1$.

Intuition

Picture δ as a spike of height $1/(2\epsilon)$ and width 2ϵ, then take $\epsilon\to 0$. All the rules below are what you get from the spike picture by taking the limit *after* doing the integral.

10 Properties (with proofs)

Key result

1. $\delta(-x) = \delta(x)$ (even).
2. $\delta(ax) = \dfrac{1}{|a|}\delta(x)$, $a \neq 0$.
3. $f(x)\delta(x-a) = f(a)\delta(x-a)$; in particular $x\,\delta(x) = 0$.
4. If $g(x)$ has simple zeros x_k (with $g'(x_k)\neq 0$): $\delta\big(g(x)\big) = \displaystyle\sum_k \frac{\delta(x-x_k)}{|g'(x_k)|}$.

5. **Derivative:** $\int f(x)\,\delta'(x-a)\mathrm{d}x = -f'(a)$; in general $\int f\,\delta^{(n)}(x-a)\mathrm{d}x = (-1)^n f^{(n)}(a)$.
6. $H'(x) = \delta(x)$ where H is the Heaviside step function.
7. $\int \delta(x-a)\delta(x-b)\mathrm{d}x = \delta(a-b)$.

Proofs.
(2): for $a>0$ put $u=ax$: $\int f(x)\delta(ax)\mathrm{d}x = \int f(u/a)\delta(u)\frac{\mathrm{d}u}{a} = \frac{f(0)}{a}$. For $a<0$, $\mathrm{d}x = \mathrm{d}u/a$ and the limits swap, giving $\frac{f(0)}{|a|}$. ■
(4): the delta only "sees" neighbourhoods of the zeros. Near x_k, $g(x)\approx g'(x_k)(x-x_k)$, so by (2) $\delta(g(x)) \approx \delta\big(g'(x_k)(x-x_k)\big) = \dfrac{\delta(x-x_k)}{|g'(x_k)|}$. Sum over the zeros. ■
(5): integrate by parts: $\int f\delta'\mathrm{d}x = [f\delta]_{-\infty}^{\infty} - \int f'\delta\,\mathrm{d}x = -f'(0)$ (the boundary term is zero since δ vanishes at infinity). Repeating n times gives $(-1)^n$. ■
(6): for a test function φ: $\int H'\varphi\,\mathrm{d}x = [H\varphi]_{-\infty}^{\infty} - \int H\varphi'\mathrm{d}x = 0 - \int_0^\infty \varphi'\mathrm{d}x = \varphi(0)$. ■
(3), $x\delta(x)=0$: $\int x\delta(x)\varphi(x)\mathrm{d}x = [x\varphi]_{x=0} = 0$ for every φ. Also $x\delta'(x) = -\delta(x)$: $\int x\delta'(x)\varphi\,\mathrm{d}x = -\frac{\mathrm{d}}{\mathrm{d}x}(x\varphi)\big|_0 = -\varphi(0)$.

Common mistake / check

$\delta(x)^2$ and $\delta(0)$ are meaningless. In $\delta(g(x))$ always divide by $|g'|$ with the *absolute value*. A delta at the end point of an integration interval gives an ambiguous half-weight: avoid it or use symmetric conventions.

10.1 Distributions: the rigorous viewpoint

A **test function** φ is infinitely differentiable and vanishes (with all derivatives) at infinity. A **distribution** T is a linear map $\varphi \mapsto \langle T,\varphi\rangle$. Ordinary integrable functions are distributions via $\langle f,\varphi\rangle = \int f\varphi\,\mathrm{d}x$. The **derivative** of a distribution is defined so that integration by parts always works:

$$\langle T',\varphi\rangle = -\langle T,\varphi'\rangle.$$

Derivatives of $|x|$ and the step function

$\dfrac{\mathrm{d}}{\mathrm{d}x}|x| = \mathrm{sgn}(x)$, and $\mathrm{sgn}(x) = 2H(x)-1$, so $\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}|x| = 2\delta(x)$. Check with a test function: $\int |x|\varphi''\mathrm{d}x = \int_0^\infty x\varphi'' - \int_{-\infty}^0 x\varphi'' = [x\varphi']_0^\infty - \int_0^\infty \varphi' - [x\varphi']_{-\infty}^0 + \int_{-\infty}^0 \varphi' = -(0-\varphi(0)) + \varphi(0) = 2\varphi(0)$. ✓(the kink in $|x|$ at 0 has "curvature" 2δ.)

11 Worked examples and the exam question

Past paper: PHY 302 Mid-sem 26.02.2026, Q1 (3 marks)

Evaluate the integral $\displaystyle\int_{-\pi/2}^{\pi/2} (x+1)^2\,\delta(\sin\pi x)\,\mathrm{d}x$.

Solution

Let $g(x) = \sin\pi x$. Zeros: $\sin\pi x = 0 \iff x = n \in \mathbb{Z}$. The integration limits are $\pm\pi/2 = \pm 1.5708\ldots$, so the zeros inside the interval are $x = -1,\ 0,\ +1$. The derivative is $g'(x) =$

$\pi\cos\pi x$, and $|g'(n)| = \pi|\cos n\pi| = \pi$ at every integer. By property (4):

$$\delta(\sin\pi x) = \frac{1}{\pi}\big[\delta(x+1) + \delta(x) + \delta(x-1)\big] \quad \text{(on this interval)}.$$

Hence

$$I = \frac{1}{\pi}\big[(x+1)^2\big]_{x=-1} + \frac{1}{\pi}\big[(x+1)^2\big]_{x=0} + \frac{1}{\pi}\big[(x+1)^2\big]_{x=1} = \frac{1}{\pi}\big(0+1+4\big) = \boxed{\frac{5}{\pi}}$$

(The points $x = \pm 2, \dots$ lie outside the range of integration.)

More evaluations

1. $\displaystyle\int_{-\infty}^{\infty} \cos x\, \delta(x^2 - \tfrac{\pi^2}{4})\,\mathrm{d}x$: zeros $x = \pm\pi/2$, $|g'| = |2x| = \pi$. Result $\frac{1}{\pi}\big[\cos\frac{\pi}{2} + \cos(-\frac{\pi}{2})\big] = 0$.
2. $\displaystyle\int_{0}^{\infty} \mathrm{e}^{-x}\delta(x^2-4)\mathrm{d}x$: only $x = 2$ lies in $(0,\infty)$; $|g'(2)| = 4$; result $\frac{1}{4}\mathrm{e}^{-2}$.
3. $\displaystyle\int_{-\infty}^{\infty} \mathrm{e}^{x}\,\delta'(x-1)\mathrm{d}x = -\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{e}^{x}\Big|_{x=1} = -\mathrm{e}$.
4. $\displaystyle\int_{-\infty}^{\infty} x^2\delta''(x-3)\mathrm{d}x = (+1)\frac{\mathrm{d}^2}{\mathrm{d}x^2}x^2\Big|_{x=3} = 2$.
5. $\delta(x^2 - a^2) = \dfrac{1}{2|a|}\big[\delta(x-a) + \delta(x+a)\big]$.
6. $\displaystyle\int_{-\infty}^{\infty} \delta(\mathrm{e}^x - 2)\mathrm{d}x = \frac{1}{\mathrm{e}^{x_0}}\Big|_{x_0=\ln 2} = \frac{1}{2}$.

12 Fourier representation and completeness

By the Fourier inversion theorem applied to $g(\omega) = \mathrm{e}^{-\mathrm{i}\omega t_0}$ (the transform of $\delta(t - t_0)$):

Key result

$$\delta(x - x') = \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}k(x-x')}\mathrm{d}k, \qquad \int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}kx}\mathrm{d}x = 2\pi\,\delta(k).$$

Check via the sinc sequence: $\frac{1}{2\pi}\int_{-L}^{L}\mathrm{e}^{\mathrm{i}kx}\mathrm{d}k = \frac{1}{2\pi}\frac{\mathrm{e}^{\mathrm{i}Lx} - \mathrm{e}^{-\mathrm{i}Lx}}{\mathrm{i}x} = \frac{\sin Lx}{\pi x} \to \delta(x)$ as $L \to \infty$.

Completeness of eigenfunctions ⇒ delta. If $\{\phi_n\}$ is a complete orthonormal set (weight w) then expanding $\delta(x - x')$ itself, $c_n = \int w\phi_n^*(x)\delta(x-x')\mathrm{d}x = w(x')\phi_n^*(x')$, so

$$\delta(x - x') = w(x')\sum_n \phi_n(x)\phi_n^*(x') \quad \text{(completeness relation)}.$$

For the Fourier series on $(-\pi, \pi)$: $\delta(x - x') = \frac{1}{2\pi}\sum_n \mathrm{e}^{\mathrm{i}n(x-x')}$ (the Dirac comb with period 2π).

Delta from the quantum-mechanical Fermi-golden-rule limit. The identity printed in the QM exam sheet, $\lim_{t\to\infty}\dfrac{\sin^2(xt/2)}{(x/2)^2} = 2\pi t\,\delta(x)$, follows from $\int_{-\infty}^{\infty}\dfrac{\sin^2(xt/2)}{(x/2)^2}\mathrm{d}x = 4\int\dfrac{\sin^2(xt/2)}{x^2}\mathrm{d}x =$

$4\cdot\dfrac{\pi t}{2}=2\pi t$ (using $\int\frac{\sin^2 au}{u^2}\mathrm{d}u=\pi a$) together with the fact that the function becomes concentrated at $x=0$ as $t\to\infty$.

13 Delta functions in several dimensions

Cartesian: $\delta^3(\boldsymbol{r}-\boldsymbol{r}_0)=\delta(x-x_0)\delta(y-y_0)\delta(z-z_0)$, with $\int\delta^3\mathrm{d}^3r=1$. In curvilinear coordinates the volume element has a Jacobian, so the delta must contain its inverse. For spherical coordinates:

$$\delta^3(\boldsymbol{r}-\boldsymbol{r}_0)=\frac{\delta(r-r_0)\,\delta(\theta-\theta_0)\,\delta(\phi-\phi_0)}{r^2\sin\theta}=\frac{\delta(r-r_0)\,\delta(\cos\theta-\cos\theta_0)\,\delta(\phi-\phi_0)}{r^2},$$

(because $\int\delta^3\,r^2\sin\theta\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\phi$ must be 1). For cylindrical coordinates $\delta^3=\dfrac{\delta(\rho-\rho_0)\delta(\phi-\phi_0)\delta(z-z_0)}{\rho}$.

A spherically symmetric delta centred at the origin can be written $\delta^3(\boldsymbol{r})=\dfrac{\delta(r)}{4\pi r^2}$ (with $\int_0^\infty\delta(r)\mathrm{d}r=1$).

Key result

Two fundamental identities of electrostatics and scattering theory:

$$\nabla\cdot\frac{\hat{\boldsymbol{r}}}{r^2}=4\pi\,\delta^3(\boldsymbol{r}),\qquad \nabla^2\frac{1}{r}=-4\pi\,\delta^3(\boldsymbol{r}),$$
$$(\nabla^2+k^2)\frac{\mathrm{e}^{\pm\mathrm{i}kr}}{r}=-4\pi\,\delta^3(\boldsymbol{r}),\qquad (\nabla^2-\mu^2)\frac{\mathrm{e}^{-\mu r}}{r}=-4\pi\,\delta^3(\boldsymbol{r}).$$

Proof of the first: for $r\neq0$, $\nabla\cdot(\hat{\boldsymbol{r}}/r^2)=\frac{1}{r^2}\partial_r(r^2\cdot r^{-2})=0$. For a ball of radius R around the origin, the divergence theorem gives $\int_V\nabla\cdot\frac{\hat{\boldsymbol{r}}}{r^2}\mathrm{d}V=\oint\frac{\hat{\boldsymbol{r}}}{R^2}\cdot R^2\hat{\boldsymbol{r}}\,\mathrm{d}\Omega=4\pi$, independent of R. A quantity that is 0 away from the origin and has integral 4π over any ball containing the origin is $4\pi\delta^3(\boldsymbol{r})$. Since $\nabla\frac{1}{r}=-\frac{\hat{\boldsymbol{r}}}{r^2}$, the second identity follows. For the third write $\mathrm{e}^{\mathrm{i}kr}/r=1/r+(\mathrm{e}^{\mathrm{i}kr}-1)/r$: the second part is continuous at the origin so produces no delta; away from the origin $(\nabla^2+k^2)\frac{\mathrm{e}^{\mathrm{i}kr}}{r}=\frac{1}{r}\frac{\mathrm{d}^2}{\mathrm{d}r^2}(r\cdot\frac{\mathrm{e}^{\mathrm{i}kr}}{r})+k^2\frac{\mathrm{e}^{\mathrm{i}kr}}{r}=\frac{-k^2+k^2}{r}\mathrm{e}^{\mathrm{i}kr}=0$.

14 Principal value and the Sokhotski–Plemelj formula

The function $1/x$ is not integrable at 0, but its **Cauchy principal value** $\mathrm{P}\frac{1}{x}$ is defined by $\int\mathrm{P}\frac{1}{x}\varphi\,\mathrm{d}x=\lim_{\epsilon\to0}\left(\int_{-\infty}^{-\epsilon}+\int_{\epsilon}^{\infty}\right)\frac{\varphi}{x}\mathrm{d}x$. Consider $\dfrac{1}{x\mp\mathrm{i}\epsilon}=\dfrac{x\pm\mathrm{i}\epsilon}{x^2+\epsilon^2}$. The real part $\frac{x}{x^2+\epsilon^2}\to\mathrm{P}\frac{1}{x}$ and the imaginary part $\pm\frac{\epsilon}{x^2+\epsilon^2}\to\pm\pi\delta(x)$ (Lorentzian):

Key result

$$\lim_{\epsilon\to0^+}\frac{1}{x-\mathrm{i}\epsilon}=\mathrm{P}\frac{1}{x}+\mathrm{i}\pi\,\delta(x),\qquad \lim_{\epsilon\to0^+}\frac{1}{x+\mathrm{i}\epsilon}=\mathrm{P}\frac{1}{x}-\mathrm{i}\pi\,\delta(x).$$

This is exactly what is needed to interpret the $\mathrm{i}\epsilon$ prescriptions in scattering theory ($1/(E-H_0+\mathrm{i}\epsilon)$) and in contour integrals with poles on the real axis (Part I: indentation gives $\pm\pi\mathrm{i}$ Res).

An application

Evaluate $\displaystyle\int_{-\infty}^{\infty}\frac{\mathrm{d}x}{(x-\mathrm{i}0)(1+x^2)}$. By Sokhotski–Plemelj this is $\mathrm{P}\int\frac{\mathrm{d}x}{x(1+x^2)}+\mathrm{i}\pi\cdot\frac{1}{1+0}=0+\mathrm{i}\pi$, the principal value vanishing because the integrand is odd. Check by residues: with $\epsilon>0$ the pole $x=\mathrm{i}\epsilon$ and the pole $x=\mathrm{i}$ lie in the upper half-plane; residues $\to1$ and $\frac{1}{(\mathrm{i})(2\mathrm{i})}=-\frac{1}{2}$; total

$2\pi\mathrm{i}\cdot\frac{1}{2}=\mathrm{i}\pi$. ✓

Practice problems

1. Evaluate $\displaystyle\int_{-\infty}^{\infty} \mathrm{e}^{-x^2}\delta(\sin x)\mathrm{d}x$ summing all zeros. *Answer:* $\sum_{n\in\mathbb{Z}}\mathrm{e}^{-n^2\pi^2} = 1 + 2\sum_{n\geq 1}\mathrm{e}^{-n^2\pi^2}\approx 1.00010$.
2. Evaluate $\displaystyle\int_0^{2\pi}\sin x\,\delta(\cos x)\mathrm{d}x$. *Answer:* zeros at $\frac{\pi}{2},\frac{3\pi}{2}$, $|g'|=1$: $\sin\frac{\pi}{2}+\sin\frac{3\pi}{2}=0$.
3. Show $\delta(x^2-1)\,x^3$ integrates to 0 over $\mathbb{R}$. *Hint:* $\frac{1}{2}[1^3+(-1)^3]=0$.
4. Show that $\dfrac{\mathrm{d}}{\mathrm{d}x}\big[\mathrm{sgn}(x)\big]=2\delta(x)$ and $\dfrac{\mathrm{d}}{\mathrm{d}x}\big[x\,H(x)\big]=H(x)$.
5. Find the derivative of $f(x)=\mathrm{e}^{-|x|}$ and f'' in the sense of distributions. *Answer:* $f'=-\mathrm{sgn}(x)\mathrm{e}^{-|x|}$, $f''=\mathrm{e}^{-|x|}-2\delta(x)$.
6. Prove $\delta(\boldsymbol{r}-\boldsymbol{r}')=\sum_{lm}\dfrac{\delta(r-r')}{r^2}Y_{lm}(\theta,\phi)Y_{lm}^*(\theta',\phi')$. *Hint:* completeness of Y_{lm}.
7. Compute the 3D Fourier transform of $\delta^3(\boldsymbol{r}-\boldsymbol{a})$. *Answer:* $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{a}}$.

Part IV
Partial Differential Equations

15 Basic notions

A **partial differential equation** relates a function $u(x, y, \dots)$ of several variables to its partial derivatives. Its **order** is that of the highest derivative; it is **linear** if u and its derivatives appear only linearly. For a linear *homogeneous* PDE the **superposition principle** holds: sums and constant multiples of solutions are solutions. This is what allows series (Fourier, Bessel, Legendre) solutions.

Name	Equation	Type
Laplace	$\nabla^2 u = 0$	elliptic
Poisson	$\nabla^2 u = -\rho/\epsilon_0$ (or $= f$)	elliptic
Helmholtz	$(\nabla^2 + k^2)u = 0$	elliptic
Diffusion / heat	$u_t = \kappa \nabla^2 u$	parabolic
Wave	$u_{tt} = c^2 \nabla^2 u$	hyperbolic
Schrödinger	$\mathrm{i}\hbar \psi_t = -\frac{\hbar^2}{2m}\nabla^2 \psi + V\psi$	parabolic-like (complex)

Boundary and initial conditions. On a boundary ∂D: **Dirichlet** (u given), **Neumann** ($\partial u/\partial n$ given), **Robin** ($\alpha u + \beta \partial_n u$ given). In time-dependent problems we also need **initial conditions**: $u(\cdot, 0)$ for diffusion; $u(\cdot, 0)$ and $u_t(\cdot, 0)$ for waves.

Key result

Well-posed problem (Hadamard): a solution exists, is unique, and depends continuously on the data. Typical well-posed problems: elliptic equations with Dirichlet or Neumann data on a closed boundary; heat equation with initial data and boundary data; wave equation with initial data and boundary data. (Laplace's equation with Cauchy data on a line is *ill-posed*.)

16 First-order PDEs and the method of characteristics

Consider $a(x, y)\, u_x + b(x, y)\, u_y = 0$. Along a curve $(x(s), y(s))$ with $\dfrac{\mathrm{d}x}{\mathrm{d}s} = a$, $\dfrac{\mathrm{d}y}{\mathrm{d}s} = b$, the chain rule gives $\dfrac{\mathrm{d}u}{\mathrm{d}s} = u_x a + u_y b = 0$: u **is constant along the characteristic curves**. The characteristics solve $\dfrac{\mathrm{d}x}{a} = \dfrac{\mathrm{d}y}{b}$, i.e. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{b}{a}$; if the integral is $\xi(x, y) = C$, the general solution is $u = f(\xi(x, y))$ with f arbitrary (fixed by data).

Key result

For $a\, u_x + b\, u_y = c$ (inhomogeneous or quasilinear), solve the system $\dfrac{\mathrm{d}x}{a} = \dfrac{\mathrm{d}y}{b} = \dfrac{\mathrm{d}u}{c}$.

Past papers: PHY 302 End-sem 27.04.2024 Q3 (5+5) and End-sem 05.05.2025 Q2 (5+5)

Given the PDE $x\dfrac{\partial u}{\partial x} - 2y\dfrac{\partial u}{\partial y} = 0$, (a) find the solution that takes the value $2y+1$ on the line $x=1$; (b) find a solution that has the value 4 at the point $(1,1)$.

Solution

Characteristics. $a = x$, $b = -2y$: $\dfrac{\mathrm{d}x}{x} = \dfrac{\mathrm{d}y}{-2y}$. Integrate: $\ln|x| = -\frac{1}{2}\ln|y| + \text{const} \Rightarrow 2\ln|x| + \ln|y| = \text{const} \Rightarrow x^2 y = C$. So the general solution is

$$u(x,y) = f\big(x^2 y\big)$$

with f arbitrary. (Check: $u_x = 2xyf'$, $u_y = x^2 f'$; $xu_x - 2yu_y = 2x^2 yf' - 2x^2 yf' = 0$. ✓)
(a) On $x=1$: $u(1,y) = f(y) = 2y+1$. So $f(\zeta) = 2\zeta + 1$ and

$$\boxed{u(x,y) = 2x^2 y + 1}.$$

(b) We need only $u(1,1) = f(1) = 4$. This fixes f at one argument only, so there are *infinitely many* solutions. Any function f with $f(1) = 4$ works, for example $f(\zeta) = 4$ giving $u = 4$ (constant), or $f(\zeta) = 4\zeta$ giving $\boxed{u = 4x^2 y}$, or $f = 3\zeta + 1$ giving $u = 3x^2 y + 1$.

Other characteristic problems

(i) Transport equation $u_t + cu_x = 0$: $\dfrac{\mathrm{d}x}{c} = \dfrac{\mathrm{d}t}{1} \Rightarrow x - ct = \xi$, $u = f(x-ct)$, a profile moving to the right at speed c without change of shape.
(ii) $yu_x - xu_y = 0$: $\dfrac{\mathrm{d}x}{y} = \dfrac{\mathrm{d}y}{-x} \Rightarrow x\,\mathrm{d}x + y\,\mathrm{d}y = 0 \Rightarrow x^2 + y^2 = C$: $u = f(x^2+y^2)$, any rotationally symmetric function.
(iii) $u_x + u_y = u$, $u(0,y) = y$: $\dfrac{\mathrm{d}x}{1} = \dfrac{\mathrm{d}y}{1} = \dfrac{\mathrm{d}u}{u}$. From the first equality $y - x = C_1$. From $\mathrm{d}x = \mathrm{d}u/u$: $u = C_2 \mathrm{e}^x$, with $C_2 = C_2(C_1)$. So $u = \mathrm{e}^x f(y-x)$; at $x=0$: $f(y) = y$. Hence $u = \mathrm{e}^x(y-x)$. Check: $u_x + u_y = \mathrm{e}^x(y-x) - \mathrm{e}^x + \mathrm{e}^x = u$. ✓
(iv) Quasilinear: $uu_x + u_y = 0$, $u(x,0) = x$. Characteristics: $\dfrac{\mathrm{d}x}{u} = \dfrac{\mathrm{d}y}{1} = \dfrac{\mathrm{d}u}{0}$: u is constant on each characteristic, which is a straight line $x = uy + x_0$ with $u(x_0,0) = x_0$, so $u = x_0$: $x = uy + u \Rightarrow \boxed{u = \dfrac{x}{1+y}}$. (Characteristics may cross: for the data $u = -x$ they cross at $y = 1$ and a shock forms.)

17 Classification of second-order PDEs; canonical forms

Consider $A\,u_{xx} + 2B\,u_{xy} + C\,u_{yy} + (\text{lower order}) = 0$ with constant (or slowly varying) A, B, C.

Key result

Let $\Delta = B^2 - AC$.

$$\Delta > 0: \text{ hyperbolic}; \qquad \Delta = 0: \text{ parabolic}; \qquad \Delta < 0: \text{ elliptic}.$$

The **characteristic curves** satisfy $A\,(\mathrm{d}y)^2 - 2B\,\mathrm{d}x\,\mathrm{d}y + C\,(\mathrm{d}x)^2 = 0$, i.e. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{B \pm \sqrt{B^2 - AC}}{A}$.
Canonical forms: hyperbolic $u_{\xi\eta} + \cdots = 0$ (or $u_{\xi\xi} - u_{\eta\eta}$); parabolic $u_{\eta\eta} + \cdots = 0$; elliptic

$u_{\xi\xi} + u_{\eta\eta} + \cdots = 0.$

Examples: wave $u_{xx} - \frac{1}{c^2}u_{tt} = 0$ ($A = 1, C = -1/c^2$, $\Delta = 1/c^2 > 0$): hyperbolic. Heat $u_t = \kappa u_{xx}$ (no u_{tt}: $A = \kappa$, $B = C = 0$, $\Delta = 0$): parabolic. Laplace $u_{xx} + u_{yy} = 0$ ($\Delta = -1$): elliptic.

Past paper: PHY 302 End-sem 27.04.2024 Q4 (5 marks)

Solve $\dfrac{\partial^2 u}{\partial x^2} + 2\dfrac{\partial^2 u}{\partial x \partial y} + \dfrac{\partial^2 u}{\partial y^2} = 0$ subject to $u(0, y) = 0$ and $u(x, 1) = x^2$.

Solution

Type. $A = 1$, $B = 1$, $C = 1$: $\Delta = B^2 - AC = 0$, so the equation is *parabolic*. The characteristic equation $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{B}{A} = 1$ gives $y - x = \text{const}$.

Change of variables. Let $\xi = x - y$ (constant on characteristics) and $\eta = y$. Chain rule: $\partial_x = \partial_\xi$, $\partial_y = -\partial_\xi + \partial_\eta$. Then

$$u_{xx} = u_{\xi\xi}, \qquad u_{xy} = \partial_\xi(-u_\xi + u_\eta) = -u_{\xi\xi} + u_{\xi\eta},$$
$$u_{yy} = (-\partial_\xi + \partial_\eta)(-u_\xi + u_\eta) = u_{\xi\xi} - 2u_{\xi\eta} + u_{\eta\eta}.$$

Sum: $u_{xx} + 2u_{xy} + u_{yy} = u_{\xi\xi} + 2(-u_{\xi\xi} + u_{\xi\eta}) + u_{\xi\xi} - 2u_{\xi\eta} + u_{\eta\eta} = u_{\eta\eta}$. So the PDE is $u_{\eta\eta} = 0$, hence $u = f(\xi) + \eta\, g(\xi)$:

$$u(x, y) = f(x - y) + y\, g(x - y).$$

Boundary condition at $x = 0$. $u(0, y) = f(-y) + y\, g(-y) = 0$ for all y. Put $s = -y$: $f(s) - s\, g(s) = 0$, so $f(s) = s\, g(s)$. Then

$$u = (x - y)\, g(x - y) + y\, g(x - y) = x\, g(x - y).$$

Condition at $y = 1$. $u(x, 1) = x\, g(x - 1) = x^2 \Rightarrow g(x - 1) = x \Rightarrow g(s) = s + 1$. Therefore $g(x - y) = x - y + 1$ and

$$\boxed{u(x, y) = x\,(x - y + 1) = x^2 - xy + x}$$

Check: $u_{xx} = 2$, $u_{xy} = -1$, $u_{yy} = 0$: $2+2(-1)+0 = 0$ ✓; $u(0, y) = 0$ ✓; $u(x, 1) = x^2 - x + x = x^2$ ✓.

Reduction of an elliptic equation: $u_{xx} + 2u_{xy} + 5u_{yy} = 0$

$\Delta = 1 - 5 = -4 < 0$. Characteristics: $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 1 \pm 2\mathrm{i}$, i.e. $y - (1 \pm 2\mathrm{i})x = \text{const}$. Take $\xi = y - x$, $\eta = 2x$ (real and imaginary parts of $y - (1 + 2\mathrm{i})x$ up to sign). Then $u_x = -u_\xi + 2u_\eta$, $u_y = u_\xi$, so $u_{xx} = u_{\xi\xi} - 4u_{\xi\eta} + 4u_{\eta\eta}$, $u_{xy} = -u_{\xi\xi} + 2u_{\xi\eta}$, $u_{yy} = u_{\xi\xi}$. Hence $u_{xx} + 2u_{xy} + 5u_{yy} = (1 - 2 + 5)u_{\xi\xi} + (-4 + 4)u_{\xi\eta} + 4u_{\eta\eta} = 4(u_{\xi\xi} + u_{\eta\eta})$: Laplace's equation in (ξ, η).

17.1 d'Alembert's solution of the wave equation

$u_{tt} = c^2 u_{xx}$ on $-\infty < x < \infty$. Put $\xi = x - ct$, $\eta = x + ct$. Then $\partial_x = \partial_\xi + \partial_\eta$, $\partial_t = c(-\partial_\xi + \partial_\eta)$, so $u_{tt} - c^2 u_{xx} = c^2\big[(\partial_\eta - \partial_\xi)^2 - (\partial_\xi + \partial_\eta)^2\big]u = -4c^2 u_{\xi\eta}$. Thus $u_{\xi\eta} = 0$ and $u = F(x - ct) + G(x + ct)$: a right-moving plus a left-moving wave. With initial data $u(x, 0) = \varphi(x)$, $u_t(x, 0) = \psi(x)$: $F + G = \varphi$, $c(-F' + G') = \psi \Rightarrow -F + G = \frac{1}{c}\int^x \psi$. Solving:

$$u(x, t) = \frac{1}{2}\big[\varphi(x - ct) + \varphi(x + ct)\big] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)\,\mathrm{d}s.$$

A plucked string released from rest

$\varphi(x) = \mathrm{e}^{-x^2}$, $\psi = 0$: $u = \frac{1}{2}[\mathrm{e}^{-(x-ct)^2} + \mathrm{e}^{-(x+ct)^2}]$: the pulse splits into two halves of half the amplitude travelling in opposite directions.

18 Separation of variables: the method

Look for solutions $u(x,y) = X(x)Y(y)$. Substituting and dividing by XY gives (function of x) = (function of y), so each equals a constant (the **separation constant**). The boundary conditions on the homogeneous sides select the allowed constants (an SL eigenvalue problem, Part I); the superposition $u = \sum c_n X_n Y_n$ is then fitted to the remaining non-homogeneous condition using orthogonality.

Procedure. (1) Separate. (2) Solve the eigenvalue problem for the variable with homogeneous conditions. (3) Solve the other ODE (choose the exponential/hyperbolic form that respects the condition at the other boundary). (4) Superpose. (5) Fit the non-homogeneous condition by an orthogonal expansion.

Laplace's equation in a rectangle

$\nabla^2 u = 0$ for $0 < x < a$, $0 < y < b$; $u(0,y) = u(a,y) = u(x,0) = 0$, $u(x,b) = f(x)$. Put $u = X(x)Y(y)$: $\dfrac{X''}{X} = -\dfrac{Y''}{Y} = -\lambda$.

- $X'' + \lambda X = 0$, $X(0) = X(a) = 0 \Rightarrow \lambda_n = (n\pi/a)^2$, $X_n = \sin\dfrac{n\pi x}{a}$.
- $Y'' = \lambda_n Y$, $Y(0) = 0 \Rightarrow Y_n = \sinh\dfrac{n\pi y}{a}$.

Superposition: $u = \sum_{n=1}^{\infty} B_n \sin\dfrac{n\pi x}{a}\sinh\dfrac{n\pi y}{a}$. At $y = b$: $f(x) = \sum \left[B_n \sinh\frac{n\pi b}{a}\right]\sin\frac{n\pi x}{a}$, so $B_n \sinh\dfrac{n\pi b}{a} = \dfrac{2}{a}\int_0^a f(x)\sin\dfrac{n\pi x}{a}\mathrm{d}x$. For $f = u_0$ constant, $\int_0^a \sin\frac{n\pi x}{a}\mathrm{d}x = \frac{a}{n\pi}(1-(-1)^n)$, so $B_n \sinh\frac{n\pi b}{a} = \frac{2u_0}{n\pi}(1-(-1)^n)$ and

$$u(x,y) = \frac{4u_0}{\pi}\sum_{n\ \text{odd}} \frac{1}{n}\frac{\sinh(n\pi y/a)}{\sinh(n\pi b/a)}\sin\frac{n\pi x}{a}.$$

19 Laplace's equation in polar coordinates: the disk

In polar coordinates (ρ,ϕ): $\nabla^2 u = \dfrac{1}{\rho}\dfrac{\partial}{\partial\rho}\Big(\rho\dfrac{\partial u}{\partial\rho}\Big) + \dfrac{1}{\rho^2}\dfrac{\partial^2 u}{\partial\phi^2} = 0$. Put $u = R(\rho)\Phi(\phi)$ and multiply by $\rho^2/(R\Phi)$:

$$\frac{\rho}{R}\frac{\mathrm{d}}{\mathrm{d}\rho}\Big(\rho\frac{\mathrm{d}R}{\mathrm{d}\rho}\Big) = -\frac{\Phi''}{\Phi} = n^2.$$

- Angular: $\Phi'' + n^2\Phi = 0$. Single-valuedness $\Phi(\phi + 2\pi) = \Phi(\phi)$ forces $n = 0, 1, 2, \ldots$ and $\Phi = A\cos n\phi + B\sin n\phi$.
- Radial (Euler equation): $\rho^2 R'' + \rho R' - n^2 R = 0$; try $R = \rho^s$: $s(s-1) + s - n^2 = 0 \Rightarrow s = \pm n$. For $n = 0$: $R = C_0 \ln\rho + D_0$.

Key result

General solution (formula sheet, eq. (4)): $u(\rho,\phi) = (C_0 \ln\rho + D_0) + \sum_{n=1}^{\infty}(A_n\cos n\phi + B_n \sin n\phi)(C_n\rho^n + D_n\rho^{-n})$.

Past papers: PHY 302 End-sem 27.04.2024 Q6 (20 marks) and End-sem 05.05.2025 Q5 (20 marks)

Solve Laplace's equation in the two-dimensional region $|\boldsymbol{r}| \le a$, subject to the boundary condition $u = f(\phi)$ on $|\boldsymbol{r}| = a$.

Solution

Step 1: regularity. The solution must be finite at $\rho = 0$, so $\ln\rho$ and ρ^{-n} are excluded ($D_n = 0$, $C_0 = 0$). Absorbing constants,

$$u(\rho,\phi) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \rho^n \big(A_n \cos n\phi + B_n \sin n\phi\big).$$

Step 2: boundary condition. At $\rho = a$: $f(\phi) = \dfrac{a_0}{2} + \sum a^n(A_n\cos n\phi + B_n \sin n\phi)$. This is the Fourier series of f on $(0, 2\pi)$, so by the orthogonality of $\cos n\phi, \sin n\phi$ (Part II):

$$a^n A_n = \frac{1}{\pi}\int_0^{2\pi} f(\phi')\cos n\phi'\,\mathrm{d}\phi' \equiv \alpha_n, \qquad a^n B_n = \frac{1}{\pi}\int_0^{2\pi} f(\phi')\sin n\phi'\,\mathrm{d}\phi' \equiv \beta_n, \qquad a_0 = \alpha_0.$$

Result (series form):

$$\boxed{u(\rho,\phi) = \frac{\alpha_0}{2} + \sum_{n=1}^{\infty}\left(\frac{\rho}{a}\right)^n \big(\alpha_n\cos n\phi + \beta_n \sin n\phi\big)}$$

Step 3: closed form (Poisson integral). Insert α_n, β_n and use $\cos n\phi\cos n\phi' + \sin n\phi \sin n\phi' = \cos n(\phi - \phi')$:

$$u = \frac{1}{2\pi}\int_0^{2\pi} f(\phi')\Big[1 + 2\sum_{n=1}^{\infty} r^n \cos n(\phi - \phi')\Big]\mathrm{d}\phi', \qquad r = \frac{\rho}{a} < 1.$$

Put $\psi = \phi - \phi'$. The bracket is $1 + 2\,\mathrm{Re}\sum_{n\ge1}(r\mathrm{e}^{\mathrm{i}\psi})^n = 1 + 2\,\mathrm{Re}\,\dfrac{r\mathrm{e}^{\mathrm{i}\psi}}{1 - r\mathrm{e}^{\mathrm{i}\psi}} = \mathrm{Re}\,\dfrac{1 + r\mathrm{e}^{\mathrm{i}\psi}}{1 - r\mathrm{e}^{\mathrm{i}\psi}}$ (geometric series, $r < 1$). Multiply numerator and denominator by $1 - r\mathrm{e}^{-\mathrm{i}\psi}$: $\dfrac{(1 + r\mathrm{e}^{\mathrm{i}\psi})(1 - r\mathrm{e}^{-\mathrm{i}\psi})}{|1 - r\mathrm{e}^{\mathrm{i}\psi}|^2} = \dfrac{1 - r^2 + r(\mathrm{e}^{\mathrm{i}\psi} - \mathrm{e}^{-\mathrm{i}\psi})}{1 - 2r\cos\psi + r^2}$, whose real part is $\dfrac{1 - r^2}{1 - 2r\cos\psi + r^2}$. Hence

$$\boxed{u(\rho,\phi) = \frac{1}{2\pi}\int_0^{2\pi} \frac{a^2 - \rho^2}{a^2 - 2a\rho\cos(\phi - \phi') + \rho^2} f(\phi')\,\mathrm{d}\phi'}$$

Checks. At the centre ($\rho = 0$): $u(0) = \frac{1}{2\pi}\int f\,\mathrm{d}\phi'$, the *mean value property*. As $\rho \to a$ the kernel tends to $2\pi\delta(\phi - \phi')$, so $u \to f$. The maximum principle: $\min f \le u \le \max f$.

Half-circle heated: $f = V_0$ on $(0,\pi)$, 0 on $(\pi, 2\pi)$

$\alpha_0 = \frac{1}{\pi}\int_0^\pi V_0 \mathrm{d}\phi' = V_0$. $\alpha_n = \frac{V_0}{\pi}\int_0^\pi \cos n\phi' \mathrm{d}\phi' = 0$. $\beta_n = \frac{V_0}{\pi}\int_0^\pi \sin n\phi' \mathrm{d}\phi' = \frac{V_0}{n\pi}(1-(-1)^n)$. So $u = \dfrac{V_0}{2} + \dfrac{2V_0}{\pi}\displaystyle\sum_{n \text{ odd}} \frac{r^n \sin n\phi}{n}$. Using $\sum_{n \text{ odd}} \dfrac{z^n}{n} = \tanh^{-1} z = \frac{1}{2}\ln\dfrac{1+z}{1-z}$ with $z = r\mathrm{e}^{\mathrm{i}\phi}$ (take imaginary part: $\frac{1}{2}\arg\dfrac{1+z}{1-z} = \frac{1}{2}\arctan\dfrac{2r\sin\phi}{1-r^2}$):

$$u(\rho,\phi) = \frac{V_0}{2} + \frac{V_0}{\pi}\arctan\frac{2a\rho\sin\phi}{a^2-\rho^2}.$$

On the boundary ($\rho \to a$, $0 < \phi < \pi$) the arctangent tends to $\pi/2$, giving $u = V_0$; for $\pi < \phi < 2\pi$ it tends to $-\pi/2$, giving 0. ✓

Exterior and annular regions

Exterior ($\rho \geq a$, u bounded at infinity): keep ρ^{-n}, discard ρ^n: $u = \frac{\alpha_0}{2} + \sum (a/\rho)^n(\alpha_n \cos n\phi + \beta_n \sin n\phi)$ (Poisson kernel with $a^2 - \rho^2 \to \rho^2 - a^2$). **Annulus** $a < \rho < b$: both powers and the $\ln\rho$ term are present; two boundary conditions fix C_n, D_n.

20 Cylindrical coordinates and Bessel functions

$\nabla^2 u = \dfrac{1}{\rho}\partial_\rho(\rho\,\partial_\rho u) + \dfrac{1}{\rho^2}\partial_\phi^2 u + \partial_z^2 u = 0$. Put $u = R(\rho)\Phi(\phi)Z(z)$ and divide by u:

$$\frac{1}{R}\Big(R'' + \frac{R'}{\rho}\Big) + \frac{1}{\rho^2}\frac{\Phi''}{\Phi} + \frac{Z''}{Z} = 0.$$

The z-part depends only on z and the rest only on ρ, ϕ, so Z''/Z is a constant; choose it as $+k^2$ (decaying/growing exponentials in z). Then multiply by ρ^2 and separate ϕ: $\Phi''/\Phi = -m^2$ (m integer by single-valuedness) and

$$\rho^2 R'' + \rho R' + \big(k^2\rho^2 - m^2\big)R = 0 \qquad \text{(Bessel's equation of order } m\text{)},$$

$$Z = E\,\mathrm{e}^{-kz} + F\,\mathrm{e}^{kz}, \qquad \Phi = C\cos m\phi + D\sin m\phi, \qquad R = A\,J_m(k\rho) + B\,Y_m(k\rho).$$

Key result

Formula sheet, eq. (5): $u(\rho,\phi,z) = \big[AJ_m(k\rho) + BY_m(k\rho)\big]\big[C\cos m\phi + D\sin m\phi\big]\big[E\mathrm{e}^{-kz} + F\mathrm{e}^{kz}\big]$.
Use J_m alone when the region contains the axis $\rho = 0$ (Y_m diverges there); use both for an annular region.

Common mistake / check

The sign of the separation constant matters. With $Z'' = +k^2 Z$ (exponential in z) the radial functions are *oscillatory* Bessel functions J_m, Y_m of $k\rho$, as above. If instead the z-dependence is oscillatory ($Z'' = -k^2 Z$, $Z = \cos kz, \sin kz$) the radial equation becomes $\rho^2 R'' + \rho R' - (k^2\rho^2 + m^2)R = 0$, whose solutions are the *modified* Bessel functions $I_m(k\rho), K_m(k\rho)$.

Past papers: PHY 302 End-sem 27.04.2024 Q5 (10 marks) and End-sem 05.05.2025 Q3 (15 marks)

A semi-infinite solid cylinder of radius a has its curved surface held at 0°C and its base held at a temperature T_0. Find the steady-state temperature distribution in the cylinder, given that the steady-state temperature distribution satisfies Laplace's equation.

Solution

Set-up. Cylindrical coordinates with the base at $z = 0$ and the cylinder extending to $z \to \infty$. The boundary conditions have no ϕ-dependence, so $T = T(\rho, z)$ ($m = 0$):

$$\frac{1}{\rho}\frac{\partial}{\partial \rho}\Big(\rho \frac{\partial T}{\partial \rho}\Big) + \frac{\partial^2 T}{\partial z^2} = 0, \quad 0 \le \rho < a,\ z > 0,$$

$$T(a, z) = 0, \qquad T(\rho, 0) = T_0, \qquad T \text{ bounded as } z \to \infty \text{ (indeed } T \to 0), \qquad T \text{ finite at } \rho = 0.$$

Separation. $T = R(\rho)Z(z)$: $\dfrac{R'' + R'/\rho}{R} = -\dfrac{Z''}{Z} = -k^2$.

- $R'' + \dfrac{1}{\rho}R' + k^2 R = 0$ is Bessel's equation of order 0 (multiply by ρ^2: $\rho^2 R'' + \rho R' + k^2\rho^2 R = 0$). Finite at the origin $\Rightarrow R = J_0(k\rho)$ (Y_0 is excluded because $Y_0 \to -\infty$ at 0).
- $Z'' = k^2 Z \Rightarrow Z = E\mathrm{e}^{-kz} + F\mathrm{e}^{kz}$. Boundedness as $z \to \infty$ forces $F = 0$.

Side boundary condition. $T(a, z) = 0 \Rightarrow J_0(ka) = 0 \Rightarrow k_n a = \alpha_n$, where α_n is the n-th positive zero of J_0 ($\alpha_1 = 2.4048$, $\alpha_2 = 5.5201$, $\alpha_3 = 8.6537$, $\alpha_4 = 11.7915$, …, from the table of zeros on the formula sheet). Therefore, superposing,

$$T(\rho, z) = \sum_{n=1}^{\infty} A_n J_0\Big(\frac{\alpha_n \rho}{a}\Big)\mathrm{e}^{-\alpha_n z/a}.$$

Base condition. At $z = 0$: $T_0 = \sum A_n J_0(\alpha_n \rho/a)$, a Fourier–Bessel expansion of the constant T_0 on $[0, a]$. The functions $J_0(\alpha_n\rho/a)$ are orthogonal with weight ρ (Part I), with norm $\int_0^a \rho J_0^2(\alpha_n\rho/a)\mathrm{d}\rho = \frac{a^2}{2}J_1^2(\alpha_n)$. Hence (formula sheet, eq. (8) with $\nu = 0$, $b = a$):

$$A_n = \frac{2}{a^2 J_1^2(\alpha_n)} \int_0^a T_0\, J_0\Big(\frac{\alpha_n \rho}{a}\Big)\rho\,\mathrm{d}\rho.$$

Evaluate the integral: let $z = \alpha_n\rho/a$, $\rho = az/\alpha_n$, $\rho\,\mathrm{d}\rho = \dfrac{a^2}{\alpha_n^2} z\,\mathrm{d}z$. Using $\dfrac{\mathrm{d}}{\mathrm{d}z}[zJ_1(z)] = zJ_0(z)$ (eq. (9) of the sheet):

$$\int_0^a J_0\Big(\frac{\alpha_n\rho}{a}\Big)\rho\,\mathrm{d}\rho = \frac{a^2}{\alpha_n^2}\int_0^{\alpha_n} zJ_0(z)\mathrm{d}z = \frac{a^2}{\alpha_n^2}\big[zJ_1(z)\big]_0^{\alpha_n} = \frac{a^2 J_1(\alpha_n)}{\alpha_n}.$$

Therefore $A_n = \dfrac{2T_0}{a^2 J_1^2(\alpha_n)} \cdot \dfrac{a^2 J_1(\alpha_n)}{\alpha_n} = \dfrac{2T_0}{\alpha_n J_1(\alpha_n)}$ and

$$\boxed{T(\rho, z) = 2T_0 \sum_{n=1}^{\infty} \frac{J_0(\alpha_n\rho/a)}{\alpha_n J_1(\alpha_n)}\,\mathrm{e}^{-\alpha_n z/a}}$$

Interpretation. Deep in the cylinder ($z \gg a$) only the slowest-decaying mode $n = 1$ survives: $T \approx \dfrac{2T_0}{2.4048\,(0.5191)} J_0(2.4048\rho/a)\mathrm{e}^{-2.4048z/a} = 1.602\,T_0 J_0(2.4048\rho/a)\mathrm{e}^{-2.4048z/a}$. The temperature falls to e^{-1} of its base value over a length $\approx a/2.4$. At $z = 0$ the series reproduces the constant T_0 (conditionally convergent; $1.602 - 1.065 + 0.851 - \cdots \to 1$ at the axis).

Variants

Finite cylinder of length h with $T=0$ on sides and top, $T=T_0$ on the base: replace $\mathrm{e}^{-\alpha_n z/a}$ by $\sinh[\alpha_n(h-z)/a]/\sinh(\alpha_n h/a)$ so that $T(\rho,h)=0$. *Non-axisymmetric data* $T(\rho,\phi,0)=g(\rho)\cos m\phi$: use J_m and the zeros of J_m. *Wave/heat in a drum or disk* (Section 23.1): the same Bessel functions appear.

21 Spherical coordinates and Legendre polynomials

$$\nabla^2 u=\frac{1}{r^2}\partial_r\big(r^2\partial_r u\big)+\frac{1}{r^2\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta u)+\frac{1}{r^2\sin^2\theta}\partial_\phi^2 u=0.$$

Put $u=R(r)\Theta(\theta)\Phi(\phi)$, multiply by r^2/u and separate: $\dfrac{1}{R}(r^2R')'=l(l+1)$, the angular part is the spherical-harmonic equation with $\Phi=\cos m\phi,\sin m\phi$ and $\Theta=P_l^m(\cos\theta)$ (the Q_l^m are singular at the poles).

- Radial: $r^2R''+2rR'-l(l+1)R=0$; try $R=r^s$: $s(s+1)=l(l+1)\Rightarrow s=l$ or $s=-(l+1)$.

Key result

Formula sheet, eq. (6): $u=(Ar^l+Br^{-(l+1)})(C\cos m\phi+D\sin m\phi)[EP_l^m(\cos\theta)+FQ_l^m(\cos\theta)]$.

Axisymmetric case ($m=0$): $u(r,\theta)=\displaystyle\sum_{l=0}^{\infty}\big(A_l r^l+B_l r^{-(l+1)}\big)P_l(\cos\theta)$. Inside a sphere ($u$ finite at $r=0$): $B_l=0$. Outside ($u\to 0$ at infinity): $A_l=0$. With $u(a,\theta)=V(\theta)$ and $x=\cos\theta$:

$$c_l\equiv\frac{2l+1}{2}\int_{-1}^{1}V(x)P_l(x)\,\mathrm{d}x\quad\text{(eq. (15))},\qquad u_{\text{in}}=\sum c_l\Big(\frac{r}{a}\Big)^l P_l,\qquad u_{\text{out}}=\sum c_l\Big(\frac{a}{r}\Big)^{l+1}P_l.$$

Surface potential $V_0\cos\theta$ and a conducting sphere in a uniform field

(i) $V=V_0\cos\theta=V_0P_1$: only $l=1$: $u_{\text{in}}=V_0\dfrac{r}{a}r\cos\theta\to V_0\dfrac{r}{a}\cos\theta$, $u_{\text{out}}=V_0\dfrac{a^2}{r^2}\cos\theta$.
(ii) A grounded conducting sphere of radius a in a uniform field $\boldsymbol{E}_0=E_0\hat{z}$: far away $u\to -E_0r\cos\theta$. Take $u=-E_0r\cos\theta+\dfrac{B}{r^2}\cos\theta$ (only $l=1$ is needed). $u(a)=0\Rightarrow -E_0a+B/a^2=0\Rightarrow B=E_0a^3$:

$$u=-E_0\Big(r-\frac{a^3}{r^2}\Big)\cos\theta,\qquad \sigma=\epsilon_0E_r\Big|_a=-\epsilon_0\frac{\partial u}{\partial r}\Big|_a=3\epsilon_0E_0\cos\theta.$$

Past paper: PHY 302 End-sem 05.05.2025, Q4 (25 marks)

A hollow conducting sphere of radius a with a vertical split is placed at the origin. If the left hemisphere surface is charged to a potential v_0 and the other hemisphere is kept at $-v_0$, find the potential v inside and outside the sphere.

Solution

Geometry. The split divides the sphere into a left and a right hemisphere. Choose the polar axis z *perpendicular to the split plane*, pointing to the right: the right hemisphere is $0\le\theta<\pi/2$

($\cos\theta > 0$) and the left hemisphere is $\pi/2 < \theta \le \pi$. Then the problem is axisymmetric:

$$V(a,\theta) = \begin{cases} -v_0, & \cos\theta > 0 \text{ (right)}, \\ +v_0, & \cos\theta < 0 \text{ (left)} \end{cases} \quad \Longleftrightarrow \quad V(x) = -v_0\,\mathrm{sgn}(x),\ x = \cos\theta.$$

General solutions. Inside ($r < a$, finite at origin): $v = \sum_l A_l r^l P_l(\cos\theta)$. Outside ($r > a$, $v \to 0$ at infinity): $v = \sum_l B_l r^{-(l+1)} P_l(\cos\theta)$. Matching to the surface potential gives $A_l a^l = B_l a^{-(l+1)} = c_l$:

$$c_l = \frac{2l+1}{2}\int_{-1}^{1} V(x)P_l(x)\,\mathrm{d}x = \frac{2l+1}{2}(-v_0)\Big[\int_0^1 P_l\,\mathrm{d}x - \int_{-1}^{0} P_l\,\mathrm{d}x\Big].$$

Parity. $P_l(-x) = (-1)^l P_l(x)$. So $\int_{-1}^{0} P_l \mathrm{d}x = (-1)^l \int_0^1 P_l \mathrm{d}x$ and the bracket is $\big(1 - (-1)^l\big)\int_0^1 P_l \mathrm{d}x$: it vanishes for even l and equals $2\int_0^1 P_l \mathrm{d}x$ for odd l. Therefore

$$c_l = -(2l+1)\,v_0 \int_0^1 P_l(x)\,\mathrm{d}x \qquad (l \text{ odd}), \qquad c_l = 0 \ (l \text{ even}).$$

The integral. Use the identity $(2l+1)P_l = P'_{l+1} - P'_{l-1}$. Then $\displaystyle\int_0^1 P_l \mathrm{d}x = \frac{1}{2l+1}\big[P_{l+1} - P_{l-1}\big]_0^1 = \frac{1}{2l+1}\Big[(1-1) - \big(P_{l+1}(0) - P_{l-1}(0)\big)\Big] = \frac{P_{l-1}(0) - P_{l+1}(0)}{2l+1}$. Hence

$$c_l = -v_0\big[P_{l-1}(0) - P_{l+1}(0)\big] \qquad (l \text{ odd}).$$

With $P_0(0) = 1$, $P_2(0) = -\frac{1}{2}$, $P_4(0) = \frac{3}{8}$, $P_6(0) = -\frac{5}{16}$, $P_8(0) = \frac{35}{128}$:

$$\begin{aligned} c_1 &= -v_0\big[1 - (-\tfrac{1}{2})\big] = -\tfrac{3}{2}v_0, \\ c_3 &= -v_0\big[-\tfrac{1}{2} - \tfrac{3}{8}\big] = +\tfrac{7}{8}v_0, \\ c_5 &= -v_0\big[\tfrac{3}{8} - (-\tfrac{5}{16})\big] = -\tfrac{11}{16}v_0, \\ c_7 &= -v_0\big[-\tfrac{5}{16} - \tfrac{35}{128}\big] = +\tfrac{75}{128}v_0. \end{aligned}$$

(Check c_1 directly: $\frac{3}{2}\int_{-1}^{1} V\,x\,\mathrm{d}x = \frac{3}{2}(-v_0)\cdot 2\int_0^1 x\,\mathrm{d}x = -\frac{3}{2}v_0$. ✓)

Key result

$$\boxed{v_{\text{in}}(r,\theta) = -v_0\Big[\tfrac{3}{2}\frac{r}{a}P_1(\cos\theta) - \tfrac{7}{8}\Big(\frac{r}{a}\Big)^3 P_3(\cos\theta) + \tfrac{11}{16}\Big(\frac{r}{a}\Big)^5 P_5(\cos\theta) - \cdots\Big]}$$

$$\boxed{v_{\text{out}}(r,\theta) = -v_0\Big[\tfrac{3}{2}\Big(\frac{a}{r}\Big)^2 P_1(\cos\theta) - \tfrac{7}{8}\Big(\frac{a}{r}\Big)^4 P_3(\cos\theta) + \tfrac{11}{16}\Big(\frac{a}{r}\Big)^6 P_5(\cos\theta) - \cdots\Big]}$$

where θ is measured from the axis pointing to the *right* (to the $-v_0$ hemisphere). Equivalently, measuring θ' from the axis pointing to the left (the $+v_0$ hemisphere), the overall signs flip: $v_{\text{in}} = v_0\big[\frac{3}{2}\frac{r}{a}P_1(\cos\theta') - \frac{7}{8}(\frac{r}{a})^3 P_3 + \cdots\big]$.

Checks. (1) At the centre $v_{\text{in}} = 0$ (the two hemispheres cancel). (2) On the plane of the split ($\theta = \pi/2$) all $P_{\text{odd}}(0) = 0$ so $v = 0$ there, inside and outside. (3) Far away the leading term is the *dipole* potential $v \approx -\frac{3}{2}v_0\frac{a^2}{r^2}\cos\theta$. (4) The coefficients follow the general rule $c_l = -v_0[P_{l-1}(0) - P_{l+1}(0)]$ for odd l; e.g. $c_9 = -v_0\big[\frac{35}{128} + \frac{63}{256}\big] = -\frac{133}{256}v_0$ (using $P_8(0) = \frac{35}{128}$, $P_{10}(0) = -\frac{63}{256}$). The signs alternate and the magnitudes decrease slowly, as expected for a discontinuous boundary potential.

22 Poisson's equation and uniqueness

$\nabla^2 u = -f$ is solved as $u = u_h + u_p$ with u_p any particular solution and u_h harmonic fixed by the boundary conditions.

Uniformly heated disk

$\nabla^2 u = -2$ (2D) in $\rho < a$, $u(a) = 0$. By symmetry $u = u(\rho)$: $\frac{1}{\rho}(\rho u')' = -2 \Rightarrow \rho u' = -\rho^2 + C$; regularity at 0 ($C = 0$) gives $u' = -\rho$, $u = -\frac{\rho^2}{2} + D$; $u(a) = 0 \Rightarrow D = \frac{a^2}{2}$: $u = \dfrac{a^2 - \rho^2}{2}$.

Key result

Uniqueness. If u_1, u_2 both satisfy $\nabla^2 u = -f$ in V with the same Dirichlet data (or Neumann data) on ∂V, let $w = u_1 - u_2$: $\nabla^2 w = 0$ and $w = 0$ (or $\partial_n w = 0$) on ∂V. Green's first identity gives $\int_V |\nabla w|^2 \mathrm{d}V = \oint w\, \partial_n w \,\mathrm{d}S - \int_V w \nabla^2 w \,\mathrm{d}V = 0$, so $\nabla w = 0$: w is constant ($= 0$ for Dirichlet; arbitrary constant for Neumann). **Maximum principle:** a harmonic function takes its maximum and minimum on the boundary.

23 The wave equation

Vibrating string. A string of tension T and mass density μ (displacement $u(x,t)$) obeys $u_{tt} = c^2 u_{xx}$, $c = \sqrt{T/\mu}$. With fixed ends $u(0,t) = u(L,t) = 0$ and initial data $u(x,0) = \varphi(x)$, $u_t(x,0) = \psi(x)$: put $u = X(x)T(t)$: $X''/X = T''/(c^2 T) = -\lambda$. $X = \sin\frac{n\pi x}{L}$, $\lambda_n = (n\pi/L)^2$; $T'' + c^2\lambda_n T = 0 \Rightarrow T = a_n \cos\omega_n t + b_n \sin\omega_n t$ with $\omega_n = \dfrac{n\pi c}{L}$:

$$u(x,t) = \sum_{n=1}^{\infty} \sin\frac{n\pi x}{L}\Big(a_n \cos\omega_n t + b_n \sin\omega_n t\Big), \quad a_n = \frac{2}{L}\int_0^L \varphi \sin\frac{n\pi x}{L}\mathrm{d}x, \quad b_n = \frac{2}{\omega_n L}\int_0^L \psi \sin\frac{n\pi x}{L}\mathrm{d}x.$$

Plucked string

Released from rest ($\psi = 0$) from the triangle $\varphi = \frac{2hx}{L}$ for $x < \frac{L}{2}$, $\varphi = \frac{2h(L-x)}{L}$ for $x > \frac{L}{2}$. $a_n = \frac{2}{L}\Big[\int_0^{L/2} \frac{2hx}{L}\sin\frac{n\pi x}{L}\mathrm{d}x + \int_{L/2}^{L} \frac{2h(L-x)}{L}\sin\frac{n\pi x}{L}\mathrm{d}x\Big]$. By symmetry about $L/2$ the two integrals are equal for odd n and cancel for even n. Evaluate one: with $k = n\pi/L$, $\int_0^{L/2} x\sin kx\,\mathrm{d}x = \big[-\frac{x\cos kx}{k}\big]_0^{L/2} + \frac{\sin(kL/2)}{k^2} = -\frac{L\cos(n\pi/2)}{2k} + \frac{\sin(n\pi/2)}{k^2}$. For odd n, $\cos(n\pi/2) = 0$, so it equals $\dfrac{L^2\sin(n\pi/2)}{n^2\pi^2}$. Hence $a_n = \frac{2}{L}\cdot 2 \cdot \frac{2h}{L}\cdot\frac{L^2\sin(n\pi/2)}{n^2\pi^2} = \dfrac{8h}{n^2\pi^2}\sin\dfrac{n\pi}{2}$:

$$u = \frac{8h}{\pi^2}\sum_{n\ \text{odd}} \frac{(-1)^{(n-1)/2}}{n^2}\sin\frac{n\pi x}{L}\cos\frac{n\pi ct}{L}.$$

Only odd harmonics appear (the pluck is symmetric), with amplitudes falling as $1/n^2$.

Energy conservation

$E = \frac{1}{2}\int_0^L (\mu u_t^2 + T u_x^2)\mathrm{d}x = \frac{\mu L}{4}\sum_n \omega_n^2(a_n^2 + b_n^2)$: each mode carries a constant energy.

23.1 The circular drum

$u_{tt} = c^2\nabla^2 u$ on the disk $\rho < a$ with $u(a,\phi,t) = 0$. Put $u = R(\rho)\Phi(\phi)\cos\omega t$: $\nabla^2 u = -k^2 u$ with $k = \omega/c$: $R'' + \frac{1}{\rho}R' + \big(k^2 - \frac{m^2}{\rho^2}\big)R = 0$, so $R = J_m(k\rho)$, and $J_m(ka) = 0 \Rightarrow k_{mn} = \alpha_{mn}/a$:

$$u_{mn} = J_m\Big(\frac{\alpha_{mn}\rho}{a}\Big)\big(C\cos m\phi + D\sin m\phi\big)\cos\Big(\frac{c\,\alpha_{mn}t}{a}\Big).$$

Using the table of zeros ($\alpha_{01} = 2.4048$, $\alpha_{11} = 3.8317$, $\alpha_{21} = 5.1356$, $\alpha_{02} = 5.5201$): the frequency ratios of the lowest modes are $1 : 1.593 : 2.136 : 2.295$ (not harmonic, which is why drums do not have a definite pitch like strings).

24 The heat (diffusion) equation

$u_t = \kappa u_{xx}$. Separating $u = X(x)T(t)$: $X''/X = T'/(\kappa T) = -\lambda$, $T = \mathrm{e}^{-\kappa\lambda t}$ (decay). This is the formula-sheet solution $u = (A\cos\lambda x + B\sin\lambda x)\mathrm{e}^{-\lambda^2\kappa t}$ (with $\lambda \to \lambda^2$).

Rod with ends held at 0: $u(x,0) = x(L-x)$

$u = \sum b_n \mathrm{e}^{-\kappa(n\pi/L)^2 t}\sin\frac{n\pi x}{L}$, $b_n = \frac{2}{L}\int_0^L x(L-x)\sin kx\,\mathrm{d}x$ with $k = n\pi/L$. Integrate by parts twice:

$$\begin{aligned}\int_0^L x(L-x)\sin kx\,\mathrm{d}x &= \Big[-\frac{x(L-x)\cos kx}{k}\Big]_0^L + \frac{1}{k}\int_0^L (L-2x)\cos kx\,\mathrm{d}x \\ &= \frac{1}{k}\Big\{\Big[\frac{(L-2x)\sin kx}{k}\Big]_0^L + \frac{2}{k}\int_0^L \sin kx\,\mathrm{d}x\Big\} = \frac{2(1-(-1)^n)}{k^3}.\end{aligned}$$

So $b_n = \dfrac{4L^2(1-(-1)^n)}{(n\pi)^3} = \dfrac{8L^2}{n^3\pi^3}$ for odd n:

$$u(x,t) = \frac{8L^2}{\pi^3}\sum_{n\ \text{odd}}\frac{1}{n^3}\mathrm{e}^{-\kappa n^2\pi^2 t/L^2}\sin\frac{n\pi x}{L}.$$

At late times only $n = 1$ survives: $u \approx \frac{8L^2}{\pi^3}\mathrm{e}^{-\kappa\pi^2 t/L^2}\sin\frac{\pi x}{L}$: the decay time is $L^2/(\kappa\pi^2)$.

Non-homogeneous boundary values: $u(0,t) = 0$, $u(L,t) = T_1$

Write $u = u_{\text{steady}} + v$ with $u_{\text{steady}} = T_1 x/L$ (satisfies the PDE and the boundary conditions). Then v obeys the homogeneous problem with $v(0) = v(L) = 0$ and $v(x,0) = u(x,0) - T_1 x/L$; expand as above.

Insulated ends

$u_x(0,t) = u_x(L,t) = 0$: eigenfunctions $\cos\frac{n\pi x}{L}$, $n = 0,1,\ldots$; $u = \frac{a_0}{2} + \sum a_n \mathrm{e}^{-\kappa(n\pi/L)^2 t}\cos\frac{n\pi x}{L}$. As $t \to \infty$, $u \to a_0/2$, the average of the initial temperature: heat is conserved.

25 Transform methods for PDEs

Key result

Formula-sheet identities (with $\mathcal{F}_s, \mathcal{F}_c$ the sine/cosine transforms, $\bar{u}$ the Laplace transform in t):

$$\mathcal{F}_s(u_{xx}) = \sqrt{\tfrac{2}{\pi}}\, s\, u(0,t) - s^2 \tilde{u}_s, \qquad \mathcal{F}_c(u_{xx}) = -\sqrt{\tfrac{2}{\pi}}\, u_x(0,t) - s^2 \tilde{u}_c,$$
$$\mathcal{F}(u_{xx}) = -s^2 \tilde{u} \quad (-\infty < x < \infty),$$

$$\mathcal{L}(u_t) = s\bar{u} - u(x,0), \qquad \mathcal{L}(u_{tt}) = s^2\bar{u} - su(x,0) - u_t(x,0).$$

Rule of thumb: use the sine transform when u is given at $x = 0$, the cosine transform when u_x is given, the full Fourier transform on the whole line, and the Laplace transform in t for initial-value problems.

Semi-infinite rod with the end held at U_0 (sine-transform method)

$u_t = \kappa u_{xx}$, $x > 0$, $u(x,0) = 0$, $u(0,t) = U_0$. Sine transform: $\dfrac{\mathrm{d}\tilde{u}_s}{\mathrm{d}t} = \kappa\Big[\sqrt{\tfrac{2}{\pi}}\, sU_0 - s^2\tilde{u}_s\Big]$. With $\tilde{u}_s(0) = 0$ the solution is $\tilde{u}_s = \sqrt{\tfrac{2}{\pi}}\dfrac{U_0}{s}\big(1 - \mathrm{e}^{-\kappa s^2 t}\big)$. Inverting, $u(x,t) = U_0 \operatorname{erfc}\Big(\dfrac{x}{2\sqrt{\kappa t}}\Big)$, the same result as the Laplace method of Part II.

Practice problems

1. Solve $u_x + 2u_y = 0$ with $u(0,y) = \sin y$. *Answer:* $u = \sin(y - 2x)$.

2. Solve $xu_x + yu_y = 2u$ with $u(x,1) = x$. *Hint:* $\mathrm{d}x/x = \mathrm{d}y/y = \mathrm{d}u/2u$; $u = y^2 f(x/y)$. *Answer:* $u = xy$.

3. Classify and reduce $u_{xx} + 4u_{xy} + 4u_{yy} = 0$. *Answer:* $\Delta = 4 - 4 = 0$ parabolic; $\xi = 2x - y$, $u_{\eta\eta} = 0$, $u = f(2x - y) + y\, g(2x - y)$.

4. Classify $u_{xx} - 3u_{xy} + 2u_{yy} = 0$ and find the general solution. *Answer:* $B = -\tfrac{3}{2}$, $\Delta = \tfrac{9}{4} - 2 = \tfrac{1}{4} > 0$: hyperbolic. Factor the operator: $(\partial_x - \partial_y)(\partial_x - 2\partial_y)u = 0$ (since $(\partial_x - a\partial_y)(\partial_x - b\partial_y) = \partial_x^2 - (a+b)\partial_{xy} + ab\,\partial_y^2$ with $a + b = 3$, $ab = 2$). Hence $u = f(x + y) + g(2x + y)$.

5. Interior of a disk, $f(\phi) = \cos^2\phi$. *Answer:* $u = \tfrac{1}{2} + \tfrac{1}{2}\big(\tfrac{\rho}{a}\big)^2 \cos 2\phi$.

6. Inside a sphere with $V(a,\theta) = V_0 \cos^2\theta$. *Answer:* $V_0\big[\tfrac{1}{3} + \tfrac{2}{3}(r/a)^2 P_2(\cos\theta)\big]$ (since $\cos^2\theta = \tfrac{1}{3}P_0 + \tfrac{2}{3}P_2$).

7. Vibrating string $u(x,0) = \sin\tfrac{\pi x}{L} + \tfrac{1}{2}\sin\tfrac{3\pi x}{L}$, $u_t(x,0) = 0$. *Answer:* $u = \sin\tfrac{\pi x}{L}\cos\tfrac{\pi c t}{L} + \tfrac{1}{2}\sin\tfrac{3\pi x}{L}\cos\tfrac{3\pi c t}{L}$.

8. Rod of length π with $u(0,t) = u(\pi,t) = 0$, $u(x,0) = \sin x + 3\sin 2x$, $\kappa = 1$. *Answer:* $\mathrm{e}^{-t}\sin x + 3\mathrm{e}^{-4t}\sin 2x$.

9. Semi-infinite cylinder: base at $T(\rho,0) = T_0(1 - \rho^2/a^2)$, curved surface at 0. *Hint:* $A_n = \dfrac{2T_0}{J_1^2(\alpha_n)}\displaystyle\int_0^1 (1 - x^2)xJ_0(\alpha_n x)\mathrm{d}x = \dfrac{2T_0}{J_1^2(\alpha_n)} \cdot \dfrac{2J_2(\alpha_n)}{\alpha_n^2}$, and $J_2(\alpha_n) = \tfrac{2}{\alpha_n}J_1(\alpha_n) - J_0(\alpha_n) = \tfrac{2}{\alpha_n}J_1(\alpha_n)$. *Answer:* $A_n = \dfrac{8T_0}{\alpha_n^3 J_1(\alpha_n)}$.

10. Show that a harmonic function inside a disk equals the average of its boundary values at the centre (mean-value property).

Part V
Green's Functions

26 The idea

Suppose we must solve $\mathcal{L}u = f$ with $\mathcal{L}$ a linear differential operator and homogeneous boundary conditions. Break the source f into point sources, $f(x) = \int f(x')\delta(x-x')\mathrm{d}x'$. If $G(x,x')$ is the response to a unit point source at x',

$$\mathcal{L}_x G(x,x') = \delta(x-x') \qquad \text{(with the same homogeneous boundary conditions)},$$

then by linearity the response to f is the superposition of the responses:

Key result

$$u(x) = \int G(x,x')\, f(x')\, \mathrm{d}x' \qquad \Longleftrightarrow \qquad \mathcal{L}u = f.$$

G is the **kernel of the inverse operator** $\mathcal{L}^{-1}$. It is the continuous-variable analogue of the matrix inverse: $A\boldsymbol{u} = \boldsymbol{f} \Rightarrow \boldsymbol{u} = A^{-1}\boldsymbol{f}$, $u_i = \sum_j G_{ij} f_j$.

Check: $\mathcal{L}u = \int \mathcal{L}_x G(x,x') f(x')\mathrm{d}x' = \int \delta(x-x')f(x')\mathrm{d}x' = f(x)$. ■

27 Green's functions for ordinary differential equations

27.1 Construction for an SL operator

Let $\mathcal{L} = \dfrac{\mathrm{d}}{\mathrm{d}x}\Big(p\dfrac{\mathrm{d}}{\mathrm{d}x}\Big) + q$ on $[a,b]$ with homogeneous boundary conditions at a and b. We want $\mathcal{L}G = \delta(x-x')$.

1. For $x \neq x'$, G solves the homogeneous equation $\mathcal{L}G = 0$.
2. Let u_1 be a solution of $\mathcal{L}u = 0$ satisfying the boundary condition at a, and u_2 one satisfying the condition at b. Then $G = A\,u_1(x)u_2(x')$ for $x < x'$ and $G = A\,u_2(x)u_1(x')$ for $x > x'$, i.e. $G = A\,u_1(x_<)\,u_2(x_>)$ with $x_< = \min(x,x')$, $x_> = \max(x,x')$. (This is symmetric: $G(x,x') = G(x',x)$.)
3. **Continuity at** $x = x'$ holds automatically. **Jump condition:** integrate $(pG')' + qG = \delta(x-x')$ over $(x'-\epsilon, x'+\epsilon)$; the qG term gives $O(\epsilon)$ and the δ gives 1:

$$p(x')\Big[G'(x'^{+}) - G'(x'^{-})\Big] = 1.$$

   Evaluate: $G'(x'^{+}) = Au_1(x')u_2'(x')$ (here $x > x'$, $x_< = x'$... careful: for $x > x'$, $G = Au_1(x')u_2(x)$ so $G' = Au_1(x')u_2'(x)$) and $G'(x'^{-}) = Au_1'(x')u_2(x')$. So the jump is $A\big[u_1u_2' - u_1'u_2\big]_{x'} = A\,W(x')$ with W the Wronskian of u_1, u_2. Abel's identity: pW = constant. Hence $A\,p\,W = 1$.

Key result

$$G(x,x') = \frac{u_1(x_<)\,u_2(x_>)}{p\,W}, \qquad W = u_1u_2' - u_1'u_2, \qquad pW = \text{const}.$$

The Green's function exists iff u_1 and u_2 are linearly independent, i.e. iff $\mathcal{L}u = 0$ has no non-trivial solution satisfying both boundary conditions (iff 0 is not an eigenvalue).

$u'' = f(x)$ on $[0,1]$, $u(0) = u(1) = 0$

$p = 1$, $q = 0$. Homogeneous solutions: $u_1 = x$ (vanishes at 0), $u_2 = x - 1$ (vanishes at 1). $W = u_1 u_2' - u_1' u_2 = x \cdot 1 - 1 \cdot (x-1) = 1$. So

$$G(x,x') = x_<(x_> - 1),$$
$$u(x) = \int_0^1 G(x,x') f(x') \mathrm{d}x' = (x-1)\int_0^x x' f(x') \mathrm{d}x' + x \int_x^1 (x'-1) f(x') \mathrm{d}x'.$$

Test with $f = 1$: $u = (x-1)\frac{x^2}{2} + x\left[\frac{x'^2}{2} - x'\right]_x^1 = (x-1)\frac{x^2}{2} + x\left(-\frac{1}{2} - \frac{x^2}{2} + x\right) = \frac{x^3 - x^2}{2} - \frac{x}{2} - \frac{x^3}{2} + x^2 = \frac{x^2 - x}{2}$. This agrees with the direct solution $u = \frac{x(x-1)}{2}$. ✓

Forced oscillator: initial-value Green's function

$\ddot{x} + \omega^2 x = F(t)$, $x(0) = \dot{x}(0) = 0$. Here the conditions are *initial* (causal), not at two end points. Require $G(t,t') = 0$ for $t < t'$ and a jump $\partial_t G = 1$ at $t = t'$: $G = \dfrac{\sin \omega(t-t')}{\omega} H(t-t')$ and $x(t) = \displaystyle\int_0^t \frac{\sin \omega(t-t')}{\omega} F(t') \mathrm{d}t'$. With damping $\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = F$: $G = \dfrac{\mathrm{e}^{-\gamma(t-t')} \sin \Omega(t-t')}{\Omega} H(t-t')$, $\Omega = \sqrt{\omega_0^2 - \gamma^2}$ (the Fourier-transform result of Part II).

27.2 Eigenfunction expansion of G

If $\mathcal{L}\phi_n = -\lambda_n w \phi_n$ (SL, ϕ_n orthonormal with weight w, all $\lambda_n \neq 0$), then expanding $G = \sum_n g_n(x')\phi_n(x)$ and using the completeness relation $\delta(x - x') = w(x') \sum \phi_n(x) \phi_n^*(x')$:

$$\mathcal{L}G = \sum_n g_n(-\lambda_n w \phi_n) = w(x') \sum \phi_n(x) \phi_n^*(x') \;\Rightarrow\; g_n = -\frac{\phi_n^*(x')}{\lambda_n}, \qquad G(x,x') = -\sum_n \frac{\phi_n(x)\phi_n^*(x')}{\lambda_n}.$$

(Same structure as $A^{-1} = \sum |v_n\rangle\langle v_n| / \lambda_n$.) For $\mathcal{L} = \dfrac{\mathrm{d}^2}{\mathrm{d}x^2}$ on $[0,1]$ with Dirichlet conditions: $\phi_n = \sqrt{2}\sin n\pi x$, $\lambda_n = n^2\pi^2$, so $G = -\sum_{n \geq 1} \dfrac{2 \sin n\pi x \sin n\pi x'}{n^2 \pi^2}$, which equals $x_<(x_> - 1)$ (negative, as it should be).

Zero modes: if some $\lambda_n = 0$ (e.g. Neumann conditions with a constant eigenfunction) G does not exist; one solves $\mathcal{L}u = f$ only if $f \perp \phi_0$ and uses a *modified Green's function* $\mathcal{L}G = \delta - \phi_0 \phi_0^*$.

28 Free-space Green's functions for ∇^2 and Helmholtz

Solve $\nabla^2 G(\boldsymbol{r}) = \delta^3(\boldsymbol{r})$ in all space. Fourier transform (Part II): $-k^2 \tilde{G} = 1$, $\tilde{G} = -1/k^2$. Invert using the radial formula $\frac{1}{(2\pi)^3} \int \mathrm{d}^3 k\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} \tilde{G}(k) = \frac{1}{2\pi^2 r} \int_0^\infty k \sin(kr) \tilde{G}(k) \mathrm{d}k$ (inverse of the radial FT):

$$G(r) = -\frac{1}{2\pi^2 r} \int_0^\infty \frac{\sin kr}{k} \mathrm{d}k = -\frac{1}{2\pi^2 r} \cdot \frac{\pi}{2} = -\frac{1}{4\pi r}.$$

Key result

Operator	Equation $\mathcal{L}G=\delta$	Green's function $G(\boldsymbol{R})$, $\boldsymbol{R}=\boldsymbol{r}-\boldsymbol{r}'$
∇^2 (3D)	$\nabla^2 G=\delta^3$	$-\dfrac{1}{4\pi R}$
∇^2 (2D)	$\nabla^2 G=\delta^2$	$\dfrac{1}{2\pi}\ln R$
∇^2 (1D)	$G''=\delta$	$\tfrac{1}{2}\lvert x-x'\rvert$
∇^2+k^2 (3D)	$(\nabla^2+k^2)G=\delta^3$	$-\dfrac{\mathrm{e}^{\pm \mathrm{i}kR}}{4\pi R}$
$\nabla^2-\mu^2$ (3D)	$(\nabla^2-\mu^2)G=\delta^3$	$-\dfrac{\mathrm{e}^{-\mu R}}{4\pi R}$

Consequently the solution of Poisson's equation $\nabla^2 u=-\rho/\epsilon_0$ in free space is $u(\boldsymbol{r})=\dfrac{1}{4\pi\epsilon_0}\displaystyle\int\frac{\rho(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\mathrm{d}^3r'$ (Coulomb's law).

Check the 2D one: $\frac{1}{\rho}\partial_\rho(\rho\cdot\frac{1}{2\pi\rho})=0$ for $\rho\neq 0$, and the flux $\oint\nabla G\cdot\mathrm{d}\boldsymbol{l}=2\pi\rho\cdot\frac{1}{2\pi\rho}=1$. **1D:** $\frac{1}{2}|x|''=\delta(x)$ (Part III).

Past paper: PHY 302 Mid-sem 4.3.2024, Q4 (10 marks)

In the Lippmann–Schwinger equation of scattering theory one needs the operator $1/(E-H_0+\mathrm{i}\epsilon)$. In position space it defines a Green's function

$$G(\vec{x}-\vec{x}')=\lim_{\epsilon\to 0^+}\left\langle\vec{x}\right|\frac{1}{E-H_0+\mathrm{i}\epsilon}\left|\vec{x}'\right\rangle,\qquad\left(E+\frac{\hbar^2\nabla^2}{2m}\right)G(\vec{x}-\vec{x}')=\delta(\vec{x}-\vec{x}').$$

Inserting the complete set of states in the momentum representation and using $H_0=\hat{p}^2/2m$, show that $G(\vec{x}-\vec{x}')=\displaystyle\int\frac{\mathrm{d}^3p}{(2\pi\hbar)^3}\frac{\mathrm{e}^{\mathrm{i}(\vec{x}-\vec{x}')\cdot\vec{p}/\hbar}}{E-p^2/2m+\mathrm{i}\epsilon}$. Now using polar coordinates and the tricks of contour integration, show that $G(\vec{r})=-\dfrac{2m}{\hbar^2}\dfrac{\mathrm{e}^{\mathrm{i}kr}}{4\pi r}$, where $\vec{r}=\vec{x}-\vec{x}'$. (FYI: this Green's function describes the wave propagating towards a point sink.)

Solution

Step 1: momentum representation. Insert $\mathbb{1}=\int\mathrm{d}^3p\,|\vec{p}\rangle\langle\vec{p}|$ on both sides of the operator, with $\langle\vec{x}|\vec{p}\rangle=\dfrac{\mathrm{e}^{\mathrm{i}\vec{p}\cdot\vec{x}/\hbar}}{(2\pi\hbar)^{3/2}}$ and $H_0|\vec{p}\rangle=\dfrac{p^2}{2m}|\vec{p}\rangle$, $\langle\vec{p}|\vec{p}'\rangle=\delta^3(\vec{p}-\vec{p}')$:

$$\begin{aligned}G&=\int\mathrm{d}^3p\int\mathrm{d}^3p'\langle\vec{x}|\vec{p}\rangle\left\langle\vec{p}\right|\frac{1}{E-H_0+\mathrm{i}\epsilon}\left|\vec{p}'\right\rangle\langle\vec{p}'|\vec{x}'\rangle=\int\mathrm{d}^3p\,\frac{\langle\vec{x}|\vec{p}\rangle\langle\vec{p}|\vec{x}'\rangle}{E-p^2/2m+\mathrm{i}\epsilon}\\&=\int\frac{\mathrm{d}^3p}{(2\pi\hbar)^3}\frac{\mathrm{e}^{\mathrm{i}(\vec{x}-\vec{x}')\cdot\vec{p}/\hbar}}{E-p^2/2m+\mathrm{i}\epsilon}.\quad\checkmark\end{aligned}$$

Step 2: rescale. Let $\vec{p}=\hbar\vec{q}$ ($\mathrm{d}^3p=\hbar^3\mathrm{d}^3q$) and $E=\hbar^2k^2/2m$ with $k>0$. Then $E-\dfrac{p^2}{2m}=\dfrac{\hbar^2}{2m}(k^2-q^2)$, and absorbing the positive constant into ϵ,

$$G(\vec{r})=\frac{1}{(2\pi)^3}\int\mathrm{d}^3q\,\frac{\mathrm{e}^{\mathrm{i}\vec{q}\cdot\vec{r}}}{\frac{\hbar^2}{2m}(k^2-q^2+\mathrm{i}\epsilon')}=-\frac{2m}{\hbar^2}\frac{1}{(2\pi)^3}\int\mathrm{d}^3q\,\frac{\mathrm{e}^{\mathrm{i}\vec{q}\cdot\vec{r}}}{q^2-k^2-\mathrm{i}\epsilon'}\equiv-\frac{2m}{\hbar^2}\frac{I}{(2\pi)^3}.$$

Step 3: angular integrals. Use spherical coordinates for $\vec{q}$ with the polar axis along $\vec{r}$:

$$I = \int_0^\infty q^2 \mathrm{d}q \int_0^{2\pi} \mathrm{d}\phi \int_{-1}^{1} \mathrm{d}(\cos\theta) \frac{\mathrm{e}^{\mathrm{i}qr\cos\theta}}{q^2 - k^2 - \mathrm{i}\epsilon'}$$
$$= 2\pi \int_0^\infty \frac{q^2 \mathrm{d}q}{q^2 - k^2 - \mathrm{i}\epsilon'} \cdot \frac{2\sin qr}{qr} = \frac{4\pi}{r} \int_0^\infty \frac{q \sin qr}{q^2 - k^2 - \mathrm{i}\epsilon'} \mathrm{d}q.$$

The integrand $q\sin(qr)/(q^2-k^2-\mathrm{i}\epsilon')$ is even in q, so $\int_0^\infty = \frac{1}{2}\int_{-\infty}^\infty$. Write $\sin qr = \frac{\mathrm{e}^{\mathrm{i}qr} - \mathrm{e}^{-\mathrm{i}qr}}{2\mathrm{i}}$. Under $q \to -q$ the term with $\mathrm{e}^{-\mathrm{i}qr}$ becomes minus the term with $\mathrm{e}^{\mathrm{i}qr}$, so $\int_{-\infty}^\infty \frac{q\sin qr}{q^2-k^2-\mathrm{i}\epsilon'}\mathrm{d}q = \frac{1}{\mathrm{i}} J$, with $J = \int_{-\infty}^\infty \frac{q\,\mathrm{e}^{\mathrm{i}qr}}{q^2-k^2-\mathrm{i}\epsilon'}\mathrm{d}q$.

Step 4: contour integral. Close the contour by a large semicircle in the *upper* half q-plane, where $\mathrm{e}^{\mathrm{i}qr} \to 0$ (Jordan's lemma, $r > 0$; the factor $q/(q^2-k^2) \to 0$). The poles are at $q^2 = k^2 + \mathrm{i}\epsilon'$, i.e. $q = \pm(k + \mathrm{i}\epsilon'')$: the pole at $q = +k + \mathrm{i}\epsilon''$ lies in the upper half-plane (inside), the other at $-k - \mathrm{i}\epsilon''$ in the lower half-plane (outside). The residue of $\frac{q\,\mathrm{e}^{\mathrm{i}qr}}{(q-q_0)(q+q_0)}$ at $q_0 = k$ is $\frac{k\,\mathrm{e}^{\mathrm{i}kr}}{2k} = \frac{\mathrm{e}^{\mathrm{i}kr}}{2}$. Hence $J = 2\pi\mathrm{i} \cdot \frac{\mathrm{e}^{\mathrm{i}kr}}{2} = \pi\mathrm{i}\,\mathrm{e}^{\mathrm{i}kr}$.

Step 5: assemble. $\int_0^\infty \frac{q\sin qr}{\dots}\mathrm{d}q = \frac{1}{2}\cdot\frac{1}{\mathrm{i}}\cdot\pi\mathrm{i}\,\mathrm{e}^{\mathrm{i}kr} = \frac{\pi}{2}\mathrm{e}^{\mathrm{i}kr}$, so $I = \frac{4\pi}{r}\cdot\frac{\pi}{2}\mathrm{e}^{\mathrm{i}kr} = \frac{2\pi^2\mathrm{e}^{\mathrm{i}kr}}{r}$, and

$$G(\vec{r}) = -\frac{2m}{\hbar^2}\cdot\frac{1}{8\pi^3}\cdot\frac{2\pi^2\mathrm{e}^{\mathrm{i}kr}}{r} = \boxed{-\frac{2m}{\hbar^2}\,\frac{\mathrm{e}^{\mathrm{i}kr}}{4\pi r}}$$

Verification. Away from $r = 0$, $(\nabla^2 + k^2)\frac{\mathrm{e}^{\mathrm{i}kr}}{r} = 0$, and near $r = 0$, $G \approx -\frac{2m}{\hbar^2}\frac{1}{4\pi r}$ so $\nabla^2 G = -\frac{2m}{\hbar^2}(-\delta^3) = \frac{2m}{\hbar^2}\delta^3$; therefore $\big(E + \frac{\hbar^2}{2m}\nabla^2\big)G = \frac{\hbar^2}{2m}(\nabla^2 + k^2)G = \delta^3$. ✓**Role of** $\mathrm{i}\epsilon$**:** the sign of ϵ selects which pole lies in the upper half-plane and hence $\mathrm{e}^{+\mathrm{i}kr}$ (outgoing wave, for time dependence $\mathrm{e}^{-\mathrm{i}Et/\hbar}$) versus $\mathrm{e}^{-\mathrm{i}kr}$ for $-\mathrm{i}\epsilon$. This is exactly the outgoing-wave boundary condition of scattering theory.

29 Green's identities and the representation formula

Green's second identity (from the divergence theorem applied to $\phi\nabla\psi - \psi\nabla\phi$):

$$\int_V \big(\phi\nabla^2\psi - \psi\nabla^2\phi\big)\mathrm{d}V = \oint_S \big(\phi\,\partial_n\psi - \psi\,\partial_n\phi\big)\mathrm{d}S, \qquad \partial_n = \hat{\boldsymbol{n}}\cdot\nabla \text{ (outward normal)}.$$

Take $\phi = u$ with $\nabla^2 u = \rho$ (the convention of the exam formula sheet) and $\psi = G(\boldsymbol{r}, \boldsymbol{r}_0)$ with $\nabla^2 G = \delta(\boldsymbol{r} - \boldsymbol{r}_0)$, $\boldsymbol{r}_0 \in V$:

$$\int_V \big(u\,\delta(\boldsymbol{r} - \boldsymbol{r}_0) - G\rho\big)\mathrm{d}V = u(\boldsymbol{r}_0) - \int_V G\rho\,\mathrm{d}V = \oint_S \big(u\,\partial_n G - G\,\partial_n u\big)\mathrm{d}S.$$

Key result

Representation formula (sheet eq. (18)):

$$u(\boldsymbol{r}_0) = \int_V G(\boldsymbol{r}, \boldsymbol{r}_0)\,\rho(\boldsymbol{r})\,\mathrm{d}V + \oint_S \Big[u(\boldsymbol{r})\frac{\partial G(\boldsymbol{r}, \boldsymbol{r}_0)}{\partial n} - G(\boldsymbol{r}, \boldsymbol{r}_0)\frac{\partial u(\boldsymbol{r})}{\partial n}\Big]\mathrm{d}S.$$

- **Dirichlet Green's function** $G_D = 0$ on S: $u(\boldsymbol{r}_0) = \int G_D\rho\,\mathrm{d}V + \oint u\,\partial_n G_D\,\mathrm{d}S$ (needs only

u on S).

- **Neumann Green's function:** $\oint \partial_n G \,\mathrm{d}S = \int \nabla^2 G \,\mathrm{d}V = 1$ forces $\partial_n G = 1/A$ (A the area of S): $u(\boldsymbol{r}_0) = \int G_N \rho \,\mathrm{d}V + \dfrac{1}{A}\oint u \,\mathrm{d}S - \oint G_N \partial_n u \,\mathrm{d}S$ (sheet eq. (20); the middle term is the average of u on S, a constant).
- $G = G_0 + h$ with G_0 the free-space function and h a harmonic function adjusted to meet the boundary condition.

30 The method of images

Choose h as the potential of fictitious "image" sources placed *outside* V, so that $G = G_0 + h$ obeys the boundary condition.

30.1 Plane

Half-space $z > 0$. For a source at $\boldsymbol{r}_0 = (x_0, y_0, z_0)$, $z_0 > 0$, take the image at $\boldsymbol{r}_0' = (x_0, y_0, -z_0)$:

$$G_D(\boldsymbol{r}, \boldsymbol{r}_0) = -\frac{1}{4\pi}\Big(\frac{1}{|\boldsymbol{r} - \boldsymbol{r}_0|} - \frac{1}{|\boldsymbol{r} - \boldsymbol{r}_0'|}\Big).$$

On $z = 0$ the two distances are equal, so $G_D = 0$. **Dirichlet problem** $\nabla^2 u = 0$, $u(x, y, 0) = f(x, y)$: the representation formula gives $u(\boldsymbol{r}_0) = \oint f \,\partial_n G_D \,\mathrm{d}S$. With $R^2 = (x - x_0)^2 + (y - y_0)^2 + z_0^2$ on the plane $z = 0$ and outward normal $-\hat{z}$: $\partial_n G_D = -\partial_z G_D|_{z=0} = \dfrac{z_0}{2\pi R^3}$. Therefore

$$u(x_0, y_0, z_0) = \frac{z_0}{2\pi} \iint \frac{f(x, y)\,\mathrm{d}x\,\mathrm{d}y}{\big[(x - x_0)^2 + (y - y_0)^2 + z_0^2\big]^{3/2}}.$$

Point charge above a grounded conducting plane

Charge q at height d: potential for $z > 0$ is $\dfrac{q}{4\pi\epsilon_0}\Big(\dfrac{1}{|\boldsymbol{r} - d\hat{z}|} - \dfrac{1}{|\boldsymbol{r} + d\hat{z}|}\Big)$. The induced surface charge density is $\sigma = -\epsilon_0 \partial_z u\big|_{z=0} = -\dfrac{qd}{2\pi(\rho^2 + d^2)^{3/2}}$, total $-q$ ($\int_0^\infty \frac{2\pi\rho\,\mathrm{d}\rho\, d}{2\pi(\rho^2+d^2)^{3/2}} = 1$). Force on the charge: $\dfrac{q^2}{4\pi\epsilon_0(2d)^2}$ attractive (the field of the image).

30.2 Sphere

For a sphere $|\boldsymbol{r}| = a$ and a source at $\boldsymbol{r}_0$ with $|\boldsymbol{r}_0| = r_0$, put the image at the *inverse point*

$$\boldsymbol{r}_1 = \frac{a^2}{r_0^2}\boldsymbol{r}_0 \qquad \text{(sheet eq. (21))}, \qquad |\boldsymbol{r} - \boldsymbol{r}_1| = \frac{a}{r_0}|\boldsymbol{r} - \boldsymbol{r}_0| \ \text{ for } |\boldsymbol{r}| = a.$$

Proof of the distance relation: $|\boldsymbol{r} - \boldsymbol{r}_1|^2 = a^2 + \dfrac{a^4}{r_0^2} - \dfrac{2a^2}{r_0^2}\,\boldsymbol{r}\cdot\boldsymbol{r}_0$ and $\dfrac{a^2}{r_0^2}|\boldsymbol{r} - \boldsymbol{r}_0|^2 = \dfrac{a^2}{r_0^2}\big(a^2 + r_0^2 - 2\boldsymbol{r}\cdot\boldsymbol{r}_0\big) = \dfrac{a^4}{r_0^2} + a^2 - \dfrac{2a^2}{r_0^2}\boldsymbol{r}\cdot\boldsymbol{r}_0$. They are equal. ✓Hence

$$G_D(\boldsymbol{r}, \boldsymbol{r}_0) = -\frac{1}{4\pi}\Big[\frac{1}{|\boldsymbol{r} - \boldsymbol{r}_0|} - \frac{a}{r_0}\frac{1}{|\boldsymbol{r} - \boldsymbol{r}_1|}\Big],$$

which vanishes on $|\boldsymbol{r}| = a$ (the second term equals $\frac{a}{r_0}\cdot\frac{r_0}{a}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_0|}$). The corresponding **Poisson integral** for the interior Dirichlet problem is

$$u(\boldsymbol{r}_0) = \frac{a^2 - r_0^2}{4\pi a}\oint_{|\boldsymbol{r}|=a}\frac{f(\boldsymbol{r})}{|\boldsymbol{r}-\boldsymbol{r}_0|^3}\,\mathrm{d}S,$$

whose value at the centre ($r_0 = 0$) is $\frac{1}{4\pi a^2}\oint f\,\mathrm{d}S$, the average over the sphere (mean value property).

Point charge outside a grounded conducting sphere

Charge q at distance $d > a$ from the centre: image charge $q' = -q\dfrac{a}{d}$ at distance $\dfrac{a^2}{d}$ from the centre (on the same line). Force on q: $F = \dfrac{qq'}{4\pi\epsilon_0\left(d - a^2/d\right)^2} = -\dfrac{q^2}{4\pi\epsilon_0}\dfrac{ad}{(d^2-a^2)^2}$ (attractive).

30.3 Disk (2D)

For the disk $|\boldsymbol{r}| < a$ the image of $\boldsymbol{r}_0$ is $\boldsymbol{r}_1 = \dfrac{a^2}{r_0^2}\boldsymbol{r}_0$ and the Dirichlet Green's function is $G_D = \dfrac{1}{2\pi}\Big[\ln|\boldsymbol{r}-\boldsymbol{r}_0| - \ln|\boldsymbol{r}-\boldsymbol{r}_1| - \ln\dfrac{r_0}{a}\Big]$. (Sheet eq. (17) omits the constant $-\ln(r_0/a)$; the constant does not affect $\partial_n G$, so the Poisson formula obtained from it is identical to the one derived in Part IV.)

31 Time-dependent Green's functions

- **Heat equation:** $\big(\partial_t - \kappa\partial_x^2\big)G = \delta(x)\delta(t)$ gives $G(x,t) = \dfrac{H(t)}{\sqrt{4\pi\kappa t}}\mathrm{e}^{-x^2/4\kappa t}$ (the heat kernel of Part II), and $u(x,t) = \displaystyle\int G(x-x',t)\varphi(x')\mathrm{d}x'$ solves the initial-value problem.
- **Wave equation in 3D:** $\big(\frac{1}{c^2}\partial_t^2 - \nabla^2\big)G = \delta^3(\boldsymbol{r})\delta(t)$ gives the *retarded* Green's function $G = \dfrac{\delta(t - r/c)}{4\pi r}$: the effect at $\boldsymbol{r}$ at time t comes from the source at the earlier time $t - r/c$. (Fourier transform in time turns it into the Helmholtz Green's function with $k = \omega/c$.) In 1D: $G = \frac{c}{2}H(ct - |x|)$.

Practice problems

1. Find the Green's function of $\dfrac{\mathrm{d}^2}{\mathrm{d}x^2} - \mu^2$ on $(-\infty,\infty)$ vanishing at $\pm\infty$. *Answer:* $G = -\dfrac{\mathrm{e}^{-\mu|x-x'|}}{2\mu}$ (jump of G' equal to 1).
2. Find G for $u'' + u = \delta$ on $[0,\pi/2]$ with $u(0) = u(\pi/2) = 0$. *Answer:* $u_1 = \sin x$, $u_2 = \cos x$, $W = -1$, $G = -\sin x_< \cos x_>$.
3. Solve $u'' = \mathrm{e}^x$ on $[0,1]$, $u(0) = u(1) = 0$ using $G = x_<(x_> - 1)$. *Answer:* $u = \mathrm{e}^x - 1 - (\mathrm{e}-1)x$.
4. Show $-\sum_{n\ge1}\dfrac{2\sin n\pi x\sin n\pi x'}{n^2\pi^2} = x_<(x_>-1)$ at $x = x' = \frac12$ gives $\sum_{n\text{ odd}}\dfrac{1}{n^2} = \dfrac{\pi^2}{8}$. *Hint:* $\sin^2(n\pi/2) = 1$ for odd n; $G(\frac12,\frac12) = -\frac14$.
5. Find the potential on the axis of a grounded sphere ($r < a$) with a point charge q inside at $\boldsymbol{r}_0$. *Hint:* image charge $-qa/r_0$ at a^2/r_0.
6. A line charge λ parallel to a grounded plane at distance d: find the image and the force per

unit length. *Answer:* image $-\lambda$ at distance d behind the plane; $F = \dfrac{\lambda^2}{4\pi\epsilon_0 d}$ per unit length.

7. Derive the 1D Helmholtz Green's function $(\partial_x^2 + k^2)G = \delta$: $G = \dfrac{\mathrm{e}^{\mathrm{i}k|x-x'|}}{2\mathrm{i}k}$ (outgoing).

Part VI
Integral Equations

32 Classification

An **integral equation** contains the unknown function $y(x)$ under an integral sign.

Key result

Type	Equation	Limits
Fredholm, 2nd kind	$y(x) = f(x) + \lambda \int_a^b K(x,t)\, y(t)\, \mathrm{d}t$	fixed a, b
Fredholm, 1st kind	$f(x) = \lambda \int_a^b K(x,t)\, y(t)\, \mathrm{d}t$	fixed
Volterra, 2nd kind	$y(x) = f(x) + \lambda \int_a^x K(x,t)\, y(t)\, \mathrm{d}t$	variable upper limit
Volterra, 1st kind	$f(x) = \lambda \int_a^x K(x,t)\, y(t)\, \mathrm{d}t$	variable

K is the **kernel**, f the (known) free term, λ a parameter. If $f \equiv 0$ the equation is **homogeneous**: it is an eigenvalue problem $y = \lambda \mathcal{K} y$ for the integral operator $(\mathcal{K}y)(x) = \int K(x,t) y(t) \mathrm{d}t$.

Special kernels. *Symmetric:* $K(x,t) = K(t,x)$ (the operator is Hermitian). *Separable (degenerate):* $K = \sum_{i=1}^{n} a_i(x) b_i(t)$. *Convolution (difference) kernel:* $K = k(x-t)$. A Volterra equation is a Fredholm equation with $K(x,t) = 0$ for $t > x$.

33 Relation to differential equations

Initial-value problems $\to$ Volterra equations. Integrate the ODE using the initial data; the integration constants are absorbed automatically.

$y' = y$, $y(0) = 1$

Integrate from 0 to x: $y(x) - 1 = \int_0^x y(t)\mathrm{d}t$, i.e. $y(x) = 1 + \int_0^x y(t)\mathrm{d}t$, a Volterra equation of the second kind ($K = 1$, $f = 1$, $\lambda = 1$). Conversely differentiating gives back the ODE and $x = 0$ gives $y(0) = 1$.

$y'' + \omega^2 y = f(x)$, $y(0) = \alpha$, $y'(0) = \beta$

Write $y'' = g(x) \equiv f(x) - \omega^2 y(x)$. Integrate once: $y'(x) = \beta + \int_0^x g(s)\mathrm{d}s$. Integrate again: $y(x) = \alpha + \beta x + \int_0^x \int_0^s g(t)\mathrm{d}t\, \mathrm{d}s$. Interchange the order of the double integral over $0 \le t \le s \le x$: $\int_0^x \mathrm{d}t\, g(t) \int_t^x \mathrm{d}s = \int_0^x (x-t) g(t) \mathrm{d}t$. Hence

$$y(x) = \alpha + \beta x + \int_0^x (x-t) f(t) \mathrm{d}t - \omega^2 \int_0^x (x-t) y(t) \mathrm{d}t,$$

a Volterra equation with kernel $K(x,t) = -\omega^2 (x-t)$ and $f_{\text{eff}} = \alpha + \beta x + \int_0^x (x-t) f$.

Boundary-value problems → Fredholm equations via the Green's function (Part V): if $\mathcal{L}y = -\lambda w y$ with homogeneous boundary conditions and G is the Green's function of $\mathcal{L}$, then $y(x) = -\lambda \int G(x,t)w(t)y(t)\,\mathrm{d}t$.

The vibrating-string eigenproblem

$y'' + \lambda y = 0$, $y(0) = y(1) = 0$. With $G = x_<(x_> - 1)$ for $\mathrm{d}^2/\mathrm{d}x^2$, $y(x) = -\lambda \int_0^1 G(x,t)y(t)\mathrm{d}t = \lambda \int_0^1 K(x,t)y(t)\mathrm{d}t$ with $K(x,t) = x_<(1 - x_>)$. Symmetric kernel; eigenvalues $\lambda_n = n^2\pi^2$, eigenfunctions $\sin n\pi x$.

34 Methods of solution

34.1 Separable kernels (Fredholm, second kind)

If $K(x,t) = \sum_i a_i(x)b_i(t)$, then $y(x) = f(x) + \lambda \sum_i a_i(x)c_i$ with the constants $c_i = \int b_i(t)y(t)\mathrm{d}t$. Multiply by $b_j(x)$ and integrate: a *linear algebraic system* for the c_j.

$$y(x) = x + \lambda \int_0^1 (1 + xt)\, y(t)\,\mathrm{d}t$$

Here $K = 1 \cdot 1 + x \cdot t$. Define $A = \int_0^1 y(t)\mathrm{d}t$, $B = \int_0^1 t\, y(t)\mathrm{d}t$. Then $y(x) = x + \lambda(A + Bx)$. Substitute into the definitions of A and B:

$$A = \int_0^1 \big[t + \lambda(A + Bt)\big]\mathrm{d}t = \tfrac{1}{2} + \lambda A + \tfrac{\lambda}{2}B,$$
$$B = \int_0^1 t\big[t + \lambda(A + Bt)\big]\mathrm{d}t = \tfrac{1}{3} + \tfrac{\lambda}{2}A + \tfrac{\lambda}{3}B.$$

Rearranged: $(1 - \lambda)A - \frac{\lambda}{2}B = \frac{1}{2}$ and $-\frac{\lambda}{2}A + (1 - \frac{\lambda}{3})B = \frac{1}{3}$. The determinant is

$$D = (1 - \lambda)\Big(1 - \frac{\lambda}{3}\Big) - \frac{\lambda^2}{4} = 1 - \frac{4\lambda}{3} + \frac{\lambda^2}{3} - \frac{\lambda^2}{4} = 1 - \frac{4\lambda}{3} + \frac{\lambda^2}{12} = \frac{\lambda^2 - 16\lambda + 12}{12}.$$

Cramer's rule: $A = \dfrac{\frac{1}{2}(1 - \frac{\lambda}{3}) + \frac{\lambda}{2}\cdot\frac{1}{3}}{D} = \dfrac{\frac{1}{2}}{D}$ and $B = \dfrac{(1-\lambda)\frac{1}{3} + \frac{\lambda}{2}\cdot\frac{1}{2}}{D} = \dfrac{\frac{1}{3} - \frac{\lambda}{12}}{D}$. Therefore

$$\boxed{y(x) = x + \frac{\lambda\big[6 + (4 - \lambda)x\big]}{\lambda^2 - 16\lambda + 12}}$$

(multiplying numerator and denominator by 12). **Checks:** as $\lambda \to 0$, $y \to x$; to first order $y \approx x + \lambda\big(\frac{1}{2} + \frac{x}{3}\big)$, which equals $x + \lambda\int_0^1 (1+xt)t\,\mathrm{d}t$. ✓**Eigenvalues:** the homogeneous equation has a non-trivial solution only when $D = 0$: $\lambda^2 - 16\lambda + 12 = 0 \Rightarrow \lambda = 8 \pm 2\sqrt{13} \approx 0.789,\ 15.21$; at these values the inhomogeneous equation is solvable only if f satisfies a compatibility condition (Fredholm alternative).

34.2 Neumann series (iteration) and the resolvent kernel

Write $y = f + \lambda\mathcal{K}y$ and iterate: $y_0 = f$, $y_{n+1} = f + \lambda\mathcal{K}y_n$:

$$y = f + \lambda\mathcal{K}f + \lambda^2\mathcal{K}^2 f + \cdots = \sum_{n=0}^{\infty} \lambda^n \mathcal{K}^n f.$$

Key result

Define the iterated kernels $K_1 = K$, $K_{n+1}(x,t) = \int K(x,s)K_n(s,t)\mathrm{d}s$ and the **resolvent kernel** $R(x,t;\lambda) = \sum_{n=0}^{\infty} \lambda^n K_{n+1}(x,t)$. Then $y(x) = f(x) + \lambda \int_a^b R(x,t;\lambda) f(t)\,\mathrm{d}t$. The series converges if $|\lambda|\,\|K\| < 1$, with $\|K\|^2 = \iint |K|^2 \mathrm{d}x\,\mathrm{d}t$ (the smallest $|\lambda_n|$ marks the radius of convergence). For Volterra equations the series converges for *all* λ.

$y(x) = 1 + \lambda \int_0^1 x\,t\,y(t)\,\mathrm{d}t$

$K = xt$: $K_2 = \int_0^1 xs \cdot st\,\mathrm{d}s = \frac{xt}{3}$, $K_3 = \frac{xt}{9}$, …, $K_n = \dfrac{xt}{3^{n-1}}$. So $R = \sum_{n\ge 0} \lambda^n \dfrac{xt}{3^n} = \dfrac{xt}{1-\lambda/3} = \dfrac{3xt}{3-\lambda}$ for $|\lambda| < 3$. Then $y = 1 + \lambda \int_0^1 \dfrac{3xt}{3-\lambda}\mathrm{d}t = 1 + \dfrac{3\lambda x}{2(3-\lambda)}$. *Direct check:* $y = 1 + \lambda x B$ with $B = \int ty = \frac{1}{2} + \frac{\lambda}{3}B \Rightarrow B = \frac{3}{2(3-\lambda)}$. ✓The singularity at $\lambda = 3$ is the eigenvalue ($y = x$: $\lambda \int_0^1 xt \cdot t\,\mathrm{d}t\cdot = \lambda x/3$, so $\lambda = 3$).

Volterra by iteration

(a) $y(x) = 1 + \int_0^x y(t)\mathrm{d}t$: $y_0 = 1$, $y_1 = 1 + x$, $y_2 = 1 + x + \frac{x^2}{2}$, …$\to \mathrm{e}^x$. **(b)** $y(x) = x + \int_0^x (x-t)y(t)\mathrm{d}t$: $y_0 = x$, $y_1 = x + \int_0^x (x-t)t\,\mathrm{d}t = x + \frac{x^3}{6}$, $y_2 = x + \frac{x^3}{6} + \frac{x^5}{120}$, …$\to \sinh x$. (Equivalent ODE: $y'' = y$, $y(0) = 0$, $y'(0) = 1$.)

34.3 Convolution kernels: Laplace and Fourier transforms

Volterra with $K = k(x-t)$. $y(t) = f(t) + \lambda \int_0^t k(t-\tau)y(\tau)\mathrm{d}\tau$ is $y = f + \lambda\, k * y$. The Laplace transform (convolution theorem, Part II) gives $Y = F + \lambda \mathcal{K} Y$, so

$$Y(s) = \frac{F(s)}{1 - \lambda \mathcal{K}(s)}, \qquad y = \mathcal{L}^{-1}\{Y\}.$$

$y(t) = t + \int_0^t \sin(t-\tau)\, y(\tau)\,\mathrm{d}\tau$

$F = \dfrac{1}{s^2}$, $\mathcal{K} = \dfrac{1}{s^2+1}$: $Y = \dfrac{1/s^2}{1 - 1/(s^2+1)} = \dfrac{1/s^2}{s^2/(s^2+1)} = \dfrac{s^2+1}{s^4} = \dfrac{1}{s^2} + \dfrac{1}{s^4}$. So $y = t + \dfrac{t^3}{6}$. (The equation is equivalent to $y'' = y + \ldots$: differentiating twice gives $y'' + \ldots$; the transform method avoids this.)

Abel's integral equation

$\int_0^x \dfrac{y(t)}{\sqrt{x-t}}\mathrm{d}t = f(x)$ (first kind, $f(0) = 0$). The kernel is $k(x) = x^{-1/2}$, $\mathcal{L}\{x^{-1/2}\} = \sqrt{\pi/s}$: $\sqrt{\dfrac{\pi}{s}}\,Y = F \Rightarrow Y = \dfrac{\sqrt{s}}{\sqrt{\pi}}F = \dfrac{s}{\sqrt{\pi s}}F$. Since $\dfrac{1}{\sqrt{s}} \leftrightarrow \dfrac{1}{\sqrt{\pi x}}$, $Y = \dfrac{1}{\pi}\, s\Big[\sqrt{\tfrac{\pi}{s}}F\Big]$, which is the transform of $\dfrac{1}{\pi}\dfrac{\mathrm{d}}{\mathrm{d}x}\displaystyle\int_0^x \frac{f(t)}{\sqrt{x-t}}\mathrm{d}t$:

$$y(x) = \frac{1}{\pi}\frac{\mathrm{d}}{\mathrm{d}x}\int_0^x \frac{f(t)}{\sqrt{x-t}}\,\mathrm{d}t.$$

For $f=x$: $\int_0^x \frac{t}{\sqrt{x-t}}\mathrm{d}t = \frac{4}{3}x^{3/2}$, so $y=\dfrac{2\sqrt{x}}{\pi}$. Check: $\int_0^x \frac{2}{\pi}\frac{\sqrt{t}}{\sqrt{x-t}}\mathrm{d}t = \frac{2}{\pi}\,x\,B(\frac{3}{2},\frac{1}{2}) = \frac{2}{\pi}x\cdot\frac{\pi}{2} = x$. ✓(Origin: the tautochrone problem.)

Fredholm on the whole line with $K=k(x-t)$. $y(x)=f(x)+\lambda\int_{-\infty}^{\infty}k(x-t)y(t)\mathrm{d}t$. The Fourier transform (convolution theorem) gives $\tilde{y}=\tilde{f}+\lambda\tilde{k}\tilde{y}$, so $\tilde{y}=\tilde{f}/(1-\lambda\tilde{k})$.

$k(x)=\mathrm{e}^{-|x|}$

$\tilde{k}=\dfrac{2}{1+\omega^2}$. $\tilde{y}=\tilde{f}\,\dfrac{1+\omega^2}{1+\omega^2-2\lambda}=\tilde{f}\Big[1+\dfrac{2\lambda}{\omega^2+\mu^2}\Big]$ with $\mu^2=1-2\lambda>0$. Since $\dfrac{1}{\omega^2+\mu^2}\leftrightarrow\dfrac{\mathrm{e}^{-\mu|x|}}{2\mu}$: $y(x)=f(x)+\dfrac{\lambda}{\mu}\displaystyle\int_{-\infty}^{\infty}\mathrm{e}^{-\mu|x-t|}f(t)\mathrm{d}t$.

34.4 Symmetric kernels: Hilbert–Schmidt theory

Let K be real and symmetric (and square-integrable). Then $y=\lambda\mathcal{K}y$ has a countable set of real eigenvalues λ_n and orthonormal eigenfunctions ϕ_n (the integral operator is Hermitian, so the SL theorems of Part I apply), and

$$K(x,t)=\sum_n \frac{\phi_n(x)\phi_n(t)}{\lambda_n}\qquad\text{(bilinear expansion).}$$

To solve $y=f+\lambda\mathcal{K}y$: expand $y=\sum y_n\phi_n$, $f=\sum f_n\phi_n$. Then $y_n=f_n+\dfrac{\lambda}{\lambda_n}y_n\Rightarrow y_n=\dfrac{\lambda_n f_n}{\lambda_n-\lambda}$:

Key result

$$y(x)=f(x)+\lambda\sum_n\frac{f_n}{\lambda_n-\lambda}\phi_n(x),\qquad f_n=\int f\phi_n\,\mathrm{d}x.$$

Fredholm alternative: (i) if λ is not an eigenvalue, the equation has a unique solution for every f; (ii) if $\lambda=\lambda_k$ is an eigenvalue, a solution exists only if $f\perp\phi_k$ (all eigenfunctions with that eigenvalue), and then it is not unique (add any multiple of ϕ_k).

Kernel $K=x_<(1-x_>)$ on $[0,1]$

Eigenfunctions $\phi_n=\sqrt{2}\sin n\pi x$, $\lambda_n=n^2\pi^2$. Bilinear expansion: $x_<(1-x_>)=\sum_n\dfrac{2\sin n\pi x\sin n\pi t}{n^2\pi^2}$. To solve $y=1+\lambda\mathcal{K}y$ ($\lambda\neq n^2\pi^2$): $f_n=\sqrt{2}\int_0^1\sin n\pi x\,\mathrm{d}x=\dfrac{\sqrt{2}\,(1-(-1)^n)}{n\pi}$ (zero for even n), so

$$y(x)=1+\lambda\sum_{n\ \text{odd}}\frac{f_n}{n^2\pi^2-\lambda}\sqrt{2}\sin n\pi x=1+\lambda\sum_{n\ \text{odd}}\frac{4}{n\pi\,(n^2\pi^2-\lambda)}\sin n\pi x.$$

Differentiating $y=1+\lambda\int Ky$ twice (using $\partial_x^2K=-\delta(x-t)$) shows that y is the solution of $y''=-\lambda y$ with $y(0)=y(1)=1$, hence $y=\cos\sqrt{\lambda}x+\dfrac{1-\cos\sqrt{\lambda}}{\sin\sqrt{\lambda}}\sin\sqrt{\lambda}x$ (finite unless $\sin\sqrt{\lambda}=0$, i.e. $\lambda=n^2\pi^2$, the eigenvalues).

Extra: first-kind equations are ill-posed

$f(x)=\int_a^b K(x,t)y(t)\mathrm{d}t$ requires f to lie in the range of $\mathcal{K}$, which is a very smooth class (the Fourier coefficients of f must decay at least as fast as the eigenvalues $1/\lambda_n$). Small errors in f are amplified without bound in y: the solution does not depend continuously on the data (not well-posed). Second-kind equations are well-posed away from eigenvalues. Practical treatments use regularisation.

Practice problems

1. Convert $y''+xy=0$, $y(0)=1$, $y'(0)=0$ to a Volterra equation. *Answer:* $y(x)=1-\int_0^x (x-t)\,t\,y(t)\,\mathrm{d}t$.

2. Solve $y(x)=x+\lambda\int_0^1 x\,t\,y(t)\,\mathrm{d}t$. *Answer:* $y=\dfrac{3x}{3-\lambda}$ ($\lambda\neq 3$); at $\lambda=3$ no solution.

3. Solve $y(x)=1+\lambda\displaystyle\int_0^{2\pi}\sin(x+t)\,y(t)\,\mathrm{d}t$. *Hint:* $\sin(x+t)=\sin x\cos t+\cos x\sin t$, so $y=1+\lambda(c_1\sin x+c_2\cos x)$ with $c_1=\int y\cos t$, $c_2=\int y\sin t$. *Answer:* $\int_0^{2\pi}\cos t\,\mathrm{d}t=\int_0^{2\pi}\sin t\,\mathrm{d}t=0$ and $\int\sin t\cos t=0$, $\int\sin^2 t=\int\cos^2 t=\pi$ give $c_1=\lambda\pi c_2$, $c_2=\lambda\pi c_1$; for $\lambda\pi\neq\pm1$, $c_1=c_2=0$ and $y\equiv 1$. The eigenvalues are $\lambda=\pm1/\pi$.

4. Solve $y(t)=\mathrm{e}^{-t}+\displaystyle\int_0^t y(\tau)\mathrm{d}\tau$ by Laplace transforms. *Answer:* $Y=\frac{1}{s+1}+\frac{Y}{s}\Rightarrow Y=\frac{s}{(s+1)(s-1)}=\frac12\left[\frac{1}{s-1}+\frac{1}{s+1}\right]$, so $y=\cosh t$.

5. Solve Abel's equation $\int_0^x\dfrac{y(t)}{\sqrt{x-t}}\mathrm{d}t=x^2$. *Answer:* $y=\dfrac{1}{\pi}\dfrac{\mathrm{d}}{\mathrm{d}x}\left[\dfrac{16}{15}x^{5/2}\right]=\dfrac{8}{3\pi}x^{3/2}$ (using $\int_0^x t^2(x-t)^{-1/2}\mathrm{d}t=x^{5/2}B(3,\tfrac12)=\tfrac{16}{15}x^{5/2}$).

6. Show that $y(x)=1+\displaystyle\int_0^x\frac{y(t)}{1+t}\mathrm{d}t$ has the solution $y=1+x$. *Hint:* differentiate: $y'=\frac{y}{1+x}$, so $\frac{\mathrm{d}\ln y}{\mathrm{d}x}=\frac{1}{1+x}$ and $y=C(1+x)$ with $y(0)=1\Rightarrow C=1$.

Part VII

Group Theory, Representations and Lie Algebras

35 Groups: definition and examples

Key result

A **group** $(G, \cdot)$ is a set with a binary operation satisfying:

(G1) **Closure:** $a, b \in G \Rightarrow ab \in G$.

(G2) **Associativity:** $(ab)c = a(bc)$.

(G3) **Identity:** there is $e \in G$ with $ea = ae = a$ for all a.

(G4) **Inverse:** each a has $a^{-1} \in G$ with $aa^{-1} = a^{-1}a = e$.

G is **abelian** if $ab = ba$ for all a, b. The **order** $|G|$ is the number of elements (finite or infinite). The identity and inverses are unique. **Rearrangement theorem:** for fixed a, the map $g \mapsto ag$ is a permutation of G (each row/column of the multiplication table contains every element exactly once).

Group	Operation	Notes
$(\mathbb{Z}, +)$, $(\mathbb{R}, +)$	addition	infinite, abelian
$(\mathbb{R} \setminus \{0\}, \times)$	multiplication	abelian
$\mathbb{Z}_n = \{0, 1, \dots, n-1\}$	addition mod n	cyclic, order n; $\cong$ n-th roots of unity
Klein four-group V_4	$\mathbb{Z}_2 \times \mathbb{Z}_2$	order 4, every element is its own inverse
S_n	composition of permutations	order $n!$, non-abelian for $n \geq 3$
D_n	symmetries of a regular n-gon	order $2n$; $D_3 \cong S_3$
$GL(n, \mathbb{K})$	matrix product	invertible $n \times n$ matrices
$SL(n)$, $O(n)$, $SO(n)$	matrix product	$\det = 1$; $R^T R = 1$; both
$U(n)$, $SU(n)$	matrix product	$U^\dagger U = 1$; $\det U = 1$

Cyclic groups. $G = \{e, g, g^2, \dots, g^{n-1}\}$ with $g^n = e$ (generator g). $\mathbb{Z}_4$ has elements $\{0, 1, 2, 3\}$ or $\{1, \mathrm{i}, -1, -\mathrm{i}\}$ under multiplication.

Multiplication tables of $\mathbb{Z}_3$ and V_4

$\mathbb{Z}_3$	e	g	g^2
e	e	g	g^2
g	g	g^2	e
g^2	g^2	e	g

V_4	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Each row and column is a permutation of the elements (Latin square).

35.1 Matrix groups: past paper questions

Past paper: PHY 302 Mid-sem 4.3.2024, Q2 (5 marks)

Let $x \in \mathbb{R}$ and $z \in \mathbb{C}$. Show that the matrices

$$g(z,x) = \begin{pmatrix} 1 & z^* & \frac{1}{2}z^*z + \mathrm{i}x \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix}$$

form a group under matrix multiplication.

Solution

We verify the four axioms, with $\det g = 1$ throughout (upper triangular with unit diagonal).
Closure. Multiply $g(z_1,x_1)\,g(z_2,x_2)$ (first matrix on the left):

$$\begin{aligned}
&(1,1) = 1, \qquad (1,2) = 1\cdot z_2^* + z_1^*\cdot 1 = z_1^* + z_2^*,\\
&(1,3) = 1\cdot\left(\tfrac{1}{2}z_2^*z_2 + \mathrm{i}x_2\right) + z_1^*\cdot z_2 + \left(\tfrac{1}{2}z_1^*z_1 + \mathrm{i}x_1\right)\cdot 1,\\
&(2,3) = 1\cdot z_2 + z_1\cdot 1 = z_1 + z_2, \quad \text{rest } (2,2) = (3,3) = 1,\ (2,1) = (3,1) = (3,2) = 0.
\end{aligned}$$

Let $z_3 = z_1 + z_2$. Then $(1,2) = z_1^* + z_2^* = z_3^*$ ✓. For the (1,3) entry, expand $\frac{1}{2}z_3^*z_3 = \frac{1}{2}|z_1+z_2|^2 = \frac{1}{2}|z_1|^2 + \frac{1}{2}|z_2|^2 + \mathrm{Re}(z_1^*z_2)$, and write $z_1^*z_2 = \mathrm{Re}(z_1^*z_2) + \mathrm{i}\,\mathrm{Im}(z_1^*z_2)$:

$$(1,3) = \tfrac{1}{2}|z_1|^2 + \tfrac{1}{2}|z_2|^2 + \mathrm{Re}(z_1^*z_2) + \mathrm{i}\,\mathrm{Im}(z_1^*z_2) + \mathrm{i}(x_1+x_2) = \tfrac{1}{2}z_3^*z_3 + \mathrm{i}\big[x_1 + x_2 + \mathrm{Im}(z_1^*z_2)\big].$$

Since $x_1 + x_2 + \mathrm{Im}(z_1^*z_2)$ is *real*, define $x_3 = x_1 + x_2 + \mathrm{Im}(z_1^*z_2) \in \mathbb{R}$. Hence

$$g(z_1,x_1)\,g(z_2,x_2) = g\big(z_1+z_2,\ x_1+x_2+\mathrm{Im}(z_1^*z_2)\big) \in G.$$

Associativity: inherited from matrix multiplication. **Identity:** $g(0,0) = \mathbb{1}_3$.
Inverse: try $g(-z,-x)$: $g(z,x)g(-z,-x) = g\big(0,\ x - x + \mathrm{Im}(z^*(-z))\big) = g(0,\ -\mathrm{Im}\,|z|^2) = g(0,0)$ since $|z|^2$ is real so its imaginary part is 0. Thus $g(z,x)^{-1} = g(-z,-x) \in G$ (and the product in the other order also gives the identity by the same formula). So G is a group. ■
Remark. It is non-abelian: $g(z_1,x_1)g(z_2,x_2)$ and $g(z_2,x_2)g(z_1,x_1)$ differ in x_3 by $\mathrm{Im}(z_1^*z_2) - \mathrm{Im}(z_2^*z_1) = 2\,\mathrm{Im}(z_1^*z_2)$. This is the (complexified) **Heisenberg group**, the group generated by x, p and the central element $\mathrm{i}\hbar$; in the oscillator language $z \sim \alpha$ is the coherent-state label, and the product rule is that of displacement operators $D(\alpha)D(\beta) = \mathrm{e}^{\mathrm{i}\,\mathrm{Im}(\alpha^*\beta)}D(\alpha+\beta)$.

Past paper: PHY 302 Mid-sem 4.3.2024, Q3 (9 marks)

Consider 2×2 real matrices M which satisfy (i) M is a rotation matrix, and (ii) $M^2 = -I_{2\times 2}$. Find all such matrices M. Together with the 2×2 identity matrix will this set form a group under matrix multiplication? Explain.

Solution

Find M. A 2×2 real rotation matrix is $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ ($R^TR = 1$, $\det R = +1$). Since $R(\theta)R(\theta') = R(\theta+\theta')$, we get $M^2 = R(2\theta)$, and $M^2 = -I = R(\pi)$ requires $2\theta = \pi \pmod{2\pi}$, i.e. $\theta = \pm\dfrac{\pi}{2}$:

$$M_+ = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad M_- = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = -M_+ = M_+^{-1}.$$

Check: $M_+^2 = \begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix} = \begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix} = -I$ ✓.
Is $S = \{I, M_+, M_-\}$ a group? Test closure: $M_+M_+ = -I$, which is *not* in S ($-I$ has $\theta = \pi$, is not one of $I, M_\pm$). So closure fails and S is **not a group**. (Also, the three elements would require $M_+M_- = I$ and $M_+^2 \in S$, impossible.)
Remedy. Adding $-I = R(\pi)$ gives $\{I, M_+, -I, M_-\} = \{R(0), R(\pi/2), R(\pi), R(3\pi/2)\}$, which is closed, contains inverses ($M_+^{-1} = M_-$, $(-I)^{-1} = -I$) and is the cyclic group $\mathbb{Z}_4$ (generated by M_+ with $M_+^4 = I$), isomorphic to $\{1, \mathrm{i}, -1, -\mathrm{i}\}$.

36 Subgroups, cosets, Lagrange's theorem

A **subgroup** $H \leq G$ is a subset that is itself a group under the same operation (check: $e \in H$; closed; inverses). For $g \in G$, the **left coset** is $gH = \{gh : h \in H\}$.

Key result

Lagrange's theorem. For finite G and $H \leq G$: $|H|$ divides $|G|$, and $|G| = |H| \cdot [G : H]$ where $[G : H]$ is the number of distinct cosets.
Proof. (i) Each coset has $|H|$ elements (the map $h \mapsto gh$ is one-to-one). (ii) Two cosets are either identical or disjoint: if $x \in g_1H \cap g_2H$ then $x = g_1h_1 = g_2h_2$, so $g_1 = g_2h_2h_1^{-1}$ and $g_1H = g_2H$. (iii) Every g lies in the coset gH. So G is partitioned into disjoint cosets of size $|H|$. ■

Consequences. The **order of an element** g (smallest $n > 0$ with $g^n = e$) equals the order of the cyclic subgroup $\langle g \rangle$, hence divides $|G|$. A group of prime order p is cyclic ($\cong \mathbb{Z}_p$): take any $g \neq e$, $\langle g \rangle$ has order $1 < n \mid p$, so $n = p$.

Worked example

$|S_3| = 6$ so subgroups have order $1, 2, 3$ or 6 (all occur): $\{e\}$; $\{e, (12)\}$, $\{e, (13)\}$, $\{e, (23)\}$; $A_3 = \{e, (123), (132)\}$; S_3. There is no subgroup of order 4 or 5.

37 Conjugacy classes, normal subgroups, homomorphisms

Key result

$a \sim b$ (conjugate) if $b = gag^{-1}$ for some $g \in G$. Conjugacy is an equivalence relation; its classes partition G. Elements of the same class have the same order. In S_n, conjugate elements have the same **cycle structure**.
A subgroup N is **normal** ($N \lhd G$) if $gNg^{-1} = N$ for all g (it is a union of classes). Then the cosets form the **quotient group** G/N with $(aN)(bN) = (ab)N$.
A **homomorphism** $\varphi : G \to G'$ satisfies $\varphi(ab) = \varphi(a)\varphi(b)$. Its **kernel** $\ker\varphi = \{g : \varphi(g) = e'\}$ is a normal subgroup, and $G/\ker\varphi \cong \text{image}(\varphi)$ (first isomorphism theorem). An **isomorphism** is a bijective homomorphism ($G \cong G'$: the same group with different labels). The **centre** $Z(G) = \{z : zg = gz \ \forall g\}$.

38 Permutation groups

S_n is the group of bijections of $\{1,\dots,n\}$, written in **cycle notation**: $(1\,2\,3)$ means $1\to 2\to 3\to 1$; disjoint cycles commute. Composition is read right to left. A k-cycle is a product of $k-1$ transpositions, so every permutation is even or odd, and the even ones form the normal subgroup A_n of order $n!/2$. **Cayley's theorem:** every finite group G is isomorphic to a subgroup of $S_{|G|}$ (via $g\mapsto(h\mapsto gh)$).

38.1 The group S_3

Elements: e, $r=(123)$, $r^2=(132)$, $a=(12)$, $b=(13)$, $c=(23)$; $|S_3|=6$, non-abelian ($ab=r^2\neq ba=r$). It is isomorphic to D_3, the symmetry group of an equilateral triangle: r = rotation by 120°, and a,b,c = reflections about the three altitudes. Multiplication table (row $\times$ column, composition right-to-left):

	e	r	r^2	a	b	c
e	e	r	r^2	a	b	c
r	r	r^2	e	b	c	a
r^2	r^2	e	r	c	a	b
a	a	c	b	e	r^2	r
b	b	a	c	r	e	r^2
c	c	b	a	r^2	r	e

Conjugacy classes (by cycle type): $\{e\}$ (cycle type 1^3), $\{r,r^2\}$ (type 3), $\{a,b,c\}$ (type $2\,1$). Three classes. $A_3=\{e,r,r^2\}\lhd S_3$ with $S_3/A_3\cong\mathbb{Z}_2$.

Past papers: PHY 302 Mid-sem 4.3.2024 Q1 (3+3) and Mid-sem 3.3.2025 Q1 (3+3)

Consider the S_3 group as the permutation group of three objects $\{x_1,x_2,x_3\}$. If we express this set of three objects as a column matrix $(x_1\ x_2\ x_3)^T$, then all the six elements of the S_3 group transformation can be represented as 3×3 matrices. (i) Construct all the six 3×3 matrices. Use just 0 and 1 as the entries of these matrices. (ii) After representing them in this manner, find out the inverse of each group element (which are all now 3×3 matrices).

Solution

(i) A permutation rearranges the entries of the column vector; the corresponding matrix has exactly one 1 in each row and column. Describe each element by its action on $(x_1,x_2,x_3)^T$:

Element	Action	Matrix	Name
E	(x_1, x_2, x_3)	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	identity
P_{12}	(x_2, x_1, x_3)	$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	swap 1,2
P_{13}	(x_3, x_2, x_1)	$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$	swap 1,3
P_{23}	(x_1, x_3, x_2)	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$	swap 2,3
C	(x_3, x_1, x_2)	$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$	cyclic (shift right)
C^2	(x_2, x_3, x_1)	$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$	cyclic (shift left)

Check the group structure: $C \cdot C\,(x_1, x_2, x_3)^T = C(x_3, x_1, x_2)^T = (x_2, x_3, x_1)^T$, so indeed C^2 is the second cyclic matrix and $C^3 = E$. Non-commutativity: $P_{12}P_{23}(x_1, x_2, x_3)^T = P_{12}(x_1, x_3, x_2)^T = (x_3, x_1, x_2)^T = C$, whereas $P_{23}P_{12}(x_1, x_2, x_3)^T = P_{23}(x_2, x_1, x_3)^T = (x_2, x_3, x_1)^T = C^2$.

(ii) Inverses. The inverse of a permutation undoes the rearrangement. Every permutation matrix is orthogonal, so $M^{-1} = M^T$. Therefore:

- $E^{-1} = E$.
- The three swaps are symmetric matrices, so $P_{ij}^{-1} = P_{ij}^T = P_{ij}$ (each swap is its own inverse).
- $C^{-1} = C^T = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} = C^2$ and $(C^2)^{-1} = \left(C^2\right)^T = C$. (Verify: $CC^2 = C^3 = E$.)

39 Representations of finite groups

Key result

A **representation** of G on a vector space V ($\dim V = d$) is a homomorphism $D : G \to GL(V)$: $D(g_1 g_2) = D(g_1)D(g_2)$, $D(e) = \mathbb{1}$, $D(g^{-1}) = D(g)^{-1}$. The **character** is $\chi(g) = \operatorname{Tr} D(g)$.

- **Equivalent** representations differ by a similarity transformation $D' = SDS^{-1}$ (same characters).
- **Reducible** if V has a proper invariant subspace (then D can be brought to block form); **irreducible (irrep)** otherwise. For finite groups every representation is a direct sum of

irreps (complete reducibility), $D = \bigoplus n_\alpha D^{(\alpha)}$.

- Every representation of a finite group is equivalent to a **unitary** one (average the inner product over the group: $\langle u, v\rangle' = \frac{1}{|G|}\sum_g \langle D(g)u, D(g)v\rangle$).
- The characters are **class functions**: $\chi(hgh^{-1}) = \chi(g)$ (trace is cyclic).

Schur's lemmas. (1) If $D^{(\alpha)}, D^{(\beta)}$ are inequivalent irreps and $AD^{(\alpha)}(g) = D^{(\beta)}(g)A$ for all g, then $A = 0$. (2) If D is an irrep over $\mathbb{C}$ and $AD(g) = D(g)A$ for all g, then $A = \lambda \mathbb{1}$. (Proof of (2): A has an eigenvector with eigenvalue λ; $A - \lambda\mathbb{1}$ commutes with D and has a non-trivial kernel, an invariant subspace; irreducibility forces the kernel to be all of V.)

Key result

Great orthogonality theorem. For inequivalent unitary irreps of dimensions d_α, d_β:

$$\sum_{g\in G} D^{(\alpha)}_{ij}(g)\, D^{(\beta)}_{kl}(g)^* = \frac{|G|}{d_\alpha}\,\delta_{\alpha\beta}\delta_{ik}\delta_{jl}.$$

Consequences:

$$\frac{1}{|G|}\sum_g \chi^{(\alpha)}(g)^*\chi^{(\beta)}(g) = \delta_{\alpha\beta} \text{ (character orthogonality)},$$

$$\sum_\alpha d_\alpha^2 = |G|, \qquad \#\text{irreps} = \#\text{conjugacy classes}.$$

Reduction formula: a reducible representation with character χ contains the irrep α exactly $n_\alpha = \frac{1}{|G|}\sum_g \chi^{(\alpha)}(g)^*\chi(g) = \frac{1}{|G|}\sum_{\text{classes } C} |C|\,\chi^{(\alpha)}(C)^*\chi(C)$ times.

39.1 Character table of S_3

Three classes, so three irreps with $d_1^2 + d_2^2 + d_3^2 = 6$: $d = 1, 1, 2$ (since $1+1+4=6$). The trivial rep A_1 ($D(g) = 1$), the sign rep A_2 ($D = \pm 1$ for even/odd permutations) and the 2-dimensional E.

S_3	$\{e\}$	$\{r, r^2\}$	$\{a, b, c\}$
	(1)	(2)	(3)
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0

Explicit E: $D(r) = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$ (rotation by 120°, trace -1), $D(a) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ (a reflection, trace 0). Check the defining relation $ara^{-1} = r^{-1}$: $D(a)D(r)D(a) = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} = D(r)^{-1} = D(r)^T$. ✓

Orthogonality checks: $\langle A_1, A_2\rangle = \frac{1}{6}[1 + 2(1)(1) + 3(1)(-1)] = 0$; $\langle E, E\rangle = \frac{1}{6}[4 + 2(1) + 0] = 1$; $\langle A_1, E\rangle = \frac{1}{6}[2 + 2(-1) + 0] = 0$.

Reducing the permutation representation

The 3×3 matrices above have characters (trace): $\chi(e)=3$, $\chi(r)=\chi(r^2)=0$ (no fixed point), $\chi(\text{swap})=1$ (one fixed point). Then

$$\begin{aligned} n_{A_1} &= \tfrac{1}{6}[3\cdot 1+2\cdot 0\cdot 1+3\cdot 1\cdot 1]=1,\\ n_{A_2} &= \tfrac{1}{6}[3+0+3\cdot 1\cdot(-1)]=0,\\ n_{E} &= \tfrac{1}{6}[3\cdot 2+2\cdot 0\cdot(-1)+3\cdot 1\cdot 0]=1. \end{aligned}$$

So the 3-dimensional permutation representation is $A_1\oplus E$ ($3=1+2$): the invariant vector $(1,1,1)^T$ (or $x_1+x_2+x_3$) and the 2D subspace $x_1+x_2+x_3=0$ on which S_3 acts as in E. This is how the normal modes of three identical masses on a triangle are classified.

Abelian groups: all irreps are 1D

For an abelian group each element is its own class, so #irreps $=|G|$ and $\sum d_\alpha^2=|G|$ forces all $d_\alpha=1$. For $\mathbb{Z}_n=\langle g\rangle$: $D^{(k)}(g^m)=\mathrm{e}^{2\pi \mathrm{i}km/n}$, $k=0,\dots,n-1$. For $\mathbb{Z}_4$ these are $1,\mathrm{i}^m,(-1)^m,(-\mathrm{i})^m$. Orthogonality is just $\sum_{m=0}^{n-1}\mathrm{e}^{2\pi\mathrm{i}(k-k')m/n}=n\delta_{kk'}$ (discrete Fourier transform!).

Extra: why representations matter in physics

(i) **Degeneracy:** if the Hamiltonian commutes with a symmetry group G, energy eigenstates transform according to irreps; the degeneracy of a level equals the dimension of the irrep (generically). (ii) **Selection rules:** a matrix element $\langle\psi_f|O|\psi_i\rangle$ vanishes unless the product $D_f^*\otimes D_O\otimes D_i$ contains the trivial irrep. (iii) **Splitting of levels** when a symmetry is lowered (crystal-field splitting). (iv) **Direct products:** $D^{(\alpha)}\otimes D^{(\beta)}=\bigoplus_\gamma n_{\alpha\beta\gamma}D^{(\gamma)}$ with $\chi_{\alpha\otimes\beta}=\chi_\alpha\chi_\beta$; e.g. in S_3: $E\otimes E=A_1\oplus A_2\oplus E$ ($\chi=4,1,0$).

40 Continuous groups and Lie groups

A **Lie group** is a group whose elements depend smoothly on a finite number n of continuous real parameters θ^a (it is a differentiable manifold, and multiplication and inversion are smooth). The number n is its **dimension**.

Group	Definition	Dimension
$U(1)=SO(2)$	$\mathrm{e}^{\mathrm{i}\theta}$; plane rotations	1
$SO(n)$	real orthogonal, $\det=+1$	$n(n-1)/2$
$SU(n)$	unitary, $\det=1$	n^2-1
$U(n)$	unitary	n^2
$SL(n,\mathbb{R})$	$\det=1$	n^2-1

$SO(n), SU(n), U(n)$ are **compact** (parameters range over a closed bounded set), so their unitary representations behave like those of finite groups with the sum over G replaced by an invariant (Haar) integral $\int \mathrm{d}\mu(g)$.

40.1 Generators and the Lie algebra

Near the identity write $g=\mathbb{1}+\mathrm{i}\,\epsilon^a T_a+\dots$ with **generators** $T_a=-\mathrm{i}\,\partial g/\partial\theta^a\big|_0$ (Hermitian for unitary groups). Finite elements are obtained by exponentiation: $g(\theta)=\exp\left(\mathrm{i}\theta^a T_a\right)$.

Key result

Closure of the group under products forces the generators to close under commutation:

$$[T_a, T_b] = \mathrm{i}\, f_{ab}{}^{c}\, T_c,$$

with real **structure constants** $f_{ab}{}^{c}$ (antisymmetric in a, b), subject to the **Jacobi identity** $[T_a,[T_b,T_c]] + [T_b,[T_c,T_a]] + [T_c,[T_a,T_b]] = 0$. The vector space spanned by the T_a with this bracket is the **Lie algebra** $\mathfrak{g}$. *The algebra determines the group locally; the global structure (e.g. $SU(2)$ vs. $SO(3)$) needs more information.*

Why commutators appear: the group commutator $g_1 g_2 g_1^{-1} g_2^{-1}$ for $g_i \approx 1 + \mathrm{i}\epsilon_i T_i$ equals $1 - \epsilon_1\epsilon_2[T_1, T_2] + \ldots$, which must again be a group element near the identity; so $[T_1, T_2]$ must be a linear combination of generators.

Casimir operators: operators built from generators that commute with all of them (e.g. J^2 for $SU(2)$); by Schur's second lemma they are multiples of the identity on each irrep, so they label irreps.

40.2 $SO(2) \cong U(1)$

$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \exp(\theta J)$ with $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $J^2 = -1$ (so $\exp(\theta J) = \cos\theta + J\sin\theta$). Abelian, one generator. All irreps are one-dimensional: $D^{(m)}(\theta) = \mathrm{e}^{\mathrm{i}m\theta}$ with $m \in \mathbb{Z}$ (single-valued since $R(2\pi) = 1$). These are exactly the **Fourier series** basis functions: Fourier analysis on the circle is representation theory of $SO(2)$.

40.3 $SO(3)$ and $su(2)$

The three generators of rotations $(J_k)_{ij} = -\mathrm{i}\epsilon_{kij}$ (Hermitian 3×3 matrices) satisfy $[J_i, J_j] = \mathrm{i}\epsilon_{ijk}J_k$ (units $\hbar = 1$), and $R(\hat{n}, \theta) = \exp(-\mathrm{i}\theta\, \hat{n}\cdot\boldsymbol{J})$. The Lie algebra $so(3)$ has dimension 3. The Pauli matrices give a second realisation: $J_k = \frac{1}{2}\sigma_k$ satisfy the same commutation relations (this is $su(2) \cong so(3)$ as Lie algebras).

40.4 $SU(2)$: the exam question

Past paper: PHY 302 Mid-sem 3.3.2025, Q7 (10 marks)

Consider a 2×2 complex matrix $H = \begin{pmatrix} u & w \\ z & v \end{pmatrix}$, $u, w, z, v \in \mathbb{C}$. (i) Under what conditions will it represent a Hermitian matrix? Show that $\mathrm{e}^{\mathrm{i}H}$ is going to be a unitary matrix. (ii) What additional condition is to be imposed such that $\mathrm{Tr}\, H = 0$. Show that with this additional condition, $\mathrm{e}^{\mathrm{i}H}$ will represent an element of $SU(2)$. (iii) Now you have a traceless Hermitian matrix H. How will it look like in terms of u, w, z, v? Represent it in terms of real parameters (only three real parameters $\theta_1, \theta_2, \theta_3$ are needed; choose the diagonal elements related to θ_3). (iv) Re-express H as $\sum_{k=1}^{3} \frac{1}{2} A_k \theta_k$. Find out the exact matrix forms of all the A_k's. (v) Find the commutation and anti-commutation relations $[A_i, A_j]$ and $\{A_i, A_j\}$. Finally write down the matrix form of $\mathrm{e}^{\mathrm{i}(A_1\phi_1 + A_2\phi_2 + A_3\phi_3)/2}$, where (ϕ_1, ϕ_2, ϕ_3) are some real parameters.

Solution

(i) Hermiticity and unitarity. $H^\dagger = \begin{pmatrix} u^* & z^* \\ w^* & v^* \end{pmatrix}$. $H = H^\dagger$ requires $u = u^*$, $v = v^*$ (diagonal entries real) and $w = z^*$ (off-diagonals complex conjugates). Then $(\mathrm{e}^{\mathrm{i}H})^\dagger = \mathrm{e}^{-\mathrm{i}H^\dagger} = \mathrm{e}^{-\mathrm{i}H}$ (the dagger of the power series flips $\mathrm{i} \to -\mathrm{i}$ and $H \to H^\dagger = H$), so

$$\mathrm{e}^{\mathrm{i}H}(\mathrm{e}^{\mathrm{i}H})^\dagger = \mathrm{e}^{\mathrm{i}H}\mathrm{e}^{-\mathrm{i}H} = \mathrm{e}^0 = \mathbb{1},$$

valid because $\mathrm{i}H$ and $-\mathrm{i}H$ commute. Hence $\mathrm{e}^{\mathrm{i}H}$ is unitary.
(ii) Traceless case. $\operatorname{Tr} H = u + v = 0 \Rightarrow v = -u$. For any square matrix $\det \mathrm{e}^A = \mathrm{e}^{\operatorname{Tr} A}$ (diagonalise: eigenvalues λ_i of A give e^{λ_i}, product $\mathrm{e}^{\sum \lambda_i}$). Hence $\det \mathrm{e}^{\mathrm{i}H} = \mathrm{e}^{\mathrm{i}\operatorname{Tr} H} = \mathrm{e}^0 = 1$. A unitary 2×2 matrix with unit determinant is in $SU(2)$. ■
(iii) Parametrisation. $H = \begin{pmatrix} u & w \\ w^* & -u \end{pmatrix}$ with $u \in \mathbb{R}$, $w \in \mathbb{C}$: three real numbers $(u, \operatorname{Re} w, \operatorname{Im} w)$. Choose $u = \frac{\theta_3}{2}$ and $w = \frac{\theta_1 - \mathrm{i}\theta_2}{2}$:

$$H = \frac{1}{2}\begin{pmatrix} \theta_3 & \theta_1 - \mathrm{i}\theta_2 \\ \theta_1 + \mathrm{i}\theta_2 & -\theta_3 \end{pmatrix}.$$

(iv) The matrices A_k. Split by the parameters: $H = \frac{1}{2}\theta_1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \frac{1}{2}\theta_2 \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix} + \frac{1}{2}\theta_3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, so

$$A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad \text{(the Pauli matrices } \sigma_1, \sigma_2, \sigma_3\text{)}.$$

(v) Algebra. Direct multiplication: $A_1 A_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix} = \begin{pmatrix} \mathrm{i} & 0 \\ 0 & -\mathrm{i} \end{pmatrix} = \mathrm{i}A_3$ and $A_2 A_1 = -\mathrm{i}A_3$; similarly cyclically. Also $A_i^2 = \mathbb{1}$. Hence

$$\boxed{[A_i, A_j] = 2\mathrm{i}\,\epsilon_{ijk} A_k, \qquad \{A_i, A_j\} = 2\delta_{ij}\mathbb{1}.}$$

(Therefore $T_k = \frac{1}{2}A_k$ satisfy $[T_i, T_j] = \mathrm{i}\epsilon_{ijk}T_k$, the $su(2)$ algebra with $f_{ijk} = \epsilon_{ijk}$.)
The exponential. Let $\phi = \sqrt{\phi_1^2 + \phi_2^2 + \phi_3^2}$, $\hat{n} = \boldsymbol{\phi}/\phi$, and $\boldsymbol{A} \cdot \hat{n} = n_1 A_1 + n_2 A_2 + n_3 A_3$. Using the anticommutator, $(\boldsymbol{A} \cdot \hat{n})^2 = \sum_{ij} n_i n_j A_i A_j = \sum n_i n_j \frac{1}{2}\{A_i, A_j\} = \hat{n} \cdot \hat{n}\,\mathbb{1} = \mathbb{1}$. Expand the exponential: with $X = \frac{\phi}{2}\boldsymbol{A} \cdot \hat{n}$, $X^2 = \frac{\phi^2}{4}\mathbb{1}$ and

$$\mathrm{e}^{\mathrm{i}X} = \sum_k \frac{(\mathrm{i}X)^k}{k!} = \cos\frac{\phi}{2}\mathbb{1} + \mathrm{i}\sin\frac{\phi}{2}(\boldsymbol{A} \cdot \hat{n})$$

(even powers give cos, odd powers give $\mathrm{i}\sin \times (\boldsymbol{A} \cdot \hat{n})$). Explicitly:

$$\boxed{\mathrm{e}^{\mathrm{i}\boldsymbol{A}\cdot\boldsymbol{\phi}/2} = \begin{pmatrix} \cos\frac{\phi}{2} + \mathrm{i}n_3 \sin\frac{\phi}{2} & \mathrm{i}(n_1 - \mathrm{i}n_2)\sin\frac{\phi}{2} \\ \mathrm{i}(n_1 + \mathrm{i}n_2)\sin\frac{\phi}{2} & \cos\frac{\phi}{2} - \mathrm{i}n_3 \sin\frac{\phi}{2} \end{pmatrix}, \qquad n_i = \frac{\phi_i}{\phi}}$$

Checks: $\det = \left(\cos\frac{\phi}{2}\right)^2 + n_3^2 \sin^2\frac{\phi}{2} + (n_1^2 + n_2^2)\sin^2\frac{\phi}{2} = 1$ ✓; unitary: the matrix has the form $\begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix}$ with $a = \cos\frac{\phi}{2} + \mathrm{i}n_3\sin\frac{\phi}{2}$, $b = \mathrm{i}(n_1 - \mathrm{i}n_2)\sin\frac{\phi}{2}$ and $|a|^2 + |b|^2 = 1$.

Key result

General element of $SU(2)$: $U = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix}$, $a, b \in \mathbb{C}$, $|a|^2 + |b|^2 = 1$. This is the 3-sphere S^3 (compact, simply connected). $U = \mathrm{e}^{-\mathrm{i}\theta\hat{n}\cdot\boldsymbol{\sigma}/2}$ represents a rotation by angle θ about $\hat{n}$.

40.5 $SU(2) \to SO(3)$: the double cover

For a vector $\boldsymbol{r}$ define the traceless Hermitian matrix $X = \boldsymbol{r}\cdot\boldsymbol{\sigma}$. For $U \in SU(2)$, $X' = UXU^\dagger$ is again traceless Hermitian, so $X' = \boldsymbol{r}'\cdot\boldsymbol{\sigma}$; and $\det X = -r^2$ is preserved, so $|\boldsymbol{r}'| = |\boldsymbol{r}|$: $\boldsymbol{r} \mapsto \boldsymbol{r}' = R(U)\boldsymbol{r}$ with $R(U) \in SO(3)$. The map $U \mapsto R(U)$ is a homomorphism, onto, with kernel $\{\mathbb{1}, -\mathbb{1}\}$ (since $UXU^\dagger = X$ for all X iff $U = \pm\mathbb{1}$): $SO(3) \cong SU(2)/\mathbb{Z}_2$. For $U = \mathrm{e}^{-\mathrm{i}\phi\sigma_3/2}$ one finds $R(U) =$ rotation about z by ϕ. **Spinor sign:** $\phi = 2\pi$ gives $R = \mathbb{1}$ but $U = -\mathbb{1}$: a 2π rotation multiplies spin-$\frac{1}{2}$ states by -1.

41 Representations of $SU(2)$ and the rotation group

Key result

The (finite-dimensional, unitary) irreducible representations of $SU(2)$ are labelled by $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$ with dimension $2j+1$. In the basis $|j,m\rangle$, $m = -j, \ldots, j$ (derived in the angular-momentum notes):

$$J_3|j,m\rangle = m|j,m\rangle, \qquad J_\pm|j,m\rangle = \sqrt{j(j+1) - m(m\pm1)}\,|j,m\pm1\rangle, \qquad \boldsymbol{J}^2 = j(j+1)\mathbb{1} \text{ (Casimir)}.$$

The group element $D^{(j)}(\hat{n},\theta) = \exp(-\mathrm{i}\theta\hat{n}\cdot\boldsymbol{J})$ has matrix elements $D^{(j)}_{m'm}$ (Wigner D-functions). Integer j: true representations of $SO(3)$. Half-integer j: representations of $SU(2)$ only (double-valued on $SO(3)$): spinors.

Characters. A rotation by angle θ about any axis is conjugate to rotation about z, so $D^{(j)} \sim \mathrm{diag}\big(\mathrm{e}^{-\mathrm{i}m\theta}\big)_{m=-j}^{j}$:

$$\chi_j(\theta) = \sum_{m=-j}^{j} \mathrm{e}^{\mathrm{i}m\theta} = \frac{\sin\left[(j+\frac{1}{2})\theta\right]}{\sin(\theta/2)}.$$

(Check: $\theta \to 0$ gives $2j+1$.) Classes of $SU(2)$ are labelled by $\theta \in [0, 2\pi]$ and the invariant measure on the class space is $\frac{1}{\pi}\sin^2\frac{\theta}{2}\,\mathrm{d}\theta$, giving the orthogonality $\dfrac{1}{\pi}\displaystyle\int_0^{2\pi} \sin^2\tfrac{\theta}{2}\,\chi_j\chi_{j'}\,\mathrm{d}\theta = \delta_{jj'}$ (for $j = j' = 0$: $\frac{1}{\pi}\int\sin^2\frac{\theta}{2} = 1$ ✓).

Key result

Clebsch–Gordan series. $D^{(j_1)} \otimes D^{(j_2)} = \displaystyle\bigoplus_{j=|j_1-j_2|}^{j_1+j_2} D^{(j)}$ (dimensions: $(2j_1+1)(2j_2+1) = \sum_j (2j+1)$). *Proof via characters:* $\chi_{j_1}\chi_{j_2} = \sum_{m_1,m_2} \mathrm{e}^{\mathrm{i}(m_1+m_2)\theta}$; the multiplicity of each total $M = m_1 + m_2$ decreases from its maximum $j_1 + j_2$, producing exactly one irrep for each j from $j_1 + j_2$ down to $|j_1 - j_2|$.

Decompositions

(a) $\frac{1}{2}\otimes\frac{1}{2}=0\oplus1$ ($4=1+3$): singlet and triplet of two spin-$\frac{1}{2}$ particles. **(b)** $1\otimes\frac{1}{2}=\frac{1}{2}\oplus\frac{3}{2}$ ($6=2+4$): the pion–nucleon system of the QM papers (isospin is also an $SU(2)$). **(c)** $1\otimes1=0\oplus1\oplus2$ ($9=1+3+5$): a Cartesian rank-2 tensor T_{ij} splits into trace ($j=0$), antisymmetric part ($j=1$, an axial vector) and symmetric traceless part ($j=2$, five components): the decomposition seen in the Cartesian-tensor chapter. **(d)** Orbital $l=1$ with spin $\frac{1}{2}$: $1\otimes\frac{1}{2}=\frac{3}{2}\oplus\frac{1}{2}$ (fine structure levels ${}^2P_{3/2},{}^2P_{1/2}$).

41.1 Other Lie groups worth knowing

- $U(1)$**:** phase transformations; charge conservation; irreps $\mathrm{e}^{\mathrm{i}q\theta}$.
- $SU(3)$**:** 8 generators $\lambda_a/2$ (Gell-Mann matrices), $[\frac{\lambda_a}{2},\frac{\lambda_b}{2}]=\mathrm{i}f_{abc}\frac{\lambda_c}{2}$, rank 2 (two commuting generators: isospin I_3 and hypercharge Y). Fundamental irreps $\mathbf{3}$ (quarks), $\bar{\mathbf{3}}$; $\mathbf{3}\otimes\bar{\mathbf{3}}=\mathbf{1}\oplus\mathbf{8}$ (mesons), $\mathbf{3}\otimes\mathbf{3}\otimes\mathbf{3}=\mathbf{1}\oplus\mathbf{8}\oplus\mathbf{8}\oplus\mathbf{10}$ (baryons).
- **Lorentz group** $SO(3,1)$**:** six generators ($\boldsymbol{J}$ and boosts $\boldsymbol{K}$), non-compact; its algebra is $su(2)\oplus su(2)$ (complexified), which classifies scalars, spinors and vectors in relativistic QM (used for the Dirac equation).

Practice problems

1. Write the multiplication table of $\mathbb{Z}_4=\{1,\mathrm{i},-1,-\mathrm{i}\}$ and list its subgroups. *Answer:* subgroups $\{1\}$, $\{1,-1\}$, $\mathbb{Z}_4$.
2. Show that $\{1,-1\}$ and $\{1,-1,\mathrm{i},-\mathrm{i}\}$ are the only subgroups of $\mathbb{Z}_4$ besides the trivial one, and use Lagrange's theorem to explain why there is none of order 3.
3. Find the conjugacy classes of D_4 (symmetries of a square, order 8) and the dimensions of its irreps. *Answer:* classes $\{e\},\{r^2\},\{r,r^3\},\{s,sr^2\},\{sr,sr^3\}$ (5 classes); irreps $1,1,1,1,2$ ($1+1+1+1+4=8$).
4. Prove that the set of rotations $R(\theta)$ is a group isomorphic to $U(1)$ via $\theta\mapsto\mathrm{e}^{\mathrm{i}\theta}$.
5. Compute the product $g(z_1,x_1)g(z_2,x_2)$ for $z_1=1$, $z_2=\mathrm{i}$ in the Heisenberg-type group of Question 2 and compare with the reverse order. *Answer:* $g(1,x_1)g(\mathrm{i},x_2)=g(1+\mathrm{i},\ x_1+x_2+\mathrm{Im}(1^*\cdot\mathrm{i}))=g(1+\mathrm{i},\ x_1+x_2+1)$, whereas in the reverse order $\mathrm{Im}(z_2^*z_1)=\mathrm{Im}(-\mathrm{i}\cdot1)=-1$ gives $g(1+\mathrm{i},\ x_1+x_2-1)$. They differ by 2 in x (non-abelian).
6. Reduce $E\otimes E$ of S_3 using characters. *Answer:* $\chi=(4,1,0)$; $n_{A_1}=\frac{1}{6}(4+2+0)=1$, $n_{A_2}=\frac{1}{6}(4+2)=1$, $n_E=\frac{1}{6}(8-2)=1$. So $E\otimes E=A_1\oplus A_2\oplus E$.
7. Show that $\mathrm{e}^{-\mathrm{i}\pi\sigma_z}=-\mathbb{1}$ and discuss the physical meaning. *Hint:* $\mathrm{e}^{-\mathrm{i}\theta\sigma_z/2}$ at $\theta=2\pi$ is $-\mathbb{1}$.
8. Verify $[A_1,A_2]=2\mathrm{i}A_3$ for the Pauli matrices and find $[A_2,A_3]$, $[A_3,A_1]$. *Answer:* $2\mathrm{i}A_1$, $2\mathrm{i}A_2$.
9. Decompose $\frac{3}{2}\otimes1$ and check the dimension count. *Answer:* $\frac{1}{2}\oplus\frac{3}{2}\oplus\frac{5}{2}$: $2+4+6=12=4\times3$.
10. Compute $\chi_1(\theta)$ for the spin-1 representation and verify it equals $1+2\cos\theta$. *Answer:* $\sum_{m=-1}^{1}\mathrm{e}^{\mathrm{i}m\theta}=1+2\cos\theta=\sin\frac{3\theta}{2}/\sin\frac{\theta}{2}$.

Part VIII

Complex-Variable Techniques: Past-Paper Problems

This part collects the residue-calculus questions from the mid-sem papers (they build on the Math Methods I material in the *Complex Analysis* booklet). Quick reminders:

Key result

Laurent expansion about an isolated singularity z_0: $f(z)=\sum_{n=-\infty}^{\infty} a_n (z-z_0)^n$. The **principal part** is the sum over $n<0$. If the principal part is absent: *removable*; if it has finitely many terms (lowest $n=-m$): *pole of order* m; if infinitely many: *essential singularity*. The **residue** is a_{-1}. For a simple pole $\mathrm{Res}=\lim (z-z_0)f$; for $f=p/q$, $\mathrm{Res}=p(z_0)/q'(z_0)$; for a pole of order m, $\mathrm{Res}=\frac{1}{(m-1)!}\lim \frac{\mathrm{d}^{m-1}}{\mathrm{d}z^{m-1}}[(z-z_0)^m f]$. **Residue theorem:** $\oint_C f\,\mathrm{d}z = 2\pi\mathrm{i}\sum_{\text{inside}} \mathrm{Res}$.

42 Singularities, principal parts and residues

Past paper: PHY 302 Mid-sem 3.3.2025, Q2 (3+3 marks)

For the following functions find their (a) singularities, (b) principal part of the function at each singularity, (c) determine the nature of singularities (i.e., whether it is a pole, a removable singularity, or an essential singularity), and (d) compute the residue of the function at each singularity: (i) $\dfrac{\sin z}{z^{2025}}$, (ii) $\dfrac{\mathrm{e}^z}{z^{2024}}$.

Solution

(i) $f(z)=\dfrac{\sin z}{z^{2025}}$.

(a) *Singularities:* the numerator is entire and the denominator vanishes only at $z=0$, so the only singularity is $z=0$.

(b) *Laurent series:* $\sin z=\sum_{k=0}^{\infty}\dfrac{(-1)^k z^{2k+1}}{(2k+1)!}$, so

$$f(z)=\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k+1)!}\,z^{2k+1-2025}=\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k+1)!}\,z^{2k-2024}.$$

The principal part consists of the terms with negative exponent: $2k-2024<0 \iff k\le 1011$:

$$\boxed{\text{P.P.}=\sum_{k=0}^{1011}\frac{(-1)^k}{(2k+1)!}\,\frac{1}{z^{2024-2k}}=\frac{1}{z^{2024}}-\frac{1}{3!\,z^{2022}}+\frac{1}{5!\,z^{2020}}-\cdots+\frac{(-1)^{1011}}{2023!\,z^{2}}}$$

(c) *Nature:* the lowest power is z^{-2024} (from $k=0$) with non-zero coefficient 1, and the principal part is finite: $z=0$ is a **pole of order 2024**.

(d) *Residue:* the coefficient of z^{-1} requires $2k-2024=-1$, i.e. $k=\frac{2023}{2}$, not an integer. Hence $\boxed{\mathrm{Res}=0}$. (Reason: $\sin z/z^{2025}$ is an *even* function of z, so only even powers appear.)

(ii) $f(z)=\dfrac{\mathrm{e}^z}{z^{2024}}$.

(a) Singularity at $z=0$ only. (b) $f=\sum_{n=0}^{\infty}\dfrac{z^{n-2024}}{n!}$; principal part ($n\le 2023$):

$$\boxed{\text{P.P.} = \sum_{n=0}^{2023} \frac{1}{n!\, z^{2024-n}} = \frac{1}{z^{2024}} + \frac{1}{z^{2023}} + \frac{1}{2!\, z^{2022}} + \cdots + \frac{1}{2023!\, z}}.$$

(c) Lowest power z^{-2024} with coefficient 1: a **pole of order 2024**. (d) The z^{-1} term has $n - 2024 = -1$, $n = 2023$: $\boxed{\text{Res} = \dfrac{1}{2023!}}$ (equivalently, $\frac{1}{2023!}\frac{\mathrm{d}^{2023}}{\mathrm{d}z^{2023}}\mathrm{e}^{z}\big|_0 = \frac{1}{2023!}$).

43 Contour integrals by the residue theorem

Past paper: PHY 302 Mid-sem 3.3.2025, Q3 (4+4 marks)

Using the residue theorem evaluate the following integrals: (i) $\oint_C \operatorname{cosec} z\, \mathrm{d}z$, (ii) $\oint_C (\mathrm{e}^{2z} - \mathrm{e}^{z})^{-1} \mathrm{d}z$, where the contour C is the circle $|z| = 3142$ oriented in the positive direction.

Solution

(i) $\operatorname{cosec} z = 1/\sin z$ has simple poles where $\sin z = 0$, i.e. at $z = n\pi$, $n \in \mathbb{Z}$. The residue is $\dfrac{1}{(\sin z)'}\Big|_{n\pi} = \dfrac{1}{\cos n\pi} = (-1)^n$.

Which poles are inside $|z| = 3142$? Need $|n\pi| < 3142 \iff |n| < \dfrac{3142}{\pi} = 1000.0\ldots$ Since $1000\pi = 3141.59\ldots < 3142$ and $1001\pi = 3144.7\ldots > 3142$, the poles inside are $n = -1000, \ldots, 1000$ (2001 poles). The sum of residues:

$$\sum_{n=-1000}^{1000} (-1)^n = \underbrace{(-1)^0}_{n=0} + 2\sum_{n=1}^{1000} (-1)^n = 1 + 2\cdot 0 = 1$$

(the sum $\sum_{n=1}^{1000}(-1)^n$ has 500 pairs $(-1+1)$, so vanishes). Therefore $\boxed{\oint_C \operatorname{cosec} z\, \mathrm{d}z = 2\pi\mathrm{i}}$.

(ii) $\mathrm{e}^{2z} - \mathrm{e}^{z} = \mathrm{e}^{z}(\mathrm{e}^{z} - 1)$. $\mathrm{e}^{z} \neq 0$ for all z, so the poles are the zeros of $\mathrm{e}^{z} - 1$: $z_k = 2\pi\mathrm{i}k$, $k \in \mathbb{Z}$. They are simple, with

$$\operatorname{Res}_{z_k} \frac{1}{\mathrm{e}^{2z} - \mathrm{e}^{z}} = \frac{1}{2\mathrm{e}^{2z} - \mathrm{e}^{z}}\Big|_{\mathrm{e}^{z}=1} = \frac{1}{2-1} = 1.$$

Poles inside: $|2\pi k| < 3142 \iff |k| < \dfrac{3142}{2\pi} = 500.06\ldots$ ($500 \cdot 2\pi = 3141.59 < 3142$, $501 \cdot 2\pi = 3147.9 > 3142$), so $k = -500, \ldots, 500$: 1001 poles, each with residue 1. Hence $\boxed{\oint_C \dfrac{\mathrm{d}z}{\mathrm{e}^{2z} - \mathrm{e}^{z}} = 2\pi\mathrm{i} \cdot 1001 = 2002\,\pi\mathrm{i}}$.

44 A real integral with a branch cut (keyhole)

Past paper: PHY 302 Mid-sem 3.3.2025, Q4 (10 marks)

Evaluate $\displaystyle\int_0^\infty \frac{x^\alpha}{1 + \sqrt{2}\,x + x^2}\,\mathrm{d}x$, $\alpha \in (-1, 1)$.

Solution

Contour. Take $f(z)=\dfrac{z^{\alpha}}{z^2+\sqrt{2}z+1}$ with the branch cut along the positive real axis and $0\le\arg z<2\pi$. Integrate around the *keyhole*: along the upper side of the cut ($\arg z=0$, $z^{\alpha}=x^{\alpha}$) from ϵ to R, round the big circle C_R anticlockwise, back along the lower side ($\arg z=2\pi$, $z^{\alpha}=x^{\alpha}\mathrm{e}^{2\pi\mathrm{i}\alpha}$) from R to ϵ, and round the small circle C_ϵ clockwise.

- On C_R: $|f|\sim R^{\alpha-2}$ and the length is $2\pi R$, so $|\int_{C_R}|\lesssim 2\pi R^{\alpha-1}\to 0$ since $\alpha<1$.
- On C_ϵ: $|f|\to\epsilon^{\alpha}$ and the length is $2\pi\epsilon$, so $|\int_{C_\epsilon}|\lesssim 2\pi\epsilon^{\alpha+1}\to 0$ since $\alpha>-1$.

These two conditions are exactly why $\alpha\in(-1,1)$. The two straight pieces give $I-\mathrm{e}^{2\pi\mathrm{i}\alpha}I=(1-\mathrm{e}^{2\pi\mathrm{i}\alpha})I$ with $I=\int_0^{\infty}\frac{x^{\alpha}}{\dots}\mathrm{d}x$.

Poles. $z^2+\sqrt{2}z+1=0\Rightarrow z=\dfrac{-\sqrt{2}\pm\mathrm{i}\sqrt{2}}{2}$, i.e. $z_1=\mathrm{e}^{3\pi\mathrm{i}/4}$, $z_2=\mathrm{e}^{5\pi\mathrm{i}/4}$ (both of modulus 1, with arguments in $(0,2\pi)$). $z_1-z_2=\dfrac{(-1+\mathrm{i})-(-1-\mathrm{i})}{\sqrt{2}}=\mathrm{i}\sqrt{2}$. Simple-pole residues ($\mathrm{Res}=z_k^{\alpha}/(z_k-z_{\text{other}})$):

$$\mathrm{Res}_{z_1}=\frac{\mathrm{e}^{3\pi\mathrm{i}\alpha/4}}{\mathrm{i}\sqrt{2}},\qquad \mathrm{Res}_{z_2}=\frac{\mathrm{e}^{5\pi\mathrm{i}\alpha/4}}{z_2-z_1}=-\frac{\mathrm{e}^{5\pi\mathrm{i}\alpha/4}}{\mathrm{i}\sqrt{2}}.$$

Residue theorem:

$$(1-\mathrm{e}^{2\pi\mathrm{i}\alpha})I=2\pi\mathrm{i}\Big[\frac{\mathrm{e}^{3\pi\mathrm{i}\alpha/4}-\mathrm{e}^{5\pi\mathrm{i}\alpha/4}}{\mathrm{i}\sqrt{2}}\Big]=\sqrt{2}\,\pi\big(\mathrm{e}^{3\pi\mathrm{i}\alpha/4}-\mathrm{e}^{5\pi\mathrm{i}\alpha/4}\big).$$

Multiply numerator and denominator by $\mathrm{e}^{-\mathrm{i}\pi\alpha}$: numerator $\mathrm{e}^{-\mathrm{i}\pi\alpha/4}-\mathrm{e}^{\mathrm{i}\pi\alpha/4}=-2\mathrm{i}\sin\frac{\pi\alpha}{4}$; denominator $\mathrm{e}^{-\mathrm{i}\pi\alpha}-\mathrm{e}^{\mathrm{i}\pi\alpha}=-2\mathrm{i}\sin\pi\alpha$. Therefore

$$\boxed{I=\frac{\sqrt{2}\,\pi\,\sin(\pi\alpha/4)}{\sin(\pi\alpha)}=\frac{\sqrt{2}\,\pi}{4\cos\frac{\pi\alpha}{4}\cos\frac{\pi\alpha}{2}}}$$

(the second form uses $\sin\pi\alpha=4\sin\frac{\pi\alpha}{4}\cos\frac{\pi\alpha}{4}\cos\frac{\pi\alpha}{2}$). **Checks:** $\alpha=0$: direct integration, $\int_0^{\infty}\frac{\mathrm{d}x}{(x+\frac{1}{\sqrt{2}})^2+\frac{1}{2}}=\sqrt{2}\big[\arctan(\sqrt{2}x+1)\big]_0^{\infty}=\sqrt{2}(\frac{\pi}{2}-\frac{\pi}{4})=\frac{\sqrt{2}\pi}{4}$, and the formula gives $\frac{\sqrt{2}\pi}{4}$. ✓As $\alpha\to\pm1$ ($\cos\frac{\pi\alpha}{2}\to0$) the integral diverges, consistent with the convergence condition.

45 Binomial coefficients and a sum by residues

Past paper: PHY 302 Mid-sem 3.3.2025, Q6 (3+7 marks)

Using residue theorem show that the binomial coefficients can be represented as ${}^{n}C_k=\dfrac{1}{2\pi\mathrm{i}}\oint_C\dfrac{(1+z)^n}{z^{k+1}}\mathrm{d}z$, where C is any positively oriented contour surrounding the origin. Afterwards using this compute the sum $S=\displaystyle\sum_{n=0}^{\infty}{}^{2n}C_n\frac{1}{5^n}$.

Solution

Representation. The integrand $\dfrac{(1+z)^n}{z^{k+1}}$ is analytic except at $z=0$. Expand $(1+z)^n = \sum_{j=0}^{n}\binom{n}{j}z^j$:

$$\frac{(1+z)^n}{z^{k+1}} = \sum_{j=0}^{n}\binom{n}{j}z^{j-k-1}.$$

The residue at 0 is the coefficient of z^{-1}, i.e. $j-k-1=-1 \Rightarrow j=k$: Res $=\binom{n}{k}$ (zero if $k>n$). By the residue theorem, $\dfrac{1}{2\pi\mathrm{i}}\oint_C = \binom{n}{k} = {}^nC_k$. ■

The sum. With $n\to 2n$, $k=n$: ${}^{2n}C_n = \dfrac{1}{2\pi\mathrm{i}}\oint \dfrac{(1+z)^{2n}}{z^{n+1}}\mathrm{d}z$. Therefore

$$S=\sum_{n=0}^{\infty}\frac{1}{5^n}\cdot\frac{1}{2\pi\mathrm{i}}\oint\frac{(1+z)^{2n}}{z^{n+1}}\mathrm{d}z = \frac{1}{2\pi\mathrm{i}}\oint\frac{1}{z}\sum_{n=0}^{\infty}\left[\frac{(1+z)^2}{5z}\right]^n\mathrm{d}z,$$

interchanging sum and integral, legitimate if the geometric series converges uniformly on C. Let $q=\dfrac{(1+z)^2}{5z}$. Choose C: $|z|=1$; then $|q|=\dfrac{|1+z|^2}{5}\le\dfrac{4}{5}<1$ ✓. Sum the series, $\sum q^n = \dfrac{1}{1-q}$:

$$S=\frac{1}{2\pi\mathrm{i}}\oint\frac{1}{z}\cdot\frac{1}{1-\frac{(1+z)^2}{5z}}\mathrm{d}z = \frac{1}{2\pi\mathrm{i}}\oint\frac{5\,\mathrm{d}z}{5z-(1+z)^2} = \frac{1}{2\pi\mathrm{i}}\oint\frac{-5\,\mathrm{d}z}{z^2-3z+1}.$$

(The factor $1/z$ cancelled with the z in the denominator, so there is no pole at 0.) The roots of z^2-3z+1 are $z_\pm=\dfrac{3\pm\sqrt{5}}{2}$: $z_-=0.382$ lies inside $|z|=1$, $z_+=2.618$ outside. Residue at z_-: $\dfrac{-5}{z_- - z_+}=\dfrac{-5}{-\sqrt{5}}=\sqrt{5}$. Hence

$$\boxed{S=\sqrt{5}\approx 2.236}$$

Check: the generating function $\sum\binom{2n}{n}x^n=(1-4x)^{-1/2}$ at $x=\frac{1}{5}$ gives $(1-\frac{4}{5})^{-1/2}=\sqrt{5}$; the first terms $1+0.4+0.24+0.16+0.112+0.081+0.059+\cdots$ approach 2.236.

46 Fibonacci numbers and the golden ratio

Past paper: PHY 302 Mid-sem 4.3.2024, Q5 (2+4+5+5+4 marks)

The n-th Fibonacci number f_n ($n\ge 0$) is defined by $f_0=1$, $f_1=1$, $f_n=f_{n-1}+f_{n-2}$ for $n\ge 2$. Consider $F(z)=\sum_{k=0}^{\infty}f_kz^k$. (i) Prove by induction that $f_n\le 2^n$ for all n. (ii) Show that the recurrence relation implies $F(z)=(1-z-z^2)^{-1}$. (*Hint:* write down the Taylor series for $zF(z)$ and $z^2F(z)$ and add.) (iii) Verify that Res $\left(\dfrac{1}{z^{n+1}(1-z-z^2)};0\right)=f_n$. (iv) Using the residue theorem show that $f_n=\dfrac{1}{\sqrt{5}}\left[\left(\dfrac{1+\sqrt{5}}{2}\right)^{n+1}-\left(\dfrac{1-\sqrt{5}}{2}\right)^{n+1}\right]$. (*Hint:* integrate $[z^{n+1}(1-z-z^2)]^{-1}$ around a circle with centre at the origin and radius R and show that the integral vanishes as $R\to\infty$.) (v) Evaluate $\Phi=\lim_{n\to\infty}\left(\dfrac{f_n}{f_{n-1}}\right)$ (up to four decimal places; $\sqrt{5}=2.23607$).

Solution

(i) Induction. *Base:* $f_0 = 1 \le 2^0 = 1$ and $f_1 = 1 \le 2^1 = 2$. *Step:* suppose $f_k \le 2^k$ for all $k < n$ ($n \ge 2$). Then

$$f_n = f_{n-1} + f_{n-2} \le 2^{n-1} + 2^{n-2} = 3 \cdot 2^{n-2} < 4 \cdot 2^{n-2} = 2^n.$$

Hence $f_n \le 2^n$ for all n. ■ (Consequence: $\sum f_k z^k$ converges for $|z| < \frac{1}{2}$ by comparison with $\sum (2z)^k$, so F is analytic near 0.)

(ii) Generating function. $zF = \sum_{k\ge 0} f_k z^{k+1} = \sum_{k\ge 1} f_{k-1} z^k$ and $z^2F = \sum_{k\ge 2} f_{k-2} z^k$. Then

$$F - zF - z^2F = f_0 + (f_1 - f_0)z + \sum_{k\ge 2} \big(f_k - f_{k-1} - f_{k-2}\big) z^k = 1 + 0 \cdot z + 0 = 1,$$

using $f_0 = f_1 = 1$ and the recurrence. So $F(z)(1 - z - z^2) = 1$, i.e. $\boxed{F(z) = \dfrac{1}{1 - z - z^2}}$.

(iii) Residue. For a small circle $|z| = \rho < \frac{1}{2}$ (inside the disk of convergence), Cauchy's formula for Taylor coefficients gives $f_n = \dfrac{1}{2\pi \mathrm{i}} \oint \dfrac{F(z)}{z^{n+1}} \mathrm{d}z = \dfrac{1}{2\pi \mathrm{i}} \oint \dfrac{\mathrm{d}z}{z^{n+1}(1 - z - z^2)}$. The only singularity inside this small circle is the pole at $z = 0$ (the zeros of $1 - z - z^2$, found below, have modulus $\ge 0.618 > \rho$). Hence $\operatorname{Res}\Big(\dfrac{1}{z^{n+1}(1 - z - z^2)}; 0\Big) = f_n$. ✓ (Equivalently: the coefficient of z^{-1} in the Laurent series $z^{-n-1} \sum f_k z^k$ is f_n.)

(iv) Explicit formula. Consider $\oint_{|z|=R} g(z)\mathrm{d}z$ with $g(z) = \dfrac{1}{z^{n+1}(1 - z - z^2)}$. For large R, $|g| \le \dfrac{1}{R^{n+1}(R^2 - R - 1)} \sim R^{-(n+3)}$, and the length is $2\pi R$: the integral is $O(R^{-(n+2)}) \to 0$ as $R \to \infty$ (for all $n \ge 0$). By the residue theorem the integral equals $2\pi\mathrm{i} \times$(sum of *all* residues inside), and since it vanishes, the sum of all residues is zero:

$$\operatorname{Res}_{z=0} + \operatorname{Res}_{z=z_1} + \operatorname{Res}_{z=z_2} = 0 \;\Rightarrow\; f_n = -\operatorname{Res}_{z_1} - \operatorname{Res}_{z_2}.$$

The other poles are the roots of $1 - z - z^2 = 0$: $z_{1,2} = \dfrac{-1 \pm \sqrt{5}}{2}$. Write $1 - z - z^2 = -(z - z_1)(z - z_2)$ and note $z_1 - z_2 = \sqrt{5}$. Then

$$\operatorname{Res}_{z_1} = \frac{1}{z_1^{n+1} \cdot \big[-(z_1 - z_2)\big]} = -\frac{1}{\sqrt{5}\, z_1^{n+1}}, \qquad \operatorname{Res}_{z_2} = \frac{1}{z_2^{n+1} \cdot \big[-(z_2 - z_1)\big]} = +\frac{1}{\sqrt{5}\, z_2^{n+1}}.$$

Therefore $f_n = \dfrac{1}{\sqrt{5}} \Big[z_1^{-(n+1)} - z_2^{-(n+1)} \Big]$. Now $z_1^{-1} = \dfrac{2}{\sqrt{5} - 1} = \dfrac{2(\sqrt{5}+1)}{4} = \dfrac{1 + \sqrt{5}}{2}$ and $z_2^{-1} = \dfrac{2}{-1 - \sqrt{5}} = -\dfrac{2(\sqrt{5}-1)}{4} = \dfrac{1 - \sqrt{5}}{2}$. Thus

$$\boxed{f_n = \frac{1}{\sqrt{5}} \Big[\Big(\frac{1 + \sqrt{5}}{2}\Big)^{n+1} - \Big(\frac{1 - \sqrt{5}}{2}\Big)^{n+1} \Big]}$$

Check: $n = 0$: $\frac{1}{\sqrt{5}} \cdot \frac{(1+\sqrt{5}) - (1-\sqrt{5})}{2} = 1 = f_0$ ✓; $n = 1$: $\frac{1}{\sqrt{5}} \frac{(1+\sqrt{5})^2 - (1-\sqrt{5})^2}{4} = \frac{1}{\sqrt{5}} \frac{4\sqrt{5}}{4} = 1 = f_1$ ✓; $n = 2$: $\varphi^3 - \psi^3 = 2\sqrt{5} \Rightarrow f_2 = 2$ ✓ ($\varphi = \frac{1+\sqrt{5}}{2}$, $\psi = \frac{1-\sqrt{5}}{2}$).

(v) Golden ratio. Since $|\psi| = 0.618 < \varphi = 1.618$, $\psi^n/\varphi^n \to 0$ and

$$\frac{f_n}{f_{n-1}} = \frac{\varphi^{n+1} - \psi^{n+1}}{\varphi^n - \psi^n} \longrightarrow \varphi = \frac{1 + \sqrt{5}}{2} = \frac{1 + 2.23607}{2} = \boxed{1.6180}$$

(equivalently $\Phi = 1 + 1/\Phi \Rightarrow \Phi^2 - \Phi - 1 = 0$).

Practice problems

1. Find the nature of the singularity of $\dfrac{1-\cos z}{z^2}$, $\dfrac{\mathrm{e}^{1/z}}{z}$ and $\dfrac{\sin z}{z^3}$ at $z=0$ and the residues. *Answer:* removable (value $\frac12$), residue 0; essential, residue 1 ($\mathrm{e}^{1/z}/z = \sum z^{-n-1}/n!$, coefficient of z^{-1} is 1); double pole, residue 0.

2. Evaluate $\displaystyle\oint_{|z|=4} \frac{\mathrm{e}^z}{z^2(z-1)}\mathrm{d}z$ and $\displaystyle\oint_{|z|=4} \tan z\,\mathrm{d}z$. *Answer:* $2\pi\mathrm{i}(\mathrm{e}-2)$; $\tan z = \sin z/\cos z$ has simple poles at $\pm\frac{\pi}{2}$ (inside, $|\pm 1.571| < 4$) and $\pm\frac{3\pi}{2} = \pm 4.712$ (outside), each of residue $\frac{\sin(\pi/2)}{-\sin(\pi/2)} = -1$: result $2\pi\mathrm{i}(-2) = -4\pi\mathrm{i}$.

3. $\displaystyle\int_0^\infty \frac{x^\alpha}{1+x^2}\mathrm{d}x$, $-1<\alpha<1$. *Answer:* $\dfrac{\pi}{2\cos(\pi\alpha/2)}$.

4. Show that $\displaystyle\int_0^{2\pi} \frac{\mathrm{d}\theta}{5+4\cos\theta} = \frac{2\pi}{3}$ and $\int_0^\infty \frac{\mathrm{d}x}{x^4+1} = \frac{\pi}{2\sqrt{2}}$.

5. The Lucas numbers satisfy $L_0=2$, $L_1=1$, $L_n = L_{n-1}+L_{n-2}$. Find the generating function and L_n. *Answer:* $\dfrac{2-z}{1-z-z^2}$; $L_n = \varphi^n + \psi^n$.

6. Using the method of this part, show $\displaystyle\sum_{n=0}^{\infty}\binom{2n}{n}t^n = \frac{1}{\sqrt{1-4t}}$ and state its radius of convergence. *Answer:* repeat the steps with $q = t(1+z)^2/z$; the series converges for $|t| < \frac14$ and diverges at $t=\frac14$ (terms $\sim 1/\sqrt{\pi n}$).

Part IX
Study Aids

47 Index of past-paper questions

Paper	Q	Topic	Where solved
Mid-sem 4.3.2024 (50)	1	S_3 as 3×3 permutation matrices, inverses	Part VII, Permutation groups
	2	Group of matrices $g(z,x)$	Part VII, Matrix groups
	3	Rotation matrices with $M^2=-I$; $\mathbb{Z}_4$	Part VII, Matrix groups
	4	Lippmann–Schwinger Green's function	Part V, Free-space Green's fn
	5	Fibonacci numbers, generating function	Part VIII, Fibonacci
Mid-sem 3.3.2025 (50)	1	S_3 again (same as above)	Part VII
	2	Singularities, principal parts, residues	Part VIII, Singularities
	3	$\oint \operatorname{cosec} z$, $\oint (\mathrm{e}^{2z}-\mathrm{e}^{z})^{-1}$	Part VIII, Contour integrals
	4	$\int_0^\infty \frac{x^\alpha}{1+\sqrt{2}x+x^2}$	Part VIII, Keyhole
	5	Hermitian SL operator, even eigenfunctions	Part I, SL theory
	6	Binomial coefficients by residues; $\sum \binom{2n}{n}5^{-n}$	Part VIII
	7	2×2 Hermitian, $SU(2)$, Pauli matrices	Part VII, $SU(2)$
Mid-sem 26.02.2026 (45)	1	$\int (x+1)^2\delta(\sin\pi x)$	Part III, Delta function
	2	$\mathcal{L}[H(t-a)\mathrm{e}^{-\mathrm{i}\omega(t-a)}]$	Part II, Laplace
	3	Convolution theorem (Laplace)	Part II, Laplace
	4a	Fourier series of Pt^2+Qt	Part II, Fourier series
	4b	$y''+3y=t^2$	Part II, Laplace/ODE
	5a	$\mathcal{L}[y'']$	Part II, Laplace
	5b	$y''+\omega^2 y=F\delta(t-t_0)$ by Laplace	Part II, Laplace
	6	3D radial Fourier transform; Yukawa	Part II, 3D FT
End-sem 27.04.2024 (70)	1	Gaussian: FT, $\langle t\rangle, \langle\omega\rangle, \Delta t, \Delta\omega$	Part II, Gaussian
	2	FT of an integral	Part II, FT of integral
	3	$xu_x-2yu_y=0$	Part IV, Characteristics
	4	$u_{xx}+2u_{xy}+u_{yy}=0$ with BC	Part IV, Classification
	5	Semi-infinite cylinder	Part IV, Bessel
	6	Laplace's equation in a disk	Part IV, Polar
End-sem 05.05.2025 (100)	1–3	(Gaussian FT; first-order PDE; semi-infinite cylinder)	same as above
	4	Split conducting sphere	Part IV, Spherical
	5	Laplace's equation in a disk	Part IV, Polar

Patterns. Every end-sem has: a Gaussian Fourier transform, a first-order characteristic PDE, a cylinder/disk/sphere Laplace problem solved by separation of variables. Every mid-sem has: S_3 or group matrices, residue calculus or Fourier/Laplace transforms, and a Green's-function or delta-function question. These are the highest-priority topics.

48 Formula sheet

Differential operators and separation solutions (exam sheet)

$$\nabla^2_{\text{polar}} = \frac{1}{\rho}\partial_\rho(\rho\partial_\rho)+\frac{1}{\rho^2}\partial^2_\phi, \quad \nabla^2_{\text{cyl}} = \nabla^2_{\text{polar}}+\partial^2_z, \quad \nabla^2_{\text{sph}} = \frac{1}{r^2}\partial_r(r^2\partial_r)+\frac{1}{r^2\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta)+\frac{1}{r^2\sin^2\theta}\partial^2_\phi.$$

Polar	$u = (C_0\ln\rho + D_0) + \sum_{n\geq 1}(A_n\cos n\phi + B_n\sin n\phi)(C_n\rho^n + D_n\rho^{-n})$
Cylindrical	$u = [AJ_m(k\rho) + BY_m(k\rho)][C\cos m\phi + D\sin m\phi][E\mathrm{e}^{-kz} + F\mathrm{e}^{kz}]$
Spherical	$u = (Ar^l + Br^{-(l+1)})(C\cos m\phi + D\sin m\phi)[EP_l^m(\cos\theta) + FQ_l^m(\cos\theta)]$
Cartesian (2D)	$u = [A\mathrm{e}^{\lambda x} + B\mathrm{e}^{-\lambda x}][C\cos\lambda y + D\sin\lambda y]$
Heat (1D)	$u = (A\cos\lambda x + B\sin\lambda x)\mathrm{e}^{-\lambda^2\kappa t}$
Fourier–Bessel	$f = \sum c_n J_\nu(\alpha_n x)$, $c_n = \dfrac{2}{b^2 J^2_{\nu+1}(\alpha_n b)}\int_0^b f(x)J_\nu(\alpha_n x)x\,\mathrm{d}x$
Legendre expansion	$f = \sum a_l P_l(x)$, $a_l = \dfrac{2l+1}{2}\int_{-1}^1 f(x)P_l(x)\mathrm{d}x$

Bessel: $\dfrac{\mathrm{d}}{\mathrm{d}z}[zJ_1(z)] = zJ_0(z)$, $\int xJ_0(x)\mathrm{d}x = xJ_1(x) + C$, $J_{\nu-1} + J_{\nu+1} = \frac{2\nu}{x}J_\nu$, $J_{\nu-1} - J_{\nu+1} = 2J'_\nu$.
Legendre: $(n+1)P_{n+1} = (2n+1)xP_n - nP_{n-1}$; $\int_{-1}^1 P_l P_{l'} = \frac{2}{2l+1}\delta_{ll'}$; $P_n(0) = 0$ (n odd), $P_0(0) = 1, P_2(0) = -\frac{1}{2}, P_4(0) = \frac{3}{8}, P_6(0) = -\frac{5}{16}, P_8(0) = \frac{35}{128}$.

Zeros of $J_n(x)$ (exam table)

k	J_0	J_1	J_2	J_3	J_4	J_5
1	2.4048	3.8317	5.1356	6.3802	7.5883	8.7715
2	5.5201	7.0156	8.4172	9.7610	11.0647	12.3386
3	8.6537	10.1735	11.6198	13.0152	14.3725	15.7002
4	11.7915	13.3237	14.7960	16.2235	17.6160	18.9801
5	14.9309	16.4706	17.9598	19.4094	20.8269	22.2178

Transforms

Fourier	$g(\omega) = \int f\mathrm{e}^{-\mathrm{i}\omega t}\mathrm{d}t$; $f' \to \mathrm{i}\omega g$; $f * h \to gh_\omega$; $\int\lvert f\rvert^2 = \frac{1}{2\pi}\int\lvert g\rvert^2$
Gaussian	$\mathrm{e}^{-at^2} \leftrightarrow \sqrt{\pi/a}\,\mathrm{e}^{-\omega^2/4a}$; $\mathrm{e}^{-a\lvert t\rvert} \leftrightarrow \frac{2a}{a^2+\omega^2}$
Integral	$\mathrm{FT}\int^t f = \frac{g}{\mathrm{i}\omega} + 2\pi c\delta(\omega)$
Laplace	$\mathcal{L}f' = sF - f(0)$; $\mathcal{L}f'' = s^2F - sf(0) - f'(0)$; $\mathcal{L}[H(t-a)f(t-a)] = \mathrm{e}^{-as}F$; $\mathcal{L}[f*g] = FG$
Inversion	$f(t) = \frac{1}{2\pi\mathrm{i}}\int_{\gamma-\mathrm{i}\infty}^{\gamma+\mathrm{i}\infty}\mathrm{e}^{st}F\,\mathrm{d}s = \sum\mathrm{Res}[\mathrm{e}^{st}F]$
Sine/cosine (PDE)	$\mathcal{F}_s(u_{xx}) = \sqrt{\frac{2}{\pi}}su(0) - s^2\tilde{u}_s$; $\mathcal{F}_c(u_{xx}) = -\sqrt{\frac{2}{\pi}}u_x(0) - s^2\tilde{u}_c$
Delta	$\delta(g(x)) = \sum\frac{\delta(x-x_k)}{\lvert g'(x_k)\rvert}$; $\int f\delta' = -f'(0)$; $\delta(x) = \frac{1}{2\pi}\int\mathrm{e}^{\mathrm{i}kx}\mathrm{d}k$
3D radial FT	$\tilde{f}(k) = \frac{4\pi}{k}\int_0^\infty rf(r)\sin kr\,\mathrm{d}r$; $\frac{\mathrm{e}^{-\alpha r}}{r} \to \frac{4\pi}{k^2+\alpha^2}$

PDEs, Green's functions, integral equations, groups

Classification	$\Delta = B^2 - AC$: >0 hyperbolic, $=0$ parabolic, <0 elliptic
d'Alembert	$u = \frac{1}{2}[\varphi(x-ct)+\varphi(x+ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct}\psi$
Poisson (disk)	$u = \frac{1}{2\pi}\int \frac{a^2-\rho^2}{a^2-2a\rho\cos(\phi-\phi')+\rho^2} f(\phi')\mathrm{d}\phi'$
Green (SL)	$G = \frac{u_1(x_<)u_2(x_>)}{pW}$; free space ∇^2: $-\frac{1}{4\pi R}$; ∇^2+k^2: $-\frac{\mathrm{e}^{\mathrm{i}kR}}{4\pi R}$
Representation	$u(\boldsymbol{r}_0) = \int G\rho\,\mathrm{d}V + \oint[u\partial_n G - G\partial_n u]\mathrm{d}S$
Images (sphere)	$\boldsymbol{r}_1 = \frac{a^2}{r_0^2}\boldsymbol{r}_0$; $\lvert\boldsymbol{r}-\boldsymbol{r}_1\rvert = \frac{a}{r_0}\lvert\boldsymbol{r}-\boldsymbol{r}_0\rvert$ on $\lvert\boldsymbol{r}\rvert = a$
Integral eq.	$y = \sum \lambda^n \mathcal{K}^n f$ (Neumann); separable $\Rightarrow$ algebraic system; Abel: $y = \frac{1}{\pi}\frac{\mathrm{d}}{\mathrm{d}x}\int_0^x \frac{f}{\sqrt{x-t}}$
Groups	Lagrange: $\lvert H\rvert \,\big\vert\, \lvert G\rvert$; $\sum_\alpha d_\alpha^2 = \lvert G\rvert$; $n_\alpha = \frac{1}{\lvert G\rvert}\sum \chi^{(\alpha)*}\chi$; $\chi_j = \frac{\sin(j+\frac{1}{2})\theta}{\sin\frac{\theta}{2}}$; $j_1 \otimes j_2 = \bigoplus_{\lvert j_1-j_2\rvert}^{j_1+j_2} j$

49 Common mistakes to avoid

1. **Forgetting the weight function** in orthogonality (Bessel: weight x; Hermite: e^{-x^2}). Normalisation integrals then come out wrong.
2. **Ignoring boundary terms** when claiming an operator is Hermitian; check $[p(vu'-uv')]_a^b = 0$.
3. **Fourier convention:** state it. The statement $\mathrm{FT}(c) = 2\pi c\,\delta(\omega)$ and Parseval's factor $\frac{1}{2\pi}$ depend on it.
4. **Fourier series at jumps:** the series converges to the midpoint, not to f.
5. **Laplace transform of derivatives:** do not forget the initial conditions $y(0), y'(0)$. The second shift theorem needs $H(t-a)f(t-a)$, not $f(t)$.
6. $\delta(g(x))$**:** include all zeros inside the range and divide by $\lvert g'\rvert$ at each.
7. **Characteristics:** the general solution of a first-order PDE contains an arbitrary *function*, not a constant; a single point value fixes nothing (Question: $f(1) = 4$).
8. **Separation constants:** choose their sign according to the boundary conditions ($\sin, \cos$ in the homogeneous direction; $\sinh$ or e^{-kz} in the other). In cylindrical problems use J_m (not Y_m) when the axis is included.
9. **Legendre series:** the coefficients use $\frac{2l+1}{2}$; interior solutions have r^l, exterior $r^{-(l+1)}$.
10. **Green's functions:** the jump condition is $pG'\big\rvert_{x'-}^{x'+} = 1$, and G must satisfy the *same homogeneous boundary conditions* as u.
11. **Integral equations:** Volterra Neumann series always converge; Fredholm ones only for $\lvert\lambda\rvert < 1/\lVert K\rVert$.
12. **Groups:** closure is the axiom most often violated (e.g. $\{I, M_+, M_-\}$); inverse of a permutation matrix is its transpose; classes of S_n are cycle types; number of irreps = number of classes.
13. $SU(2)$ **vs** $SO(3)$**:** $\mathrm{e}^{-\mathrm{i}2\pi\sigma_z/2} = -\mathbb{1}$; spinors are double-valued on $SO(3)$.

End of PHY 302 booklet.